

STAT 5511 Midterm

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Question 1 (Analysis of dat1)

Summary

For the series $Y_t = X_t$ I chose an ARIMA(1,1,2) model excluding the intercept term. The parameter estimates are $\phi_1 = -0.582$, $\theta_1 = 0.5722$, and $\theta_2 = 0.4589$.

```
load("mid-data.rsav")
library(astsa)
```

```
xx <- dat1
sarima(xx, 1,1,2,no.constant=T,details=F)
```

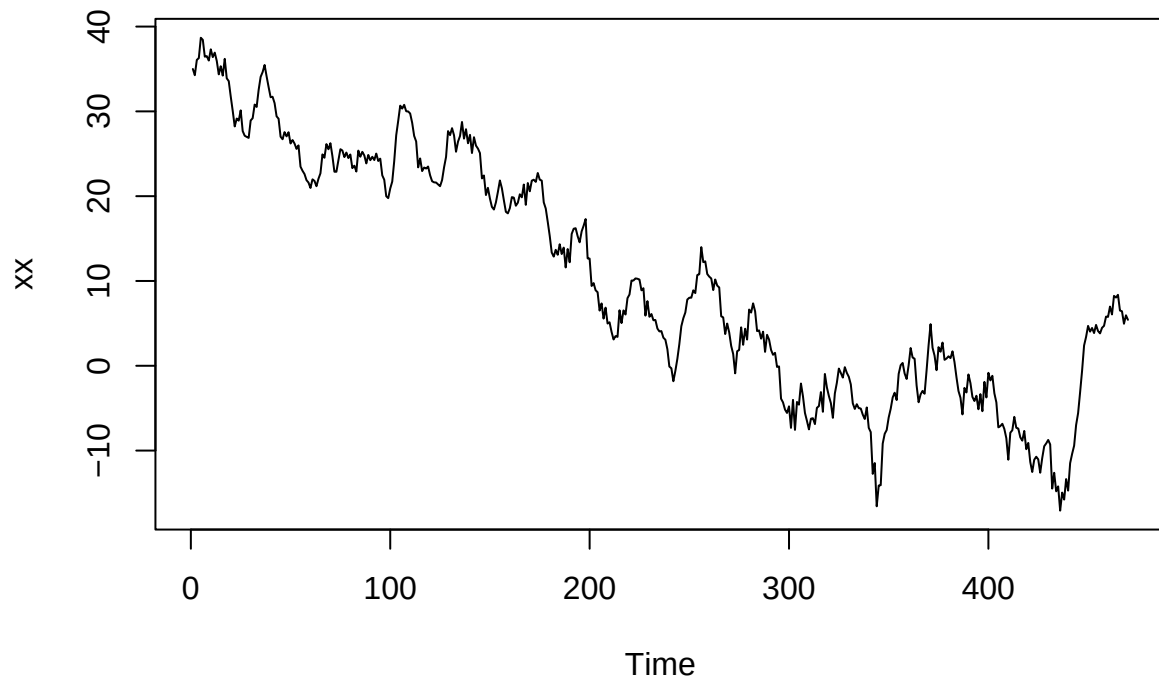
```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1  -0.5802 0.0682 -8.5025      0
## ma1   0.5722 0.0678  8.4337      0
## ma2   0.4589 0.0405 11.3233      0
##
## sigma^2 estimated as 1.783545 on 466 degrees of freedom
##
## AIC = 3.434944  AICc = 3.435054  BIC = 3.470343
##
```

Explanation

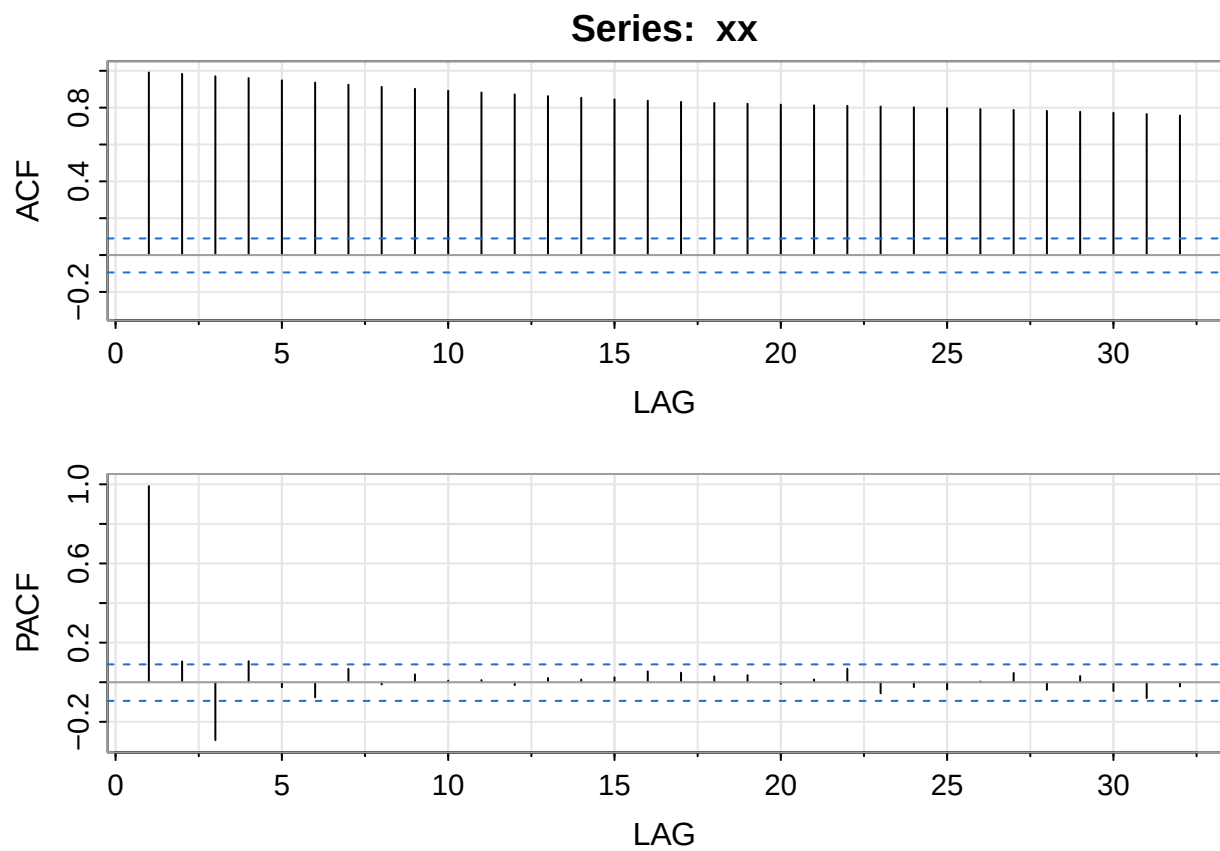
The initial plot and ACF and PACF of X_t suggest that we need to perform a first difference because it is not stationary. After the first difference, the ACF and PACF suggest that there is no seasonality. The significant lags of the ACF cut off for the most part after lag 4 and those of the PACF cut off after lag 5, so we fit different models on the first differenced data with a maximum p of 5 and a maximum q of 4. 15 of the 30 models had an AICc multiplied by the length of the series within 5 of the minimum, and only five had a BIC multiplied by the length of the series within 10 of the minimum. The five which met both of these criteria were ARMA(1,2), ARMA(1,3), ARMA(2,2), ARMA(2,3), and ARMA(5,0). Of these, the ARMA(1,2) was chosen because it has the second best AICc, the best BIC, and the highest LBP values. So the final model on the original series X_t is ARIMA(1,1,2). The intercept was not included because it was not significant at any level.

Output

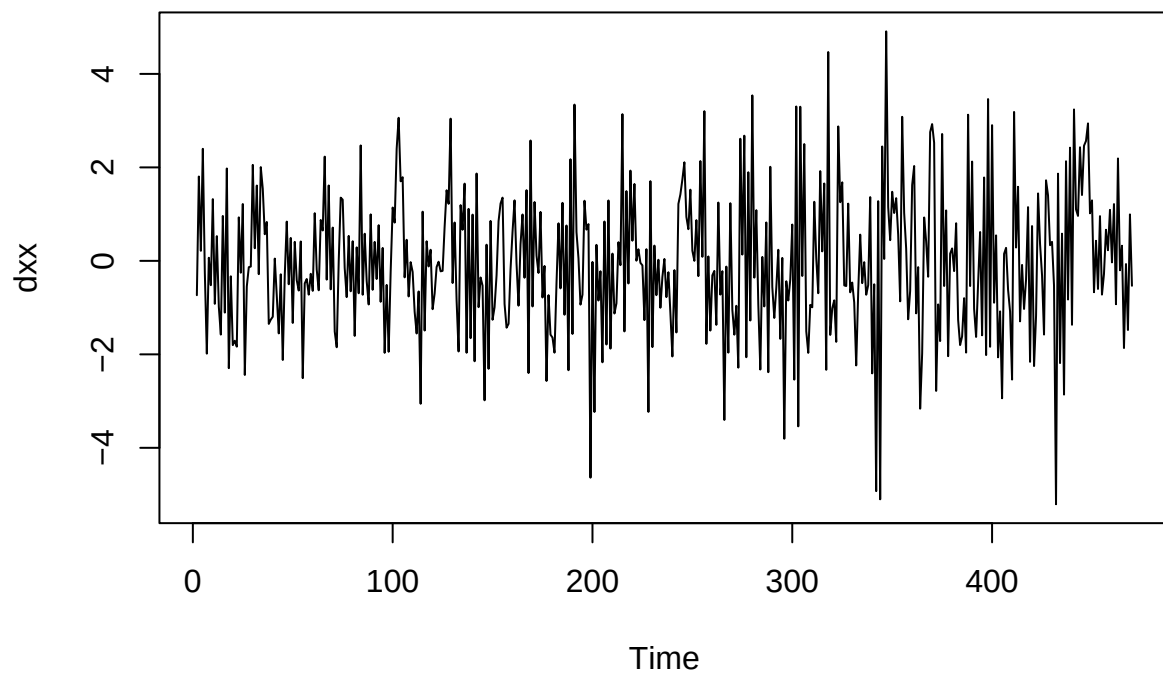
```
plot(xx)
```



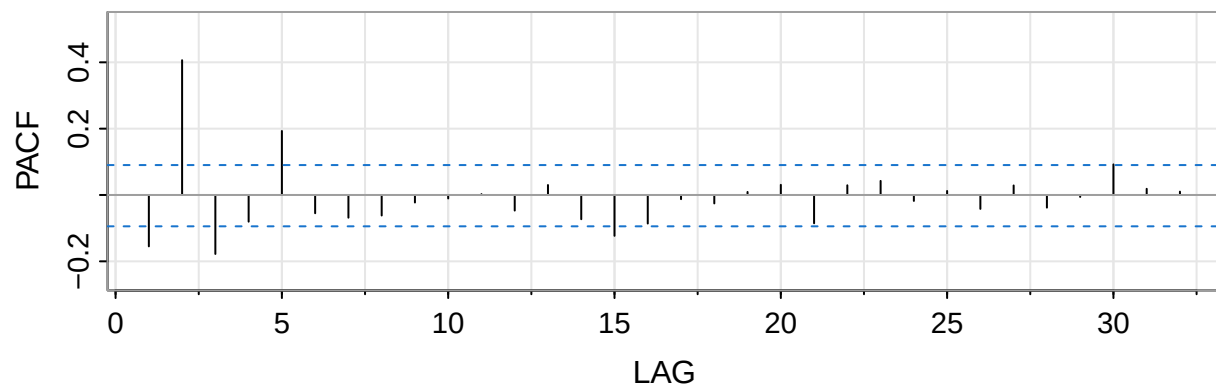
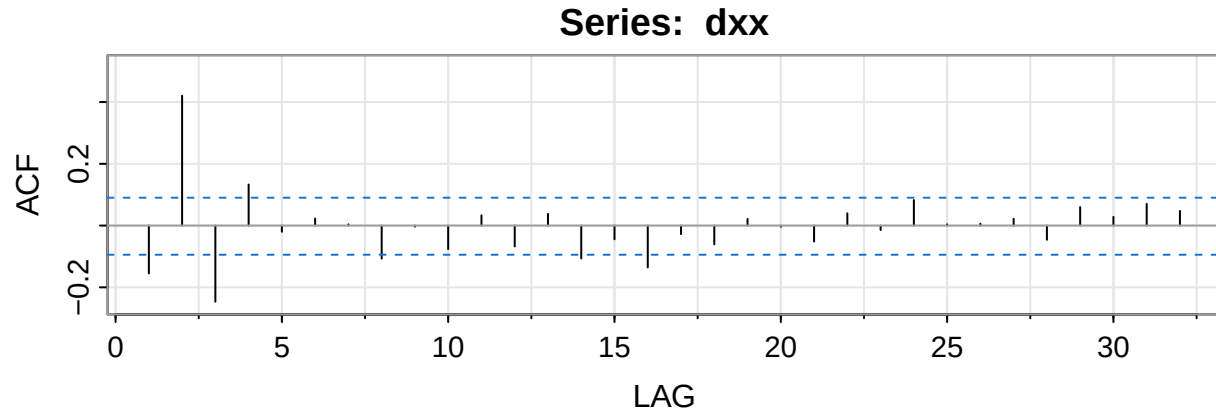
```
invisible(acf2(xx))
```



```
dxx <- diff(xx)
plot(dxx)
```



```
invisible(acf2(dxx))
```



```
maxAR<-5+1
maxMA<-4+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
  fits_all[[ii]]<-vector("list",length=maxMA)
  for (jj in 1:maxMA){
    fits_all[[ii]][[jj]]<-invisible(sarima(dxx,p=ii-1,d=0,q=jj-1,no.constant=FALSE, details=FALSE))
    if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
    if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
  }
}
```

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## xmean  -0.0631 0.0708 -0.8907  0.3735
##
## sigma^2 estimated as 2.351619 on 468 degrees of freedom
##
## AIC = 3.70151  AICc = 3.701528  BIC = 3.71921
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
```

```

## ma1      -0.0875 0.0349 -2.5067  0.0125
## xmean    -0.0629 0.0642 -0.9806  0.3273
##
## sigma^2 estimated as 2.320013 on 467 degrees of freedom
##
## AIC = 3.69226  AICc = 3.692314  BIC = 3.718809
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0861 0.0479  1.7964  0.0731
## ma2      0.4554 0.0454 10.0201  0.0000
## xmean   -0.0630 0.0990 -0.6356  0.5254
##
## sigma^2 estimated as 1.941816 on 466 degrees of freedom
##
## AIC = 3.519557  AICc = 3.519667  BIC = 3.554956
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1     -0.0056 0.0446 -0.1246  0.9009
## ma2      0.4097 0.0403 10.1708  0.0000
## ma3     -0.2267 0.0396 -5.7292  0.0000
## xmean   -0.0621 0.0733 -0.8465  0.3977
##
## sigma^2 estimated as 1.82052 on 465 degrees of freedom
##
## AIC = 3.459542  AICc = 3.459726  BIC = 3.503791
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1     -0.0105 0.0463 -0.2276  0.8201
## ma2      0.4246 0.0442  9.6069  0.0000
## ma3     -0.2268 0.0405 -5.6008  0.0000
## ma4      0.1215 0.0516  2.3538  0.0190
## xmean   -0.0609 0.0810 -0.7517  0.4526
##
## sigma^2 estimated as 1.799092 on 464 degrees of freedom
##
## AIC = 3.451988  AICc = 3.452265  BIC = 3.505088
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.1547 0.0456 -3.3938  0.0007
## xmean   -0.0627 0.0606 -1.0355  0.3010
##
## sigma^2 estimated as 2.295122 on 467 degrees of freedom

```

```

##
## AIC = 3.681508  AICc = 3.681563  BIC = 3.708058
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.8091 0.0468 -17.2788  0.0000
## ma1       0.6043 0.0565  10.7043  0.0000
## xmean    -0.0621 0.0593  -1.0473  0.2955
##
## sigma^2 estimated as 2.096233 on 466 degrees of freedom
##
## AIC = 3.59545  AICc = 3.59556  BIC = 3.630849
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.5811 0.0681 -8.5302  0.0000
## ma1       0.5721 0.0677  8.4452  0.0000
## ma2       0.4581 0.0406 11.2804  0.0000
## xmean    -0.0612 0.0791 -0.7736  0.4396
##
## sigma^2 estimated as 1.781286 on 465 degrees of freedom
##
## AIC = 3.437939  AICc = 3.438123  BIC = 3.482188
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.5606 0.1038 -5.3986  0.0000
## ma1       0.5481 0.1126  4.8655  0.0000
## ma2       0.4490 0.0518  8.6706  0.0000
## ma3      -0.0200 0.0725 -0.2753  0.7832
## xmean    -0.0611 0.0780 -0.7827  0.4342
##
## sigma^2 estimated as 1.781008 on 464 degrees of freedom
##
## AIC = 3.442043  AICc = 3.442319  BIC = 3.495142
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.5483 0.1610 -3.4050  0.0007
## ma1       0.5360 0.1648  3.2530  0.0012
## ma2       0.4500 0.0529  8.5124  0.0000
## ma3      -0.0237 0.0819 -0.2898  0.7721
## ma4       0.0071 0.0691  0.1027  0.9182
## xmean    -0.0610 0.0783 -0.7792  0.4362
##
## sigma^2 estimated as 1.780965 on 463 degrees of freedom

```

```

##
## AIC = 3.446285  AICc = 3.446673  BIC = 3.508234
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.0922 0.0421 -2.1908 0.0290
## ar2       0.4067 0.0421  9.6560 0.0000
## xmean    -0.0605 0.0930 -0.6506 0.5156
##
## sigma^2 estimated as 1.913955 on 466 degrees of freedom
##
## AIC = 3.50493  AICc = 3.50504  BIC = 3.540329
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.2842 0.0735 -3.8647 0.0001
## ar2       0.3859 0.0452  8.5438 0.0000
## ma1       0.2365 0.0722  3.2752 0.0011
## xmean    -0.0608 0.0870 -0.6996 0.4845
##
## sigma^2 estimated as 1.877439 on 465 degrees of freedom
##
## AIC = 3.490036  AICc = 3.49022  BIC = 3.534285
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.6059 0.1142 -5.3068 0.0000
## ar2      -0.0270 0.0972 -0.2776 0.7815
## ma1       0.5932 0.1034  5.7386 0.0000
## ma2       0.4750 0.0720  6.5944 0.0000
## xmean    -0.0610 0.0780 -0.7815 0.4349
##
## sigma^2 estimated as 1.780994 on 464 degrees of freedom
##
## AIC = 3.442035  AICc = 3.442311  BIC = 3.495134
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.4001 0.0703  5.6901 0.0000
## ar2       0.5509 0.0702  7.8487 0.0000
## ma1      -0.4281 0.0701 -6.1034 0.0000
## ma2      -0.1038 0.0790 -1.3133 0.1897
## ma3      -0.4682 0.0405 -11.5480 0.0000
## xmean    -0.0837 0.0160 -5.2372 0.0000
##
## sigma^2 estimated as 1.756609 on 463 degrees of freedom

```



```

##
## AIC = 3.437041  AICc = 3.437429  BIC = 3.49899
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3914 0.1053  3.7187 0.0002
## ar2      0.5585 0.0986  5.6667 0.0000
## ma1     -0.4183 0.1140 -3.6699 0.0003
## ma2     -0.1097 0.0947 -1.1584 0.2473
## ma3     -0.4640 0.0562 -8.2499 0.0000
## ma4     -0.0080 0.0756 -0.1063 0.9154
## xmean   -0.0838 0.0159 -5.2683 0.0000
##
## sigma^2 estimated as 1.756518 on 462 degrees of freedom
##
## AIC = 3.441281  AICc = 3.441799  BIC = 3.51208
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0192 0.0454 -0.4233 0.6723
## ar2      0.3905 0.0416  9.3782 0.0000
## ar3     -0.1788 0.0454 -3.9351 0.0001
## xmean   -0.0615 0.0778 -0.7905 0.4297
##
## sigma^2 estimated as 1.852407 on 465 degrees of freedom
##
## AIC = 3.476717  AICc = 3.4769  BIC = 3.520966
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1235 0.1504  0.8211 0.4120
## ar2      0.4031 0.0433  9.3041 0.0000
## ar3     -0.2398 0.0711 -3.3737 0.0008
## ma1     -0.1462 0.1503 -0.9725 0.3313
## xmean   -0.0617 0.0751 -0.8213 0.4119
##
## sigma^2 estimated as 1.848547 on 464 degrees of freedom
##
## AIC = 3.47891  AICc = 3.479186  BIC = 3.532009
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.7340 0.2358 -3.1128 0.0020
## ar2     -0.0519 0.1060 -0.4900 0.6244
## ar3      0.0673 0.1205  0.5587 0.5766
## ma1      0.7216 0.2314  3.1179 0.0019

```

```

## ma2      0.5007 0.0833  6.0143  0.0000
## xmean   -0.0610 0.0796 -0.7668  0.4436
##
## sigma^2 estimated as 1.780213 on 463 degrees of freedom
##
## AIC = 3.445885  AICc = 3.446273  BIC = 3.507834
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4099 0.1107  3.7032  0.0002
## ar2      0.5518 0.0704  7.8426  0.0000
## ar3     -0.0110 0.0947 -0.1158  0.9079
## ma1     -0.4366 0.1018 -4.2885  0.0000
## ma2     -0.1022 0.0802 -1.2740  0.2033
## ma3     -0.4613 0.0722 -6.3870  0.0000
## xmean   -0.0838 0.0159 -5.2769  0.0000
##
## sigma^2 estimated as 1.7565 on 462 degrees of freedom
##
## AIC = 3.441277  AICc = 3.441795  BIC = 3.512076
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5092 0.0784  6.4972  0.0000
## ar2     -0.3743 0.0793 -4.7204  0.0000
## ar3     -0.5479 0.0738 -7.4275  0.0000
## ma1     -0.5375 0.0778 -6.9107  0.0000
## ma2      0.8347 0.0884  9.4367  0.0000
## ma3      0.0263 0.0846  0.3107  0.7562
## ma4      0.4553 0.0432 10.5494  0.0000
## xmean   -0.0605 0.0769 -0.7865  0.4320
##
## sigma^2 estimated as 1.751939 on 461 degrees of freedom
##
## AIC = 3.439445  AICc = 3.440113  BIC = 3.519095
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0334 0.0459 -0.7260  0.4682
## ar2      0.4213 0.0452  9.3165  0.0000
## ar3     -0.1803 0.0453 -3.9814  0.0001
## ar4     -0.0793 0.0461 -1.7218  0.0858
## xmean   -0.0620 0.0719 -0.8629  0.3886
##
## sigma^2 estimated as 1.840673 on 464 degrees of freedom
##
## AIC = 3.47468  AICc = 3.474956  BIC = 3.527779
##

```

```

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.7017 0.1382 -5.0765 0.0000
## ar2     0.4264 0.0561  7.5976 0.0000
## ar3     0.0769 0.0764  1.0064 0.3147
## ar4    -0.2423 0.0449 -5.4006 0.0000
## ma1     0.6794 0.1384  4.9107 0.0000
## xmean  -0.0620 0.0722 -0.8582 0.3912
##
## sigma^2 estimated as 1.799823 on 463 degrees of freedom
##
## AIC = 3.45674  AICc = 3.457128  BIC = 3.518689
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.1892 0.4910 -0.3854 0.7001
## ar2    -0.1002 0.2439 -0.4107 0.6815
## ar3    -0.1875 0.2074 -0.9039 0.3665
## ar4     0.1069 0.2433  0.4394 0.6605
## ma1     0.1748 0.4889  0.3576 0.7208
## ma2     0.5496 0.2382  2.3074 0.0215
## xmean  -0.0614 0.0776 -0.7917 0.4289
##
## sigma^2 estimated as 1.784601 on 462 degrees of freedom
##
## AIC = 3.452598  AICc = 3.453116  BIC = 3.523397
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     0.2022 0.2991  0.6760 0.4994
## ar2     0.0078 0.2230  0.0351 0.9720
## ar3    -0.1779 0.1225 -1.4529 0.1469
## ar4     0.1787 0.1174  1.5224 0.1286
## ma1    -0.2179 0.2990 -0.7289 0.4664
## ma2     0.4505 0.2173  2.0728 0.0387
## ma3    -0.1826 0.1496 -1.2207 0.2228
## xmean  -0.0612 0.0818 -0.7472 0.4553
##
## sigma^2 estimated as 1.780252 on 461 degrees of freedom
##
## AIC = 3.454448  AICc = 3.455116  BIC = 3.534097
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     0.5580 0.1114  5.0091 0.0000
## ar2    -0.3739 0.0777 -4.8143 0.0000

```

```

## ar3      -0.5534 0.0770 -7.1881  0.0000
## ar4       0.0596 0.1078  0.5528  0.5807
## ma1      -0.5800 0.1040 -5.5796  0.0000
## ma2       0.8455 0.0884  9.5625  0.0000
## ma3       0.0156 0.0900  0.1732  0.8626
## ma4       0.4144 0.0938  4.4161  0.0000
## xmean    -0.0604 0.0790 -0.7647  0.4448
##
## sigma^2 estimated as 1.750994 on 460 degrees of freedom
##
## AIC = 3.44303  AICc = 3.443866  BIC = 3.531529
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.0181 0.0452 -0.3997  0.6895
## ar2       0.4557 0.0451 10.1041  0.0000
## ar3      -0.2607 0.0483 -5.3956  0.0000
## ar4      -0.0727 0.0452 -1.6075  0.1086
## ar5       0.1917 0.0453  4.2328  0.0000
## xmean    -0.0611 0.0871 -0.7015  0.4834
##
## sigma^2 estimated as 1.772331 on 463 degrees of freedom
##
## AIC = 3.441506  AICc = 3.441893  BIC = 3.503455
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.1957 0.1904 -1.0278  0.3046
## ar2       0.4502 0.0465  9.6718  0.0000
## ar3      -0.1864 0.0947 -1.9690  0.0495
## ar4      -0.1045 0.0568 -1.8381  0.0667
## ar5       0.1794 0.0502  3.5739  0.0004
## ma1       0.1850 0.1918  0.9643  0.3354
## xmean    -0.0610 0.0847 -0.7201  0.4719
##
## sigma^2 estimated as 1.768736 on 462 degrees of freedom
##
## AIC = 3.443764  AICc = 3.444282  BIC = 3.514563
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.2320 0.2345 -0.9892  0.3231
## ar2       0.2826 0.1835  1.5398  0.1243
## ar3      -0.1757 0.1104 -1.5917  0.1121
## ar4      -0.0442 0.0945 -0.4680  0.6400
## ar5       0.1510 0.0642  2.3535  0.0190
## ma1       0.2198 0.2371  0.9270  0.3544
## ma2       0.1745 0.1835  0.9509  0.3422

```

```

## xmean  -0.0609 0.0839 -0.7261  0.4682
##
## sigma^2 estimated as 1.765604 on 461 degrees of freedom
##
## AIC = 3.446279  AICc = 3.446946  BIC = 3.525928
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.5669 0.4193 -1.3518 0.1771
## ar2       0.0908 0.2913  0.3118 0.7553
## ar3      -0.3211 0.1469 -2.1862 0.0293
## ar4      -0.0565 0.0800 -0.7063 0.4804
## ar5       0.2118 0.0826  2.5627 0.0107
## ma1       0.5547 0.4212  1.3171 0.1885
## ma2       0.3621 0.2872  1.2606 0.2081
## ma3       0.3070 0.2666  1.1515 0.2501
## xmean    -0.0609 0.0829 -0.7355 0.4624
##
## sigma^2 estimated as 1.760556 on 460 degrees of freedom
##
## AIC = 3.447734  AICc = 3.448571  BIC = 3.536233
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.2463 0.1994  1.2351 0.2174
## ar2      -0.0959 0.1744 -0.5498 0.5827
## ar3      -0.6203 0.0766 -8.0968 0.0000
## ar4      -0.1285 0.1321 -0.9722 0.3315
## ar5       0.1655 0.1009  1.6409 0.1015
## ma1      -0.2673 0.1968 -1.3583 0.1750
## ma2       0.5639 0.1758  3.2081 0.0014
## ma3       0.2373 0.1544  1.5364 0.1251
## ma4       0.4111 0.0885  4.6428 0.0000
## xmean    -0.0600 0.0827 -0.7251 0.4688
##
## sigma^2 estimated as 1.746183 on 459 degrees of freedom
##
## AIC = 3.444249  AICc = 3.445273  BIC = 3.541598
##

```

```
nn <- length(xx)
```

```
#look at AICs and BICs
```

```
print("check AICcs")
```

```
## [1] "check AICcs"
```

```
## [1] "check AICcs"
```

```
for (ii in 1:maxAR){
```

```
  for(jj in 1:maxMA){
```

```
    print(paste0(ii-1,"",jj-1))
```

```
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
```

```
}  
}
```

```
## [1] "00"  
##      AICc  
## 124.1267  
## [1] "01"  
##      AICc  
## 119.7963  
## [1] "02"  
##      AICc  
## 38.65197  
## [1] "03"  
##      AICc  
## 10.47958  
## [1] "04"  
##      AICc  
## 6.972866  
## [1] "10"  
##      AICc  
## 114.7431  
## [1] "11"  
##      AICc  
## 74.32155  
## [1] "12"  
##      AICc  
## 0.3261034  
## [1] "13"  
##      AICc  
## 2.298646  
## [1] "14"  
##      AICc  
## 4.344679  
## [1] "20"  
##      AICc  
## 31.77713  
## [1] "21"  
##      AICc  
## 24.81174  
## [1] "22"  
##      AICc  
## 2.294828  
## [1] "23"  
## AICc  
##      0  
## [1] "24"  
##      AICc  
## 2.054053  
## [1] "30"  
##      AICc  
## 18.55169  
## [1] "31"  
##      AICc  
## 19.62593
```

```
## [1] "32"
##      AICc
## 4.15676
## [1] "33"
##      AICc
## 2.052217
## [1] "34"
##      AICc
## 1.26162
## [1] "40"
##      AICc
## 17.63778
## [1] "41"
##      AICc
## 9.258574
## [1] "42"
##      AICc
## 7.372911
## [1] "43"
##      AICc
## 8.31292
## [1] "44"
##      AICc
## 3.025711
## [1] "50"
##      AICc
## 2.098339
## [1] "51"
##      AICc
## 3.22087
## [1] "52"
##      AICc
## 4.473289
## [1] "53"
##      AICc
## 5.236726
## [1] "54"
##      AICc
## 3.686852
```

```
print("check BICs")
```

```
## [1] "check BICs"
```

```
## [1] "check BICs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
  }
}
```

```
## [1] "00"
##      BIC
## 111.4001
```

```

## [1] "01"
##      BIC
## 111.2119
## [1] "02"
##      BIC
## 34.20109
## [1] "03"
##      BIC
## 10.15347
## [1] "04"
##      BIC
## 10.76274
## [1] "10"
##      BIC
## 106.1587
## [1] "11"
##      BIC
## 69.87066
## [1] "12"
## BIC
##      0
## [1] "13"
##      BIC
## 6.08852
## [1] "14"
##      BIC
## 12.24167
## [1] "20"
##      BIC
## 27.32624
## [1] "21"
##      BIC
## 24.48563
## [1] "22"
##      BIC
## 6.084702
## [1] "23"
##      BIC
## 7.896987
## [1] "24"
##      BIC
## 14.04923
## [1] "30"
##      BIC
## 18.22558
## [1] "31"
##      BIC
## 23.4158
## [1] "32"
##      BIC
## 12.05375
## [1] "33"
##      BIC
## 14.04739

```



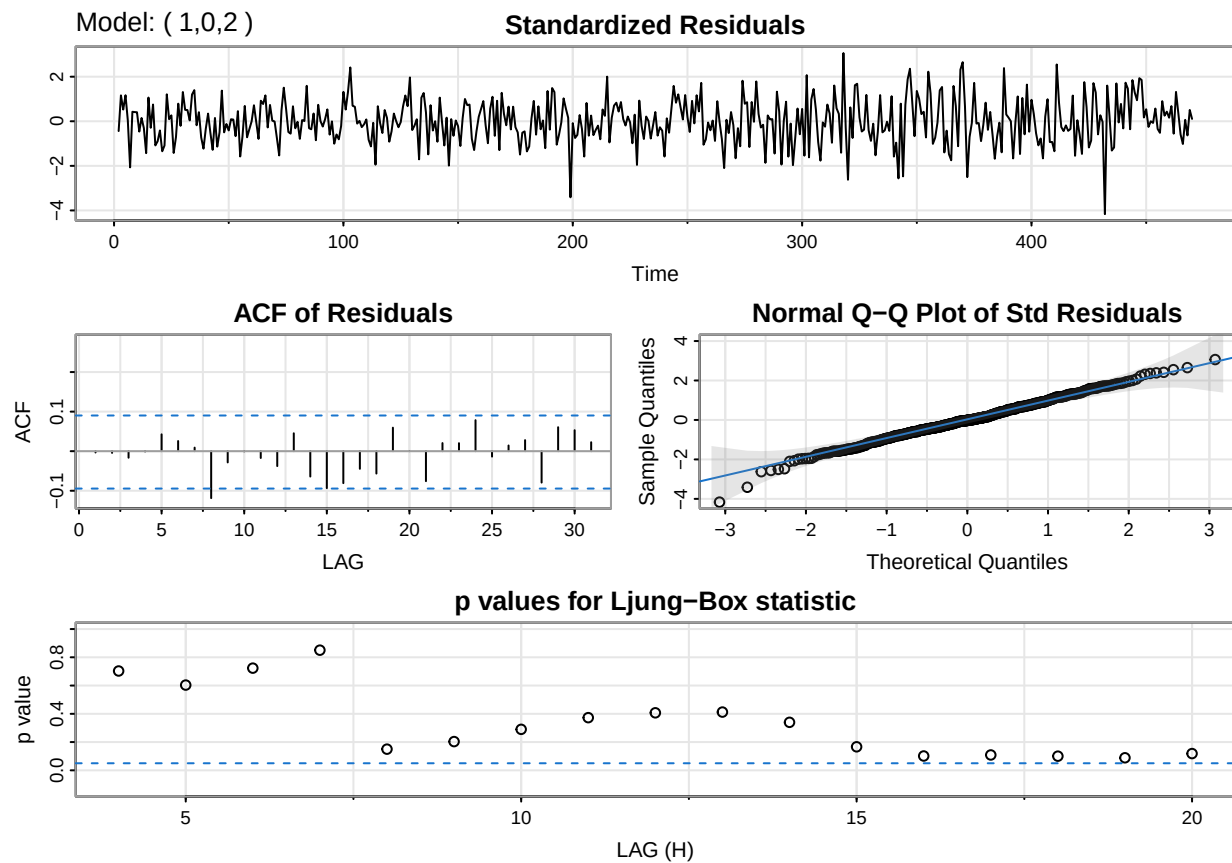
```
## [1] "34"
##      BIC
## 17.34601
## [1] "40"
##      BIC
## 21.42766
## [1] "41"
##      BIC
## 17.15556
## [1] "42"
##      BIC
## 19.36809
## [1] "43"
##      BIC
## 24.39731
## [1] "44"
##      BIC
## 23.19027
## [1] "50"
##      BIC
## 9.995325
## [1] "51"
##      BIC
## 15.21605
## [1] "52"
##      BIC
## 20.55768
## [1] "53"
##      BIC
## 25.40128
## [1] "54"
##      BIC
## 27.92248
```

ARMA(1,2) on differenced data:

```
sarima(dxx, 1,0,2,no.constant=F,details=T)
```

```
## initial  value 0.428415
## iter    2 value 0.388221
## iter    3 value 0.330827
## iter    4 value 0.322476
## iter    5 value 0.303432
## iter    6 value 0.290933
## iter    7 value 0.289853
## iter    8 value 0.289681
## iter    9 value 0.289659
## iter   10 value 0.289659
## iter   10 value 0.289659
## final   value 0.289659
## converged
## initial  value 0.289373
## iter    2 value 0.289372
## iter    3 value 0.289370
## iter    4 value 0.289370
```

```
## iter 4 value 0.289370
## iter 4 value 0.289370
## final value 0.289370
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.5811  0.0681 -8.5302  0.0000
## ma1     0.5721  0.0677  8.4452  0.0000
## ma2     0.4581  0.0406 11.2804  0.0000
## xmean   -0.0612  0.0791 -0.7736  0.4396
##
## sigma^2 estimated as 1.781286 on 465 degrees of freedom
##
## AIC = 3.437939  AICc = 3.438123  BIC = 3.482188
##
```

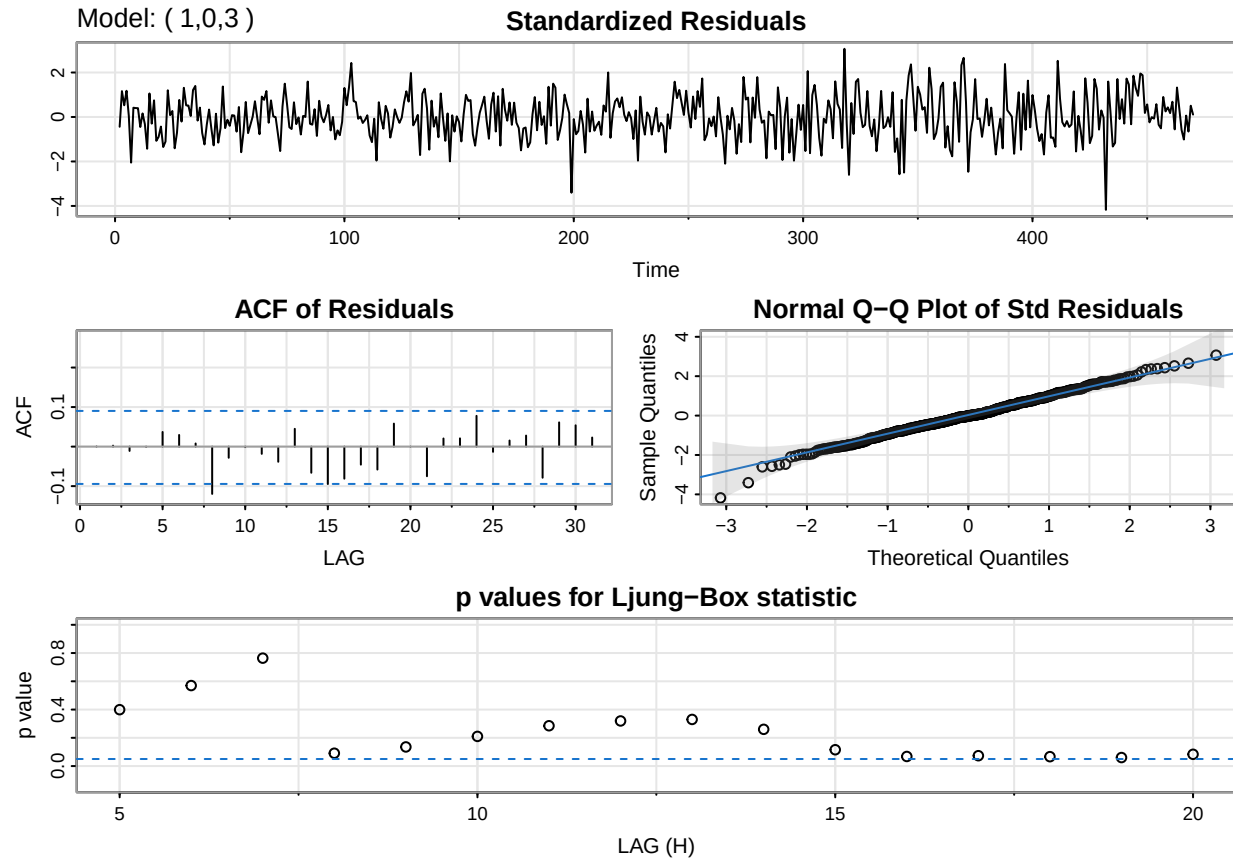


ARMA(1,3) on differenced data:

```
sarima(dxx, 1,0,3,no.constant=F,details=T)
```

```
## initial value 0.428415
## iter 2 value 0.343708
## iter 3 value 0.326062
## iter 4 value 0.302234
## iter 5 value 0.300813
```

```
## iter    6 value 0.300534
## iter    7 value 0.299048
## iter    8 value 0.294672
## iter    9 value 0.293332
## iter   10 value 0.292335
## iter   11 value 0.290039
## iter   12 value 0.289699
## iter   13 value 0.289654
## iter   14 value 0.289629
## iter   15 value 0.289627
## iter   16 value 0.289627
## iter   17 value 0.289626
## iter   18 value 0.289626
## iter   19 value 0.289625
## iter   20 value 0.289625
## iter   21 value 0.289625
## iter   22 value 0.289625
## iter   22 value 0.289625
## iter   22 value 0.289625
## final   value 0.289625
## converged
## initial  value 0.289303
## iter     2 value 0.289301
## iter     3 value 0.289298
## iter     4 value 0.289292
## iter     5 value 0.289292
## iter     6 value 0.289291
## iter     7 value 0.289290
## iter     8 value 0.289290
## iter     9 value 0.289290
## iter    10 value 0.289290
## iter    10 value 0.289290
## iter    10 value 0.289290
## final   value 0.289290
## converged
## <><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1       -0.5606 0.1038 -5.3986 0.0000
## ma1        0.5481 0.1126  4.8655 0.0000
## ma2        0.4490 0.0518  8.6706 0.0000
## ma3       -0.0200 0.0725 -0.2753 0.7832
## xmean     -0.0611 0.0780 -0.7827 0.4342
##
## sigma^2 estimated as 1.781008 on 464 degrees of freedom
##
## AIC = 3.442043  AICc = 3.442319  BIC = 3.495142
##
```



ARMA(2,2) on differenced data:

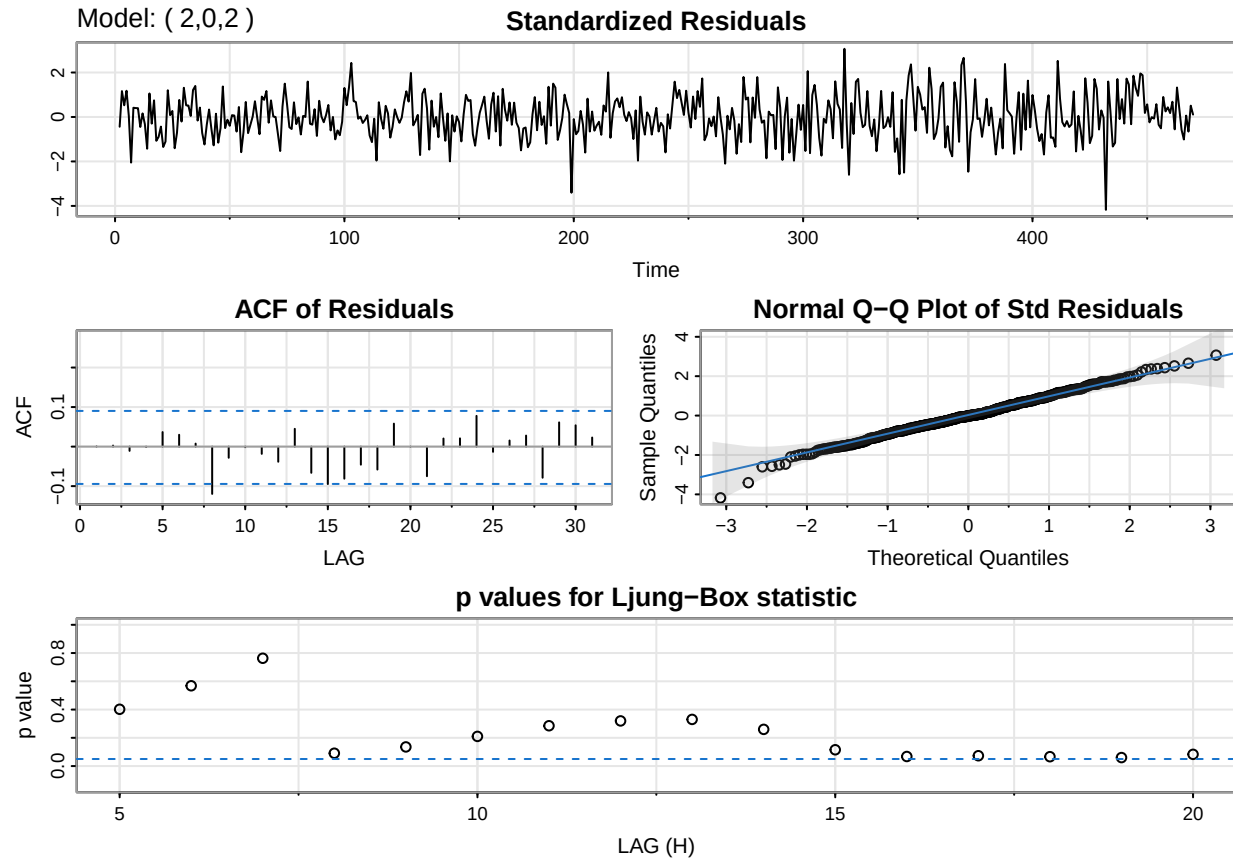
```
sarima(dxx, 2,0,2,no.constant=F,details=T)
```

```
## initial value 0.427899
## iter 2 value 0.364890
## iter 3 value 0.332297
## iter 4 value 0.317684
## iter 5 value 0.309664
## iter 6 value 0.301404
## iter 7 value 0.298544
## iter 8 value 0.293711
## iter 9 value 0.292508
## iter 10 value 0.290400
## iter 11 value 0.290120
## iter 12 value 0.290028
## iter 13 value 0.290026
## iter 14 value 0.290025
## iter 15 value 0.290024
## iter 16 value 0.290024
## iter 17 value 0.290024
## iter 18 value 0.290024
## iter 19 value 0.290023
## iter 20 value 0.290023
## iter 20 value 0.290023
## iter 20 value 0.290023
## final value 0.290023
```

```

## converged
## initial value 0.289327
## iter 2 value 0.289320
## iter 3 value 0.289308
## iter 4 value 0.289300
## iter 5 value 0.289294
## iter 6 value 0.289291
## iter 7 value 0.289288
## iter 8 value 0.289286
## iter 9 value 0.289286
## iter 10 value 0.289286
## iter 10 value 0.289286
## iter 10 value 0.289286
## final value 0.289286
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.6059 0.1142 -5.3068 0.0000
## ar2    -0.0270 0.0972 -0.2776 0.7815
## ma1     0.5932 0.1034  5.7386 0.0000
## ma2     0.4750 0.0720  6.5944 0.0000
## xmean  -0.0610 0.0780 -0.7815 0.4349
##
## sigma^2 estimated as 1.780994 on 464 degrees of freedom
##
## AIC = 3.442035  AICc = 3.442311  BIC = 3.495134
##

```



ARMA(2,3) on differenced data:

```
sarima(dxx, 2,0,3,no.constant=F,details=T)
```

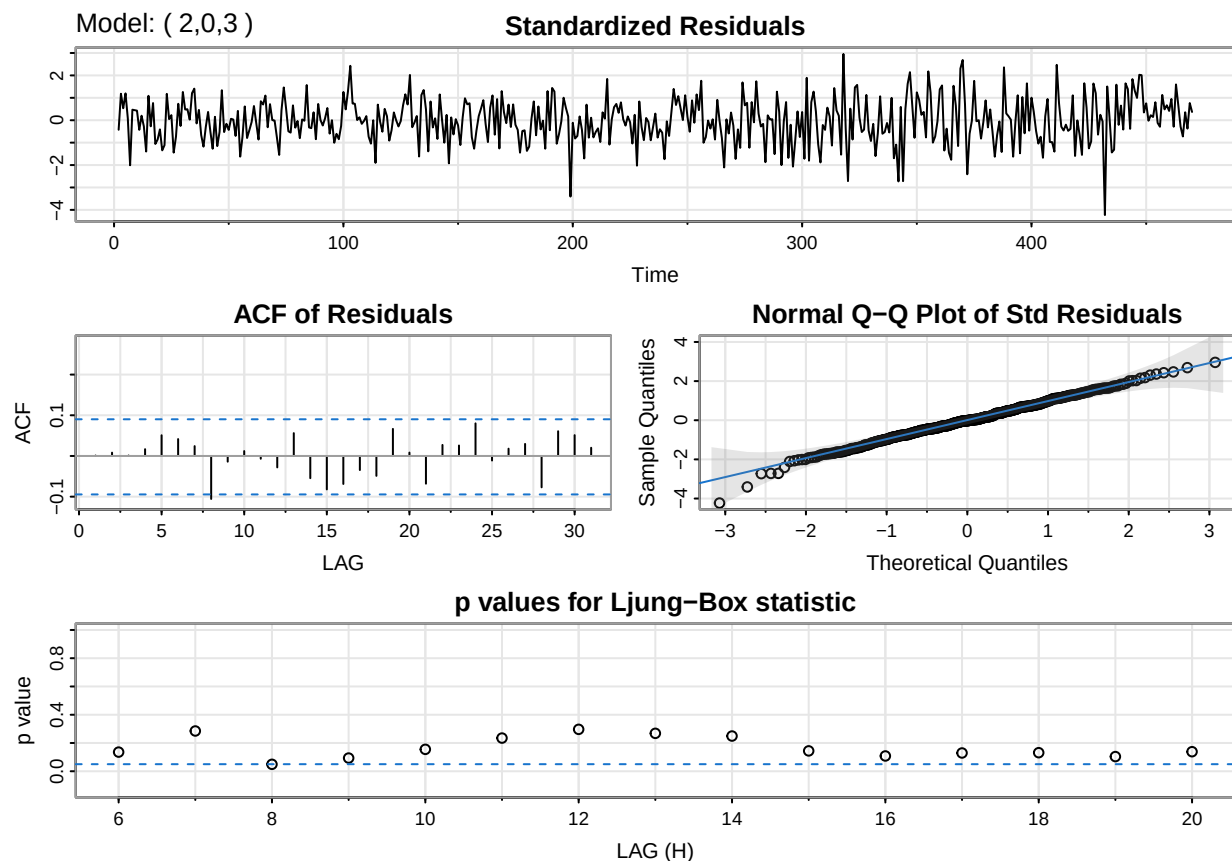
```
## initial value 0.427899
## iter 2 value 0.395214
## iter 3 value 0.297894
## iter 4 value 0.293292
## iter 5 value 0.290555
## iter 6 value 0.290274
## iter 7 value 0.290042
## iter 8 value 0.289872
## iter 9 value 0.289869
## iter 10 value 0.289869
## iter 11 value 0.289866
## iter 12 value 0.289861
## iter 13 value 0.289849
## iter 14 value 0.289826
## iter 15 value 0.289823
## iter 16 value 0.289804
## iter 17 value 0.289802
## iter 18 value 0.289800
## iter 19 value 0.289798
## iter 20 value 0.289798
## iter 21 value 0.289796
## iter 22 value 0.289792
## iter 23 value 0.289784
```

```
## iter 24 value 0.289775
## iter 25 value 0.289768
## iter 26 value 0.289766
## iter 27 value 0.289766
## iter 28 value 0.289766
## iter 29 value 0.289766
## iter 30 value 0.289766
## iter 31 value 0.289766
## iter 31 value 0.289766
## iter 31 value 0.289766
## final value 0.289766
## converged
## initial value 0.289303
## iter 2 value 0.289291
## iter 3 value 0.289269
## iter 4 value 0.289263
## iter 5 value 0.289262
## iter 6 value 0.289261
## iter 7 value 0.289258
## iter 8 value 0.289255
## iter 9 value 0.289249
## iter 10 value 0.289236
## iter 11 value 0.289177
## iter 12 value 0.289091
## iter 13 value 0.288946
## iter 14 value 0.288789
## iter 15 value 0.288708
## iter 16 value 0.288582
## iter 17 value 0.288395
## iter 18 value 0.288320
## iter 19 value 0.288238
## iter 20 value 0.288062
## iter 21 value 0.287843
## iter 22 value 0.287453
## iter 23 value 0.287161
## iter 24 value 0.286772
## iter 25 value 0.286325
## iter 26 value 0.285542
## iter 27 value 0.285158
## iter 28 value 0.285154
## iter 29 value 0.285145
## iter 30 value 0.285139
## iter 31 value 0.285137
## iter 32 value 0.285128
## iter 33 value 0.285118
## iter 34 value 0.285104
## iter 35 value 0.285073
## iter 36 value 0.284992
## iter 37 value 0.284988
## iter 38 value 0.284911
## iter 39 value 0.284874
## iter 40 value 0.284805
## iter 41 value 0.284725
## iter 42 value 0.284713
```

```

## iter 43 value 0.284705
## iter 44 value 0.284693
## iter 45 value 0.284671
## iter 46 value 0.284660
## iter 47 value 0.284657
## iter 48 value 0.284657
## iter 49 value 0.284657
## iter 49 value 0.284657
## iter 49 value 0.284657
## final value 0.284657
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4001 0.0703  5.6901 0.0000
## ar2      0.5509 0.0702  7.8487 0.0000
## ma1     -0.4281 0.0701 -6.1034 0.0000
## ma2     -0.1038 0.0790 -1.3133 0.1897
## ma3     -0.4682 0.0405 -11.5480 0.0000
## xmean   -0.0837 0.0160 -5.2372 0.0000
##
## sigma^2 estimated as 1.756609 on 463 degrees of freedom
##
## AIC = 3.437041  AICc = 3.437429  BIC = 3.49899
##

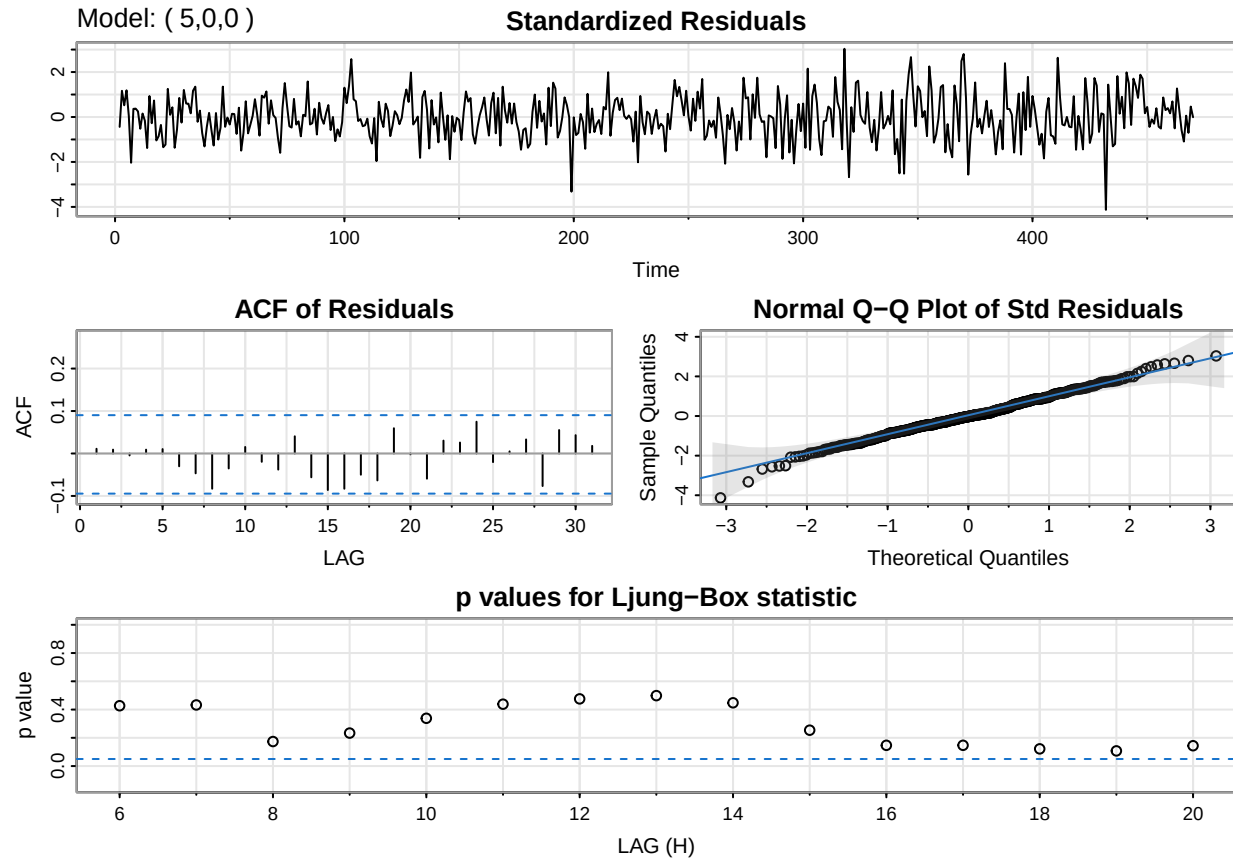
```



ARMA(5,0) on differenced data:

```
sarima(dxx, 5,0,0,no.constant=F,details=T)
```

```
## initial value 0.428308
## iter 2 value 0.366483
## iter 3 value 0.292684
## iter 4 value 0.288231
## iter 5 value 0.288052
## iter 6 value 0.288029
## iter 7 value 0.288027
## iter 8 value 0.288027
## iter 9 value 0.288025
## iter 10 value 0.288025
## iter 10 value 0.288025
## iter 10 value 0.288025
## final value 0.288025
## converged
## initial value 0.286914
## iter 2 value 0.286905
## iter 3 value 0.286898
## iter 4 value 0.286890
## iter 5 value 0.286889
## iter 6 value 0.286889
## iter 7 value 0.286889
## iter 8 value 0.286889
## iter 8 value 0.286889
## iter 8 value 0.286889
## final value 0.286889
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.0181 0.0452 -0.3997 0.6895
## ar2     0.4557 0.0451 10.1041 0.0000
## ar3    -0.2607 0.0483 -5.3956 0.0000
## ar4    -0.0727 0.0452 -1.6075 0.1086
## ar5     0.1917 0.0453 4.2328 0.0000
## xmean  -0.0611 0.0871 -0.7015 0.4834
##
## sigma^2 estimated as 1.772331 on 463 degrees of freedom
##
## AIC = 3.441506 AICc = 3.441893 BIC = 3.503455
##
```



Final choice: ARMA(1,2) (ARIMA(1,1,2) on before differencing) with no intercept: ARMA(1,2) on differenced data:

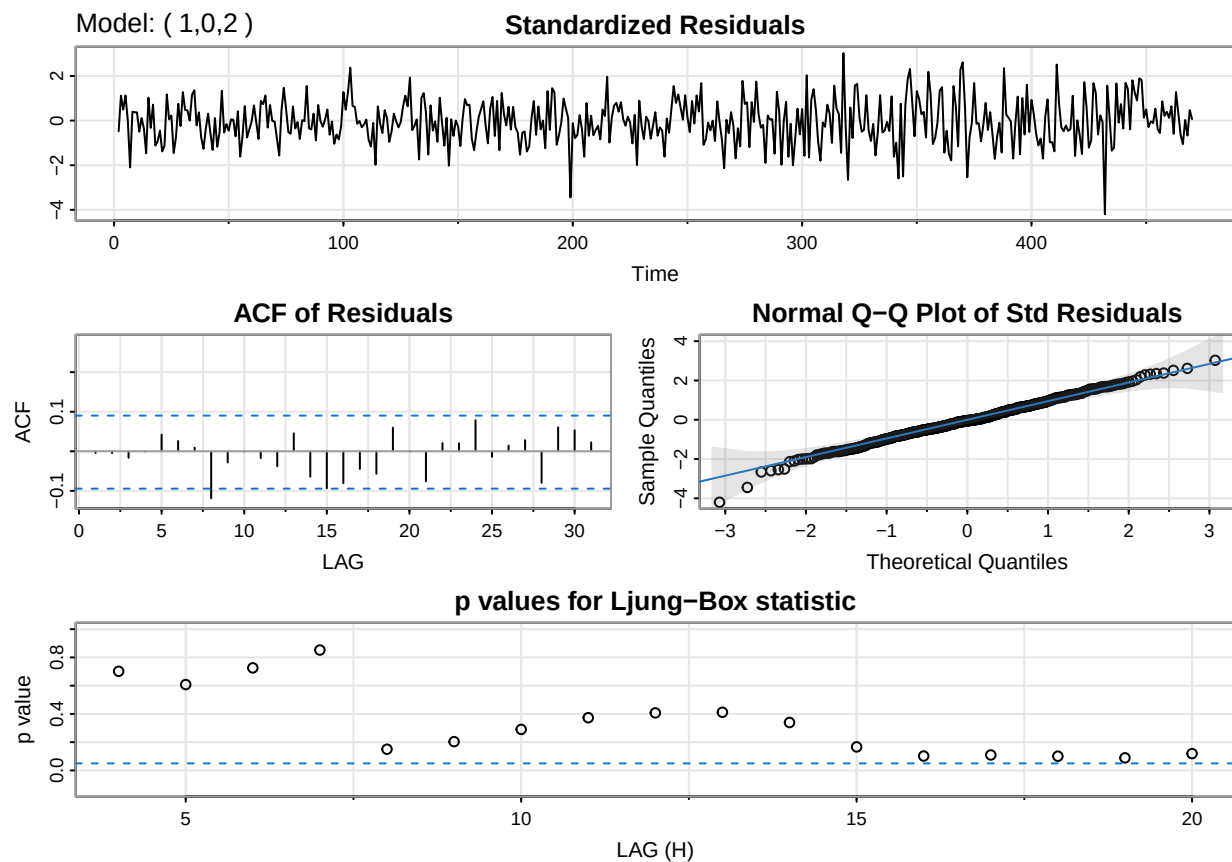
```
sarima(dxx, 1,0,2,no.constant=T,details=T)
```

```
## initial value 0.429220
## iter 2 value 0.387834
## iter 3 value 0.331235
## iter 4 value 0.322963
## iter 5 value 0.304229
## iter 6 value 0.291858
## iter 7 value 0.290449
## iter 8 value 0.290262
## iter 9 value 0.290257
## iter 10 value 0.290256
## iter 11 value 0.290255
## iter 12 value 0.290255
## iter 13 value 0.290255
## iter 13 value 0.290255
## iter 13 value 0.290255
## final value 0.290255
## converged
## initial value 0.290007
## iter 2 value 0.290007
## iter 3 value 0.290005
## iter 4 value 0.290005
## iter 5 value 0.290005
```

```

## iter    5 value 0.290005
## iter    5 value 0.290005
## final   value 0.290005
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1   -0.5802  0.0682  -8.5025     0
## ma1    0.5722  0.0678   8.4337     0
## ma2    0.4589  0.0405  11.3233     0
##
## sigma^2 estimated as 1.783545 on 466 degrees of freedom
##
## AIC = 3.434944  AICc = 3.435054  BIC = 3.470344
##

```



Question 2 (Analysis of dat2)

Summary

For the series $Y_t = X_t$ I chose an ARIMA(1,0,3) model including the intercept term. The parameter estimates are $\phi_1 = 0.2522$, $\theta_1 = -0.1692$, $\theta_2 = -0.0476$, $\theta_3 = -0.3471$, and $xmean = 3.1994$.

```
xx <- dat2
sarima(xx, 1,0,3,no.constant=F,details=F)

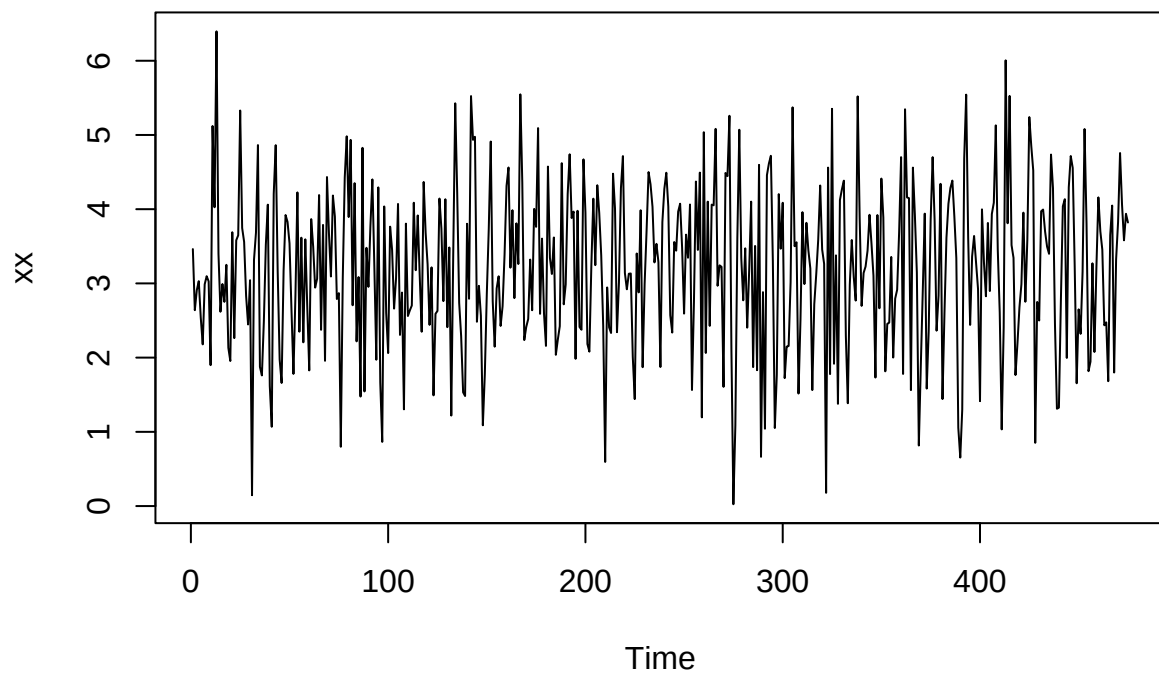
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2522 0.1024   2.4640  0.0141
## ma1     -0.1692 0.0954  -1.7731  0.0769
## ma2     -0.0476 0.0449  -1.0606  0.2894
## ma3     -0.3471 0.0412  -8.4227  0.0000
## xmean     3.1994 0.0272 117.6565  0.0000
##
## sigma^2 estimated as 1.021186 on 470 degrees of freedom
##
## AIC = 2.88502  AICc = 2.885289  BIC = 2.937609
##
```

Explanation

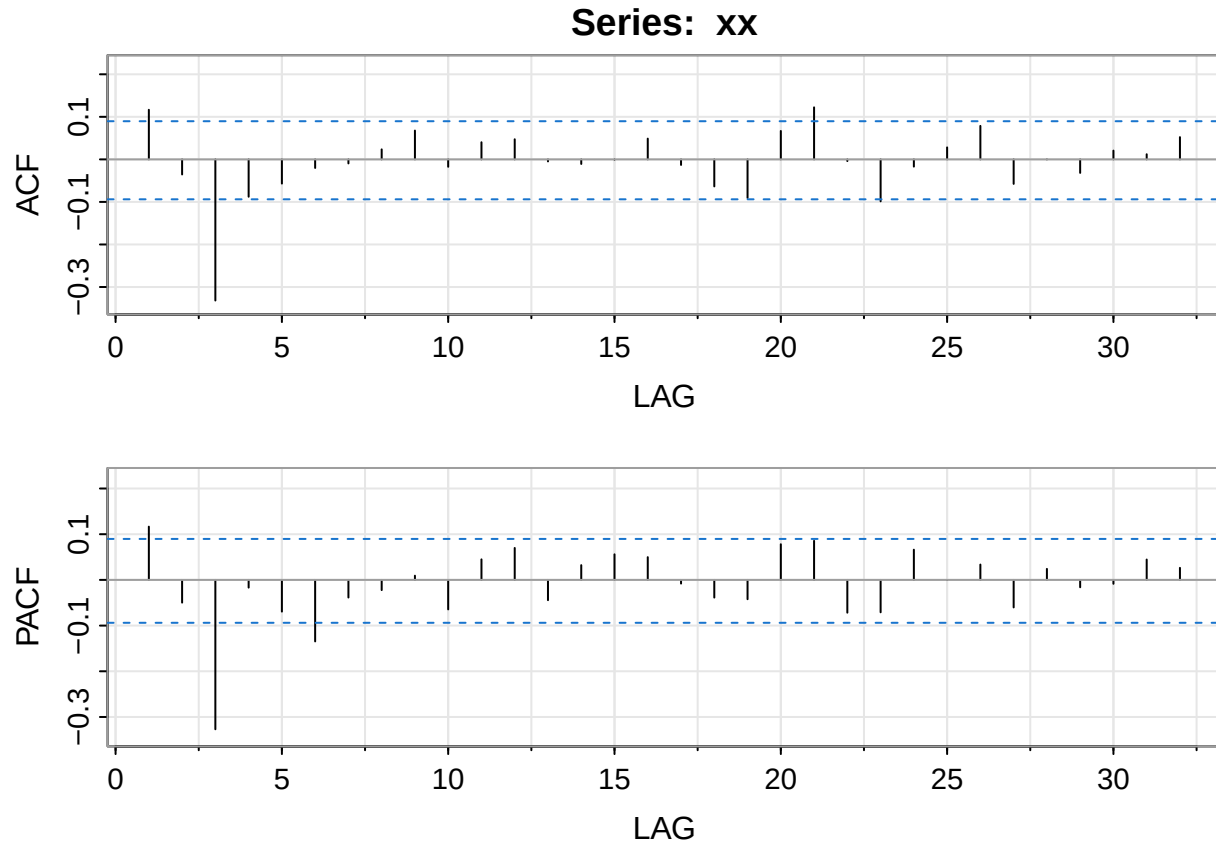
The initial plot and ACF and PACF of X_t suggest that the series is stationary. The ACF and PACF suggest that there is no seasonality. The significant lags of the ACF cut off for the most part after lag 3 and those of the PACF cut off after lag 3, so we fit different models on the first differenced data with a maximum p of 3 and a maximum q of 3. 7 of the 16 models had an AICc multiplied by the length of the series within 10 of the minimum, and six had a BIC multiplied by the length of the series within 10 of the minimum. The six which met both of these criteria were ARMA(0,3), ARMA(1,3), ARMA(2,3), ARMA(3,0), ARMA(3,1), and ARMA(3,2). Of these, the ARMA(1,3) was chosen because it has the best AICc, the second best BIC, and the highest LBP values. So the final model on the original series X_t is ARIMA(1,0,3). The intercept was included because it was significant at all levels.

Output

```
plot(xx)
```



```
invisible(acf2(xx))
```



```
library(astsa)
maxAR<-3+1
maxMA<-3+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
  fits_all[[ii]]<-vector("list",length=maxMA)
  for (jj in 1:maxMA){
    fits_all[[ii]][[jj]]<-invisible(sarima(xx,p=ii-1,d=0,q=jj-1,no.constant=FALSE, details=FALSE))
    if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
    if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
  }
}
```

```
## <><><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## xmean    3.2009 0.0499 64.1191      0
##
## sigma^2 estimated as 1.183749 on 474 degrees of freedom
##
## AIC = 3.014985  AICc = 3.015002  BIC = 3.032514
##
## <><><><><><><><><><><><><><>
##
```

```

## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.1136 0.0426  2.6693  0.0079
## xmean    3.2011 0.0552 57.9920  0.0000
##
## sigma^2 estimated as 1.167493 on 473 degrees of freedom
##
## AIC = 3.005395  AICc = 3.005449  BIC = 3.03169
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.1657 0.0614  2.6981  0.0072
## ma2      0.0791 0.0654  1.2093  0.2272
## xmean    3.2014 0.0616 51.9816  0.0000
##
## sigma^2 estimated as 1.163948 on 472 degrees of freedom
##
## AIC = 3.006613  AICc = 3.006721  BIC = 3.041673
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0509 0.0444  1.1478  0.2516
## ma2     -0.0576 0.0486 -1.1842  0.2369
## ma3     -0.3513 0.0397 -8.8404  0.0000
## xmean    3.1997 0.0300 106.5081  0.0000
##
## sigma^2 estimated as 1.032038 on 471 degrees of freedom
##
## AIC = 2.891309  AICc = 2.891488  BIC = 2.935133
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1164 0.0455  2.5562  0.0109
## xmean    3.2011 0.0561 57.0651  0.0000
##
## sigma^2 estimated as 1.167656 on 473 degrees of freedom
##
## AIC = 3.005536  AICc = 3.00559  BIC = 3.031831
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0539 0.1581  0.3407  0.7334
## ma1      0.0647 0.1523  0.4250  0.6710
## xmean    3.2011 0.0558 57.3965  0.0000
##
## sigma^2 estimated as 1.167216 on 472 degrees of freedom

```

```

##
## AIC = 3.00937  AICc = 3.009477  BIC = 3.04443
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.6270 0.1020 -6.1456      0
## ma1       0.8230 0.0979  8.4059      0
## ma2       0.2862 0.0508  5.6279      0
## xmean     3.2015 0.0630 50.8167      0
##
## sigma^2 estimated as 1.123072 on 471 degrees of freedom
##
## AIC = 2.975342  AICc = 2.975521  BIC = 3.019166
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.2522 0.1024  2.4640 0.0141
## ma1      -0.1692 0.0954 -1.7731 0.0769
## ma2      -0.0476 0.0449 -1.0606 0.2894
## ma3      -0.3471 0.0412 -8.4227 0.0000
## xmean     3.1994 0.0272 117.6565 0.0000
##
## sigma^2 estimated as 1.021186 on 470 degrees of freedom
##
## AIC = 2.88502  AICc = 2.885289  BIC = 2.937609
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.1222 0.0458  2.6679 0.0079
## ar2      -0.0496 0.0458 -1.0840 0.2789
## xmean     3.2010 0.0534 59.9557 0.0000
##
## sigma^2 estimated as 1.164763 on 472 degrees of freedom
##
## AIC = 3.007276  AICc = 3.007383  BIC = 3.042335
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.8671 0.0644 13.4669      0
## ar2      -0.2528 0.0454 -5.5711      0
## ma1      -0.7827 0.0519 -15.0902      0
## xmean     3.1993 0.0271 117.8552      0
##
## sigma^2 estimated as 1.091106 on 471 degrees of freedom
##
## AIC = 2.946645  AICc = 2.946825  BIC = 2.99047

```



```

##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.3366 0.1292 -2.6055 0.0095
## ar2      -0.5749 0.0943 -6.0973 0.0000
## ma1       0.5297 0.1090  4.8607 0.0000
## ma2       0.7403 0.0797  9.2878 0.0000
## xmean     3.2011 0.0570 56.1364 0.0000
##
## sigma^2 estimated as 1.095667 on 470 degrees of freedom
##
## AIC = 2.955055  AICc = 2.955324  BIC = 3.007644
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.3090 0.1300  2.3774 0.0178
## ar2      -0.0876 0.1307 -0.6704 0.5029
## ma1      -0.2241 0.1229 -1.8241 0.0688
## ma2       0.0292 0.1209  0.2415 0.8093
## ma3      -0.3460 0.0432 -8.0175 0.0000
## xmean     3.1994 0.0275 116.4534 0.0000
##
## sigma^2 estimated as 1.020294 on 469 degrees of freedom
##
## AIC = 2.888397  AICc = 2.888775  BIC = 2.949751
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.1058 0.0433  2.4443 0.0149
## ar2      -0.0100 0.0436 -0.2306 0.8177
## ar3      -0.3257 0.0433 -7.5295 0.0000
## xmean     3.2004 0.0381 83.9925 0.0000
##
## sigma^2 estimated as 1.039963 on 471 degrees of freedom
##
## AIC = 2.898862  AICc = 2.899041  BIC = 2.942687
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.2173 0.2337  0.9301 0.3528
## ar2      -0.0239 0.0533 -0.4482 0.6542
## ar3      -0.3188 0.0481 -6.6331 0.0000
## ma1      -0.1249 0.2570 -0.4861 0.6271
## xmean     3.2003 0.0364 87.8231 0.0000
##
## sigma^2 estimated as 1.039292 on 470 degrees of freedom

```

```
##
## AIC = 2.902435 AICc = 2.902704 BIC = 2.955024
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2687 0.1142  2.3525 0.0191
## ar2      0.2229 0.1047  2.1284 0.0338
## ar3     -0.3410 0.0454 -7.5090 0.0000
## ma1     -0.1970 0.1192 -1.6535 0.0989
## ma2     -0.2818 0.1045 -2.6976 0.0072
## xmean    3.1995 0.0286 111.7306 0.0000
##
## sigma^2 estimated as 1.026713 on 469 degrees of freedom
##
## AIC = 2.894629 AICc = 2.895007 BIC = 2.955984
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2256 0.1363  1.6550 0.0986
## ar2      0.0769 0.1648  0.4668 0.6409
## ar3     -0.1413 0.1222 -1.1563 0.2482
## ma1     -0.1407 0.1342 -1.0486 0.2949
## ma2     -0.1275 0.1589 -0.8025 0.4226
## ma3     -0.2325 0.1120 -2.0761 0.0384
## xmean    3.1995 0.0277 115.5103 0.0000
##
## sigma^2 estimated as 1.018293 on 468 degrees of freedom
##
## AIC = 2.890668 AICc = 2.891173 BIC = 2.960787
##

nn <- length(xx)
#look at AICs and BICs
print("check AICcs")

## [1] "check AICcs"

## [1] "check AICcs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
  }
}

## [1] "00"
##      AICc
## 61.61379
## [1] "01"
##      AICc
## 57.07571
```

```

## [1] "02"
##      AICc
## 57.67995
## [1] "03"
##      AICc
## 2.944417
## [1] "10"
##      AICc
## 57.14272
## [1] "11"
##      AICc
## 58.98948
## [1] "12"
##      AICc
## 42.86016
## [1] "13"
## AICc
##      0
## [1] "20"
##      AICc
## 57.99458
## [1] "21"
##      AICc
## 29.22933
## [1] "22"
##      AICc
## 33.26678
## [1] "23"
##      AICc
## 1.655914
## [1] "30"
##      AICc
## 6.532347
## [1] "31"
##      AICc
## 8.272222
## [1] "32"
##      AICc
## 4.616172
## [1] "33"
##      AICc
## 2.794717

print("check BICs")

## [1] "check BICs"
## [1] "check BICs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
  }
}

```

```

## [1] "00"
##      BIC
## 46.25608
## [1] "01"
##      BIC
## 45.86435
## [1] "02"
##      BIC
## 50.60637
## [1] "03"
## BIC
##      0
## [1] "10"
##      BIC
## 45.93135
## [1] "11"
##      BIC
## 51.9159
## [1] "12"
##      BIC
## 39.91574
## [1] "13"
##      BIC
## 1.176073
## [1] "20"
##      BIC
## 50.921
## [1] "21"
##      BIC
## 26.28492
## [1] "22"
##      BIC
## 34.44285
## [1] "23"
##      BIC
## 6.943746
## [1] "30"
##      BIC
## 3.587931
## [1] "31"
##      BIC
## 9.448295
## [1] "32"
##      BIC
## 9.904004
## [1] "33"
##      BIC
## 12.18552

```

ARMA(0,3):

```
sarima(xx, 0,0,3,no.constant=F,details=T)
```

```

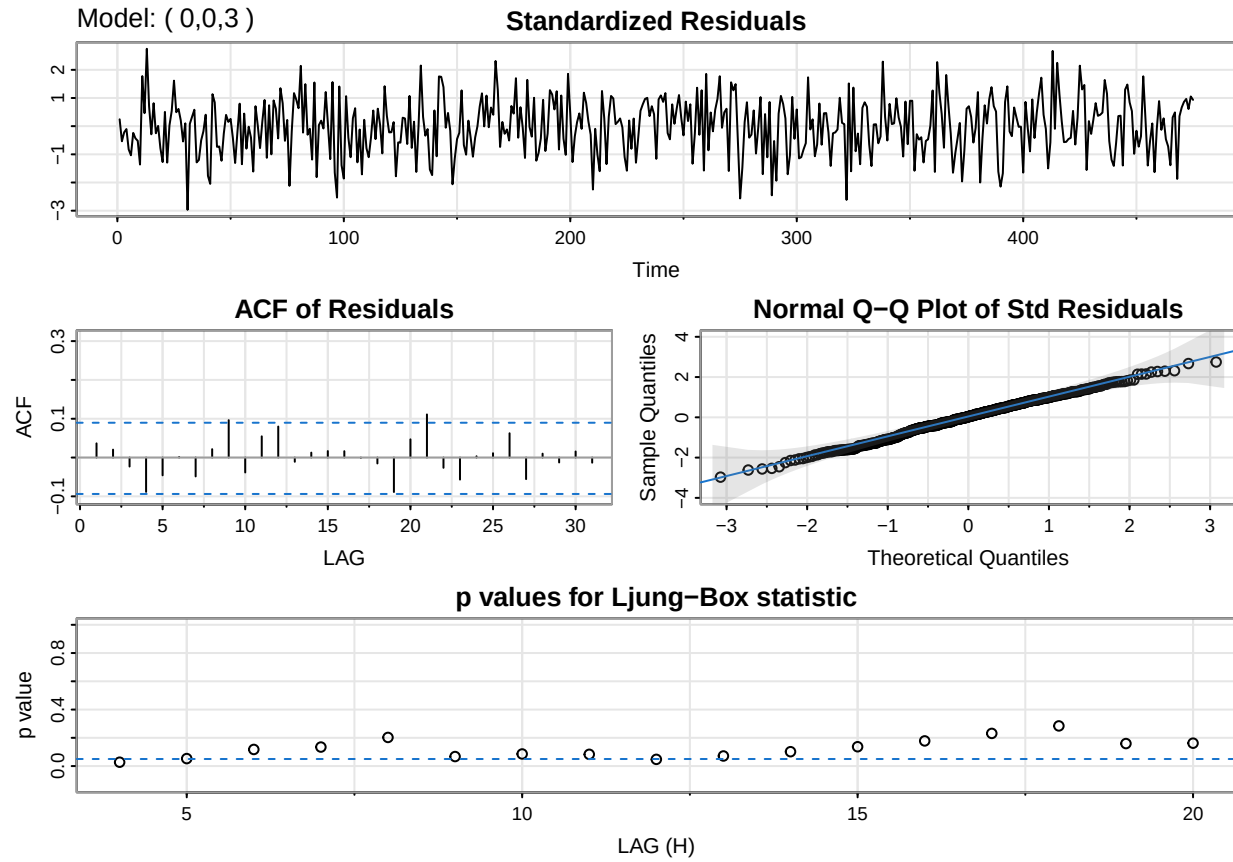
## initial value 0.084343
## iter 2 value 0.018296

```

```

## iter    3 value 0.015990
## iter    4 value 0.015915
## iter    5 value 0.015913
## iter    6 value 0.015912
## iter    7 value 0.015912
## iter    8 value 0.015912
## iter    8 value 0.015912
## iter    8 value 0.015912
## final   value 0.015912
## converged
## initial  value 0.016190
## iter    2 value 0.016190
## iter    3 value 0.016190
## iter    4 value 0.016190
## iter    5 value 0.016190
## iter    5 value 0.016190
## iter    5 value 0.016190
## final   value 0.016190
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0509 0.0444   1.1478  0.2516
## ma2     -0.0576 0.0486  -1.1842  0.2369
## ma3     -0.3513 0.0397  -8.8404  0.0000
## xmean     3.1997 0.0300 106.5081  0.0000
##
## sigma^2 estimated as 1.032038 on 471 degrees of freedom
##
## AIC = 2.891309  AICc = 2.891488  BIC = 2.935133
##

```

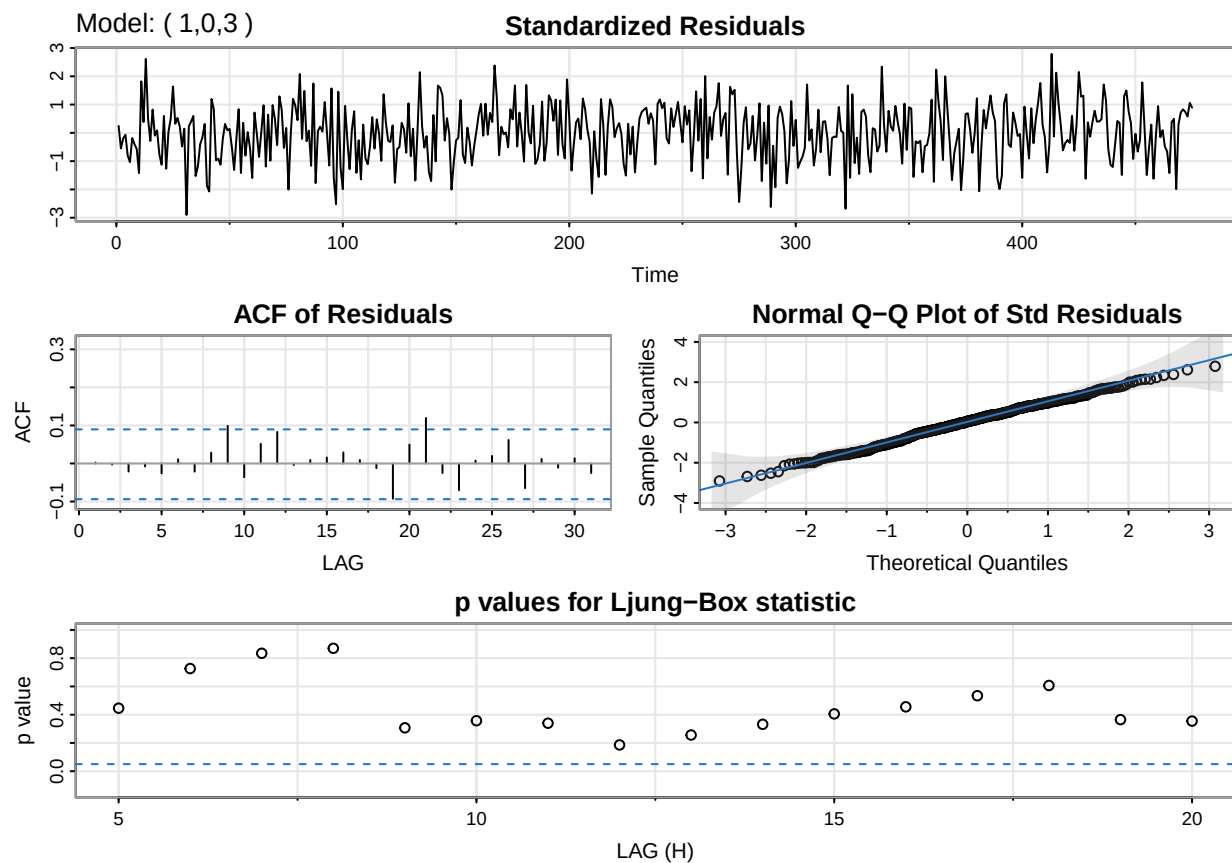


ARMA(1,3):

```
sarima(xx, 1,0,3,no.constant=F,details=T)
```

```
## initial value 0.085336
## iter 2 value 0.028944
## iter 3 value 0.016854
## iter 4 value 0.015155
## iter 5 value 0.014450
## iter 6 value 0.013167
## iter 7 value 0.012302
## iter 8 value 0.011893
## iter 9 value 0.011859
## iter 10 value 0.011857
## iter 11 value 0.011857
## iter 12 value 0.011857
## iter 12 value 0.011857
## final value 0.011857
## converged
## initial value 0.010943
## iter 2 value 0.010942
## iter 3 value 0.010941
## iter 4 value 0.010941
## iter 5 value 0.010940
## iter 6 value 0.010940
## iter 7 value 0.010940
## iter 8 value 0.010940
```

```
## iter    8 value 0.010940
## iter    8 value 0.010940
## final   value 0.010940
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2522 0.1024   2.4640 0.0141
## ma1     -0.1692 0.0954  -1.7731 0.0769
## ma2     -0.0476 0.0449  -1.0606 0.2894
## ma3     -0.3471 0.0412  -8.4227 0.0000
## xmean     3.1994 0.0272 117.6565 0.0000
##
## sigma^2 estimated as 1.021186 on 470 degrees of freedom
##
## AIC = 2.88502  AICc = 2.885289  BIC = 2.937609
##
```



ARMA(2,3):

```
sarima(xx, 2,0,3,no.constant=F,details=T)
```

```
## initial value 0.086111
## iter    2 value 0.030552
## iter    3 value 0.017877
## iter    4 value 0.015630
```

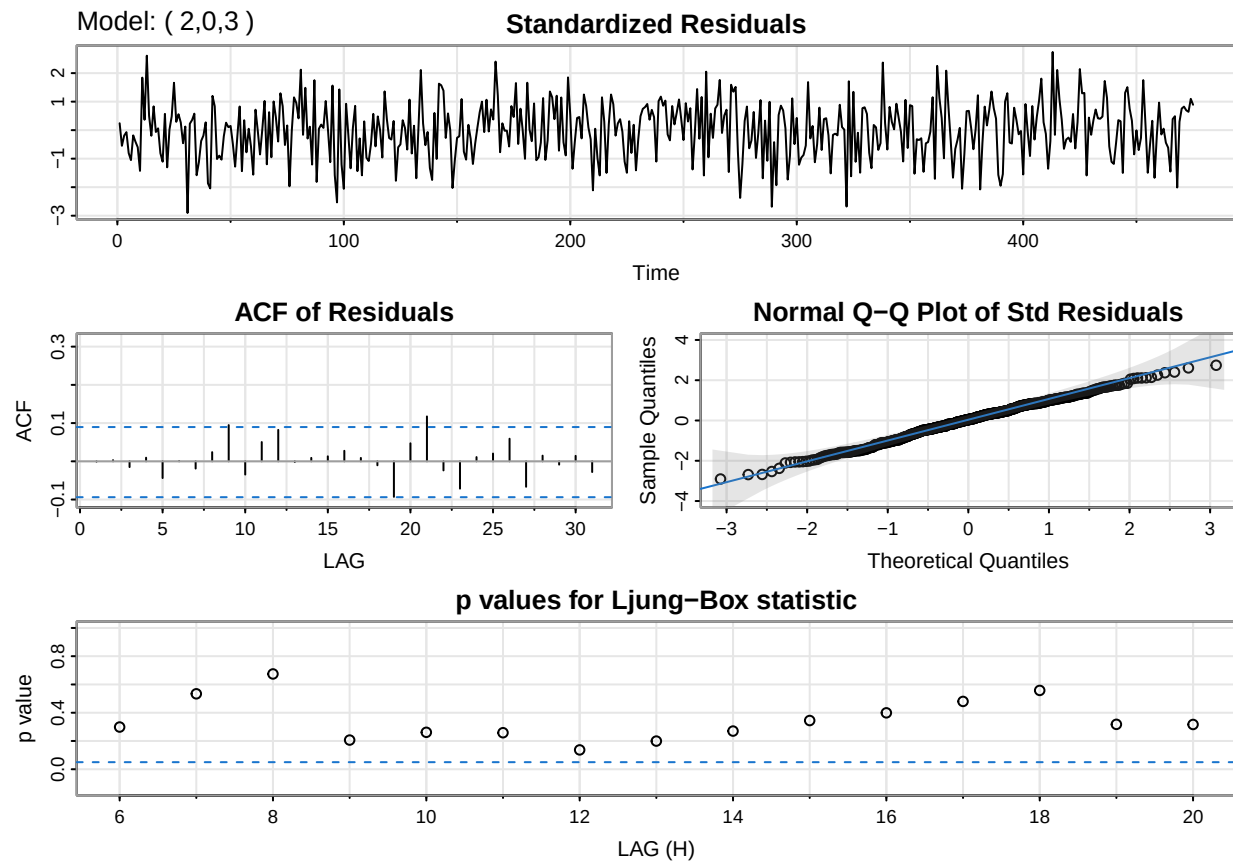
```

## iter    5 value 0.014634
## iter    6 value 0.013222
## iter    7 value 0.012714
## iter    8 value 0.012449
## iter    9 value 0.012348
## iter   10 value 0.012245
## iter   11 value 0.011583
## iter   12 value 0.011473
## iter   13 value 0.011442
## iter   14 value 0.011442
## iter   15 value 0.011442
## iter   16 value 0.011442
## iter   17 value 0.011442
## iter   18 value 0.011442
## iter   19 value 0.011442
## iter   20 value 0.011442
## iter   21 value 0.011442
## iter   22 value 0.011442
## iter   23 value 0.011442
## iter   23 value 0.011442
## iter   23 value 0.011442
## final   value 0.011442
## converged
## initial  value 0.010531
## iter    2 value 0.010528
## iter    3 value 0.010527
## iter    4 value 0.010526
## iter    5 value 0.010526
## iter    6 value 0.010525
## iter    7 value 0.010523
## iter    8 value 0.010523
## iter    9 value 0.010523
## iter   10 value 0.010523
## iter   11 value 0.010523
## iter   12 value 0.010523
## iter   13 value 0.010523
## iter   13 value 0.010523
## iter   13 value 0.010523
## final   value 0.010523
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3090 0.1300  2.3774 0.0178
## ar2     -0.0876 0.1307 -0.6704 0.5029
## ma1     -0.2241 0.1229 -1.8241 0.0688
## ma2      0.0292 0.1209  0.2415 0.8093
## ma3     -0.3460 0.0432 -8.0175 0.0000
## xmean    3.1994 0.0275 116.4534 0.0000
##
## sigma^2 estimated as 1.020294 on 469 degrees of freedom
##
## AIC = 2.888397  AICc = 2.888775  BIC = 2.949751

```



```
##
```

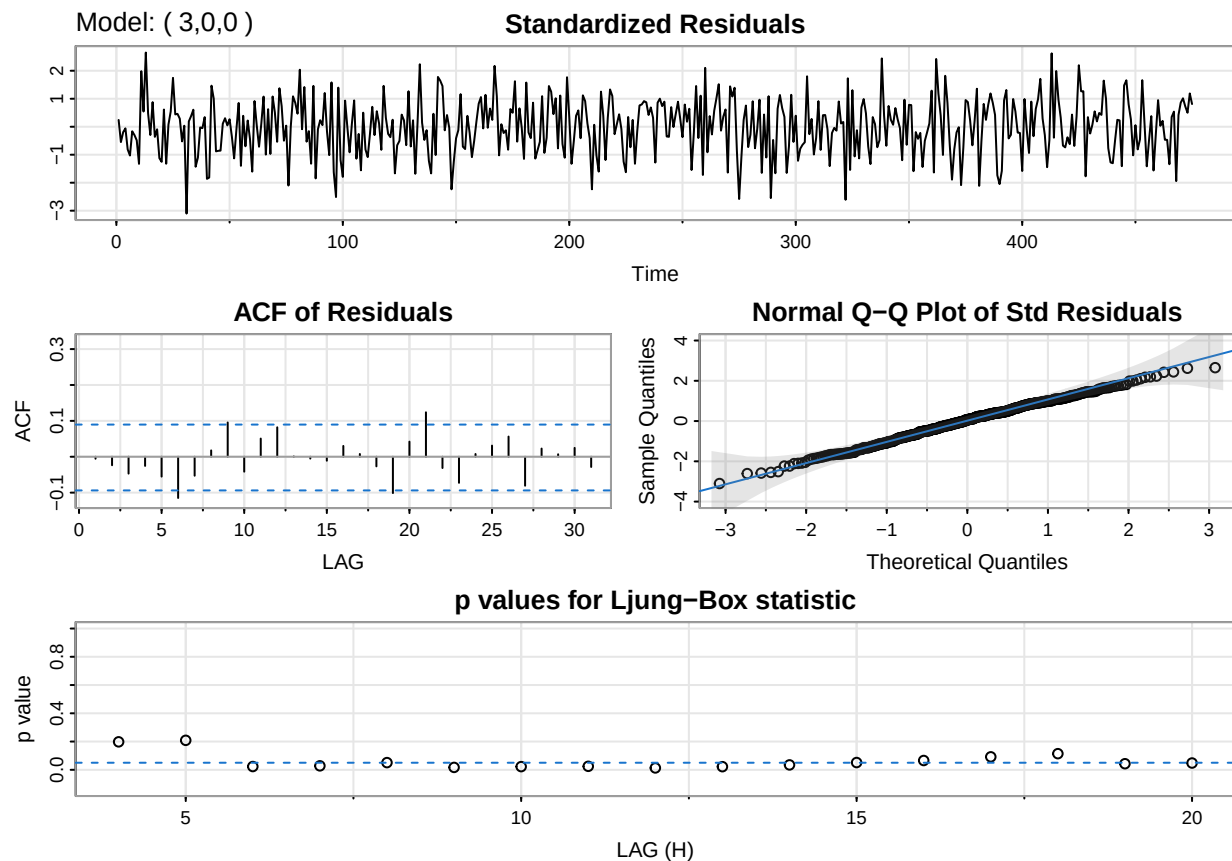


ARMA(3,0):

```
sarima(xx, 3,0,0,no.constant=F,details=T)
```

```
## initial value 0.087094
## iter 2 value 0.022767
## iter 3 value 0.022350
## iter 4 value 0.022341
## iter 5 value 0.022341
## iter 6 value 0.022341
## iter 7 value 0.022341
## iter 7 value 0.022341
## iter 7 value 0.022341
## final value 0.022341
## converged
## initial value 0.019968
## iter 2 value 0.019966
## iter 3 value 0.019966
## iter 4 value 0.019966
## iter 5 value 0.019966
## iter 6 value 0.019966
## iter 6 value 0.019966
## iter 6 value 0.019966
## final value 0.019966
## converged
```

```
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1058 0.0433  2.4443 0.0149
## ar2     -0.0100 0.0436 -0.2306 0.8177
## ar3     -0.3257 0.0433 -7.5295 0.0000
## xmean     3.2004 0.0381 83.9925 0.0000
##
## sigma^2 estimated as 1.039963 on 471 degrees of freedom
##
## AIC = 2.898862  AICc = 2.899041  BIC = 2.942687
##
```



ARMA(3,1):

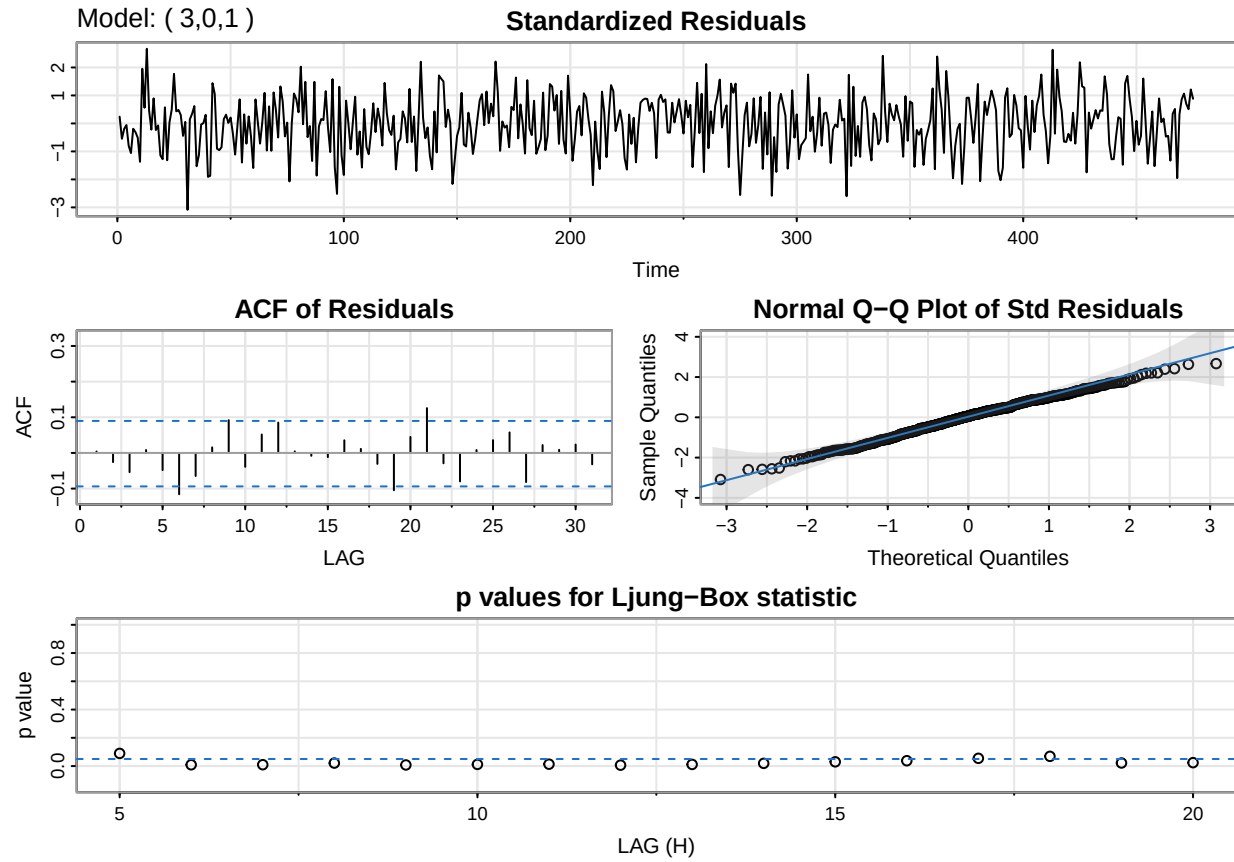
```
sarima(xx, 3,0,1,no.constant=F,details=T)
```

```
## initial value 0.087094
## iter 2 value 0.031564
## iter 3 value 0.025221
## iter 4 value 0.022735
## iter 5 value 0.022634
## iter 6 value 0.022582
## iter 7 value 0.022246
## iter 8 value 0.022088
## iter 9 value 0.022014
```

```

## iter 10 value 0.022011
## iter 11 value 0.022010
## iter 12 value 0.022010
## iter 12 value 0.022010
## final value 0.022010
## converged
## initial value 0.019650
## iter 2 value 0.019648
## iter 3 value 0.019648
## iter 4 value 0.019648
## iter 5 value 0.019648
## iter 6 value 0.019648
## iter 7 value 0.019648
## iter 8 value 0.019647
## iter 9 value 0.019647
## iter 10 value 0.019647
## iter 11 value 0.019647
## iter 11 value 0.019647
## iter 11 value 0.019647
## final value 0.019647
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2173 0.2337  0.9301  0.3528
## ar2     -0.0239 0.0533 -0.4482  0.6542
## ar3     -0.3188 0.0481 -6.6331  0.0000
## ma1     -0.1249 0.2570 -0.4861  0.6271
## xmean     3.2003 0.0364 87.8231  0.0000
##
## sigma^2 estimated as 1.039292 on 470 degrees of freedom
##
## AIC = 2.902435  AICc = 2.902704  BIC = 2.955024
##

```



ARMA(3,2):

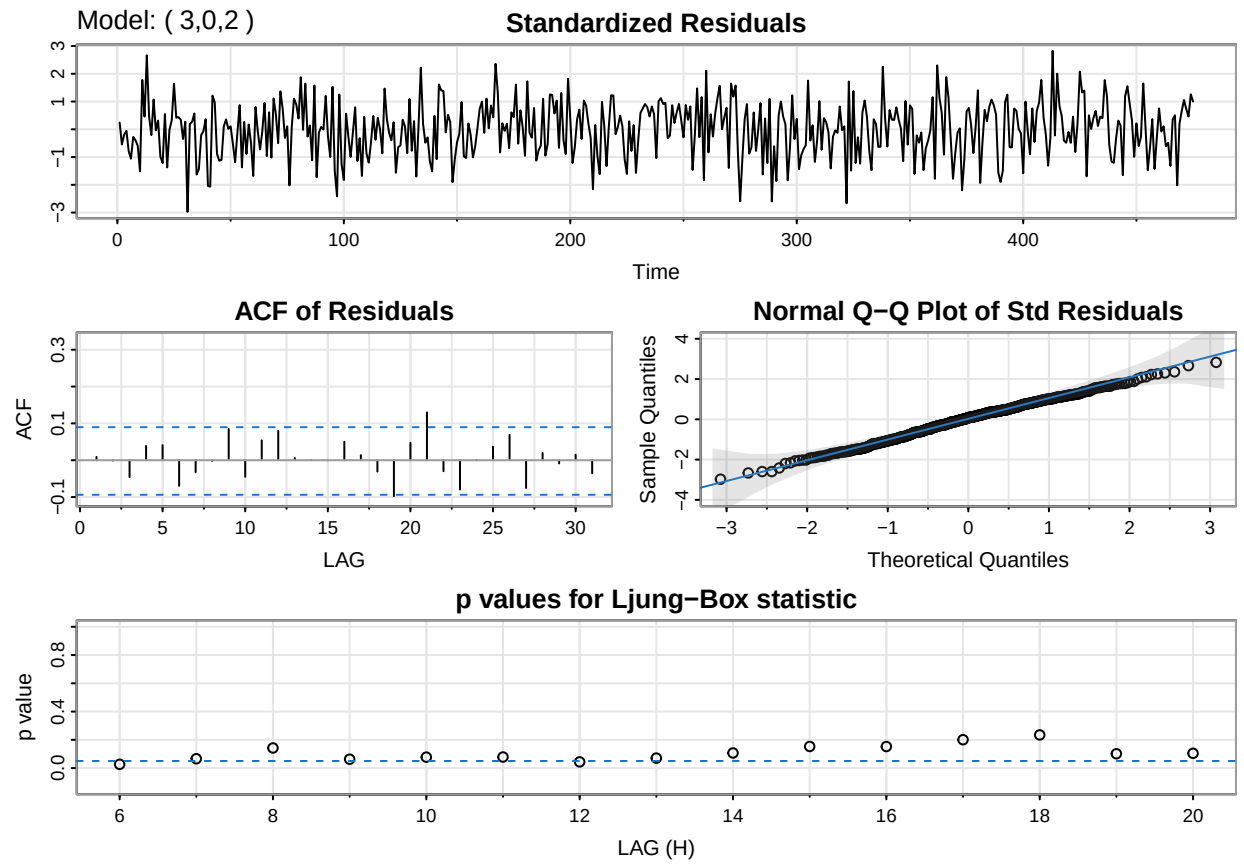
```
sarima(xx, 3,0,2,no.constant=F,details=T)
```

```
## initial value 0.087094
## iter 2 value 0.033094
## iter 3 value 0.026943
## iter 4 value 0.022314
## iter 5 value 0.021896
## iter 6 value 0.021305
## iter 7 value 0.018577
## iter 8 value 0.017163
## iter 9 value 0.016585
## iter 10 value 0.015872
## iter 11 value 0.015805
## iter 12 value 0.015763
## iter 13 value 0.015735
## iter 14 value 0.015657
## iter 15 value 0.015655
## iter 16 value 0.015651
## iter 17 value 0.015646
## iter 18 value 0.015643
## iter 19 value 0.015642
## iter 20 value 0.015641
## iter 21 value 0.015641
## iter 22 value 0.015641
## iter 23 value 0.015641
```

```

## iter 24 value 0.015641
## iter 25 value 0.015640
## iter 25 value 0.015640
## iter 25 value 0.015640
## final value 0.015640
## converged
## initial value 0.013647
## iter 2 value 0.013646
## iter 3 value 0.013641
## iter 4 value 0.013641
## iter 5 value 0.013640
## iter 6 value 0.013639
## iter 7 value 0.013639
## iter 8 value 0.013639
## iter 9 value 0.013639
## iter 10 value 0.013639
## iter 11 value 0.013639
## iter 12 value 0.013639
## iter 13 value 0.013639
## iter 13 value 0.013639
## iter 13 value 0.013639
## final value 0.013639
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2687 0.1142  2.3525 0.0191
## ar2      0.2229 0.1047  2.1284 0.0338
## ar3     -0.3410 0.0454 -7.5090 0.0000
## ma1     -0.1970 0.1192 -1.6535 0.0989
## ma2     -0.2818 0.1045 -2.6976 0.0072
## xmean     3.1995 0.0286 111.7306 0.0000
##
## sigma^2 estimated as 1.026713 on 469 degrees of freedom
##
## AIC = 2.894629  AICc = 2.895007  BIC = 2.955984
##

```



Question 3 (Analysis of dat3)

Summary

For the series $Y_t = X_t$ I chose an SARIMA(1,0,2)x(1,1,1) model excluding the intercept term. The parameter estimates are $\phi_1 = 0.4565$, $\theta_1 = 0.0006$, $\theta_2 = 0.2974$, $\Phi_1 = 0.4469$, and $\Theta_1 = -0.9728$.

```
xx <- dat3
sarima(xx, p = 1, d = 0, q = 2, P = 1, D = 1,
Q = 1, S = 10,no.constant=F,details=F)

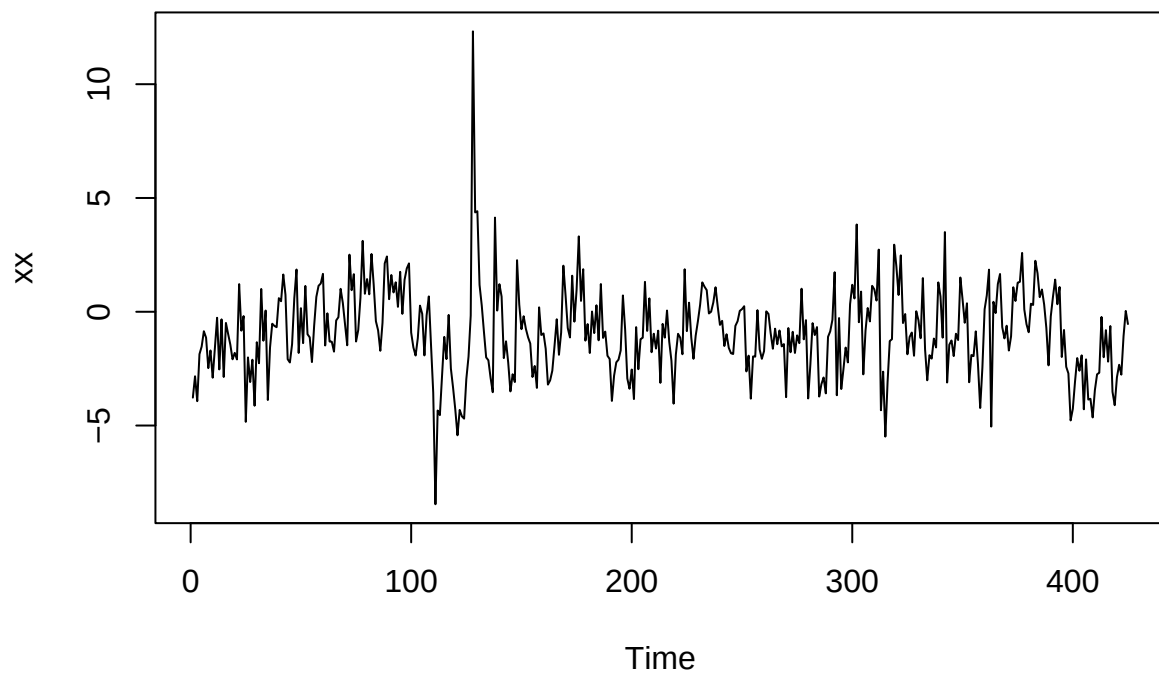
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE  t.value p.value
## ar1          0.4565 0.1006   4.5382  0.0000
## ma1          0.0006 0.1025   0.0060  0.9952
## ma2          0.2974 0.0605   4.9163  0.0000
## sar1          0.4469 0.0565   7.9086  0.0000
## sma1         -0.9728 0.0564 -17.2566  0.0000
## constant    -0.0003 0.0024  -0.1250  0.9006
##
## sigma^2 estimated as 2.037676 on 409 degrees of freedom
##
## AIC = 3.630854  AICc = 3.631351  BIC = 3.698801
##
```

Explanation

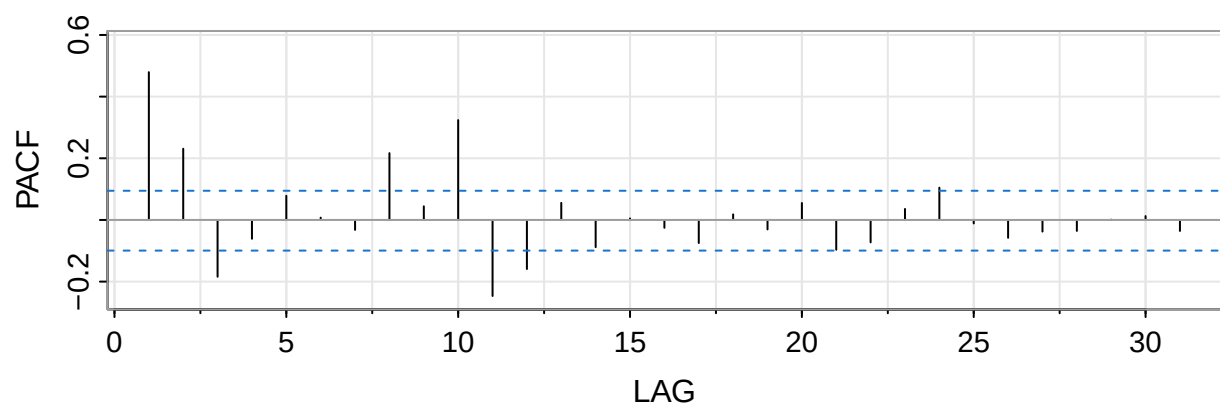
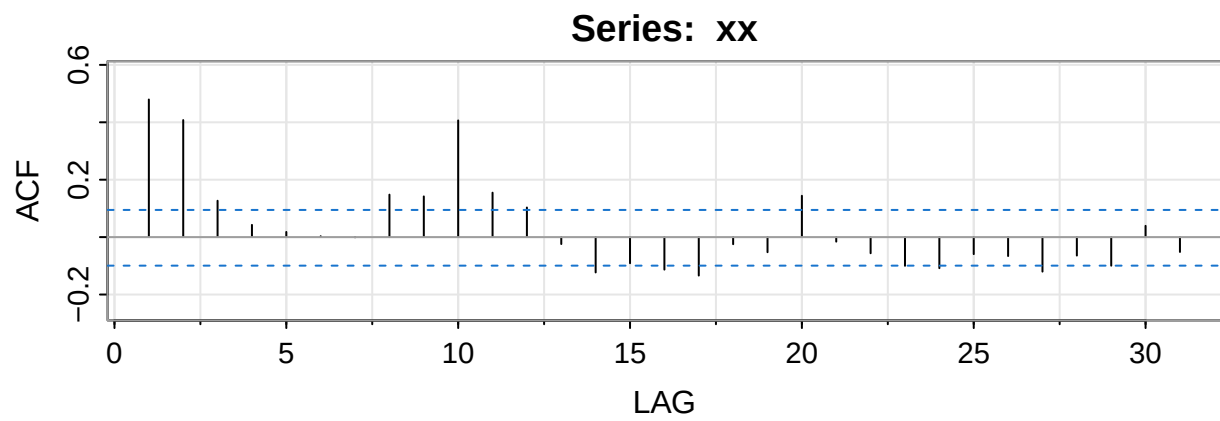
The initial plot and ACF and PACF of X_t suggest that the series is stationary so we don't need to take a non-seasonal difference but there is a seasonality of 10 so we should take a seasonal difference at lag 10. After the seasonal difference, the significant non seasonal lags of the ACF cut off for the most part after lag 4 and those of the PACF cut off after lag 3, while the significant seasonal lags of the ACF cut off for the most part after lag 2 and those of the PACF cut off after lag 2, so we fit different models on the first seasonally differenced data with a maximum p of 3, a maximum q of 4, a maximum P of 2 and a maximum Q of 2. Many of the models had AICs very close to the minimum so only models with a BIC multiplied by the length of the series within 5 of the minimum were considered. The ones which met this criteria were SARMA(1,2)x(1,1), SARMA(1,2)x(1,2), SARMA(1,2)x(2,1), SARMA(2,0)x(1,1), SARMA(2,1)x(1,1), SARMA(3,0)x(1,1), SARMA(3,0)x(1,2) and SARMA(3,0)x(2,1). Of these, the SARMA(1,2)x(1,1) was chosen because it has the best BIC and the highest LBP values. So the final model on the original series X_t is SARIMA(1,0,2)x(1,1,1)[10]. The intercept was not included because it was not significant at any level. One additional note is that all models had a heavy right tail in the normal QQ plot of the standardized residuals, but there is not much we can do about this as it is caused by the large spike in the series.

Output

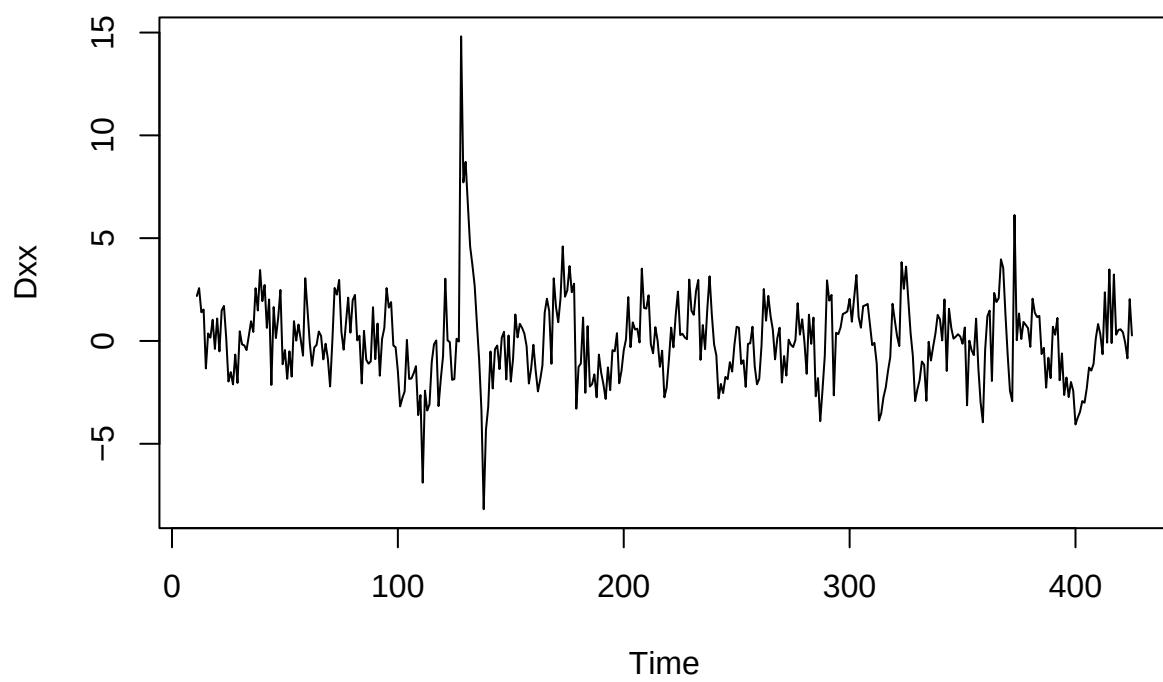
```
plot(xx)
```



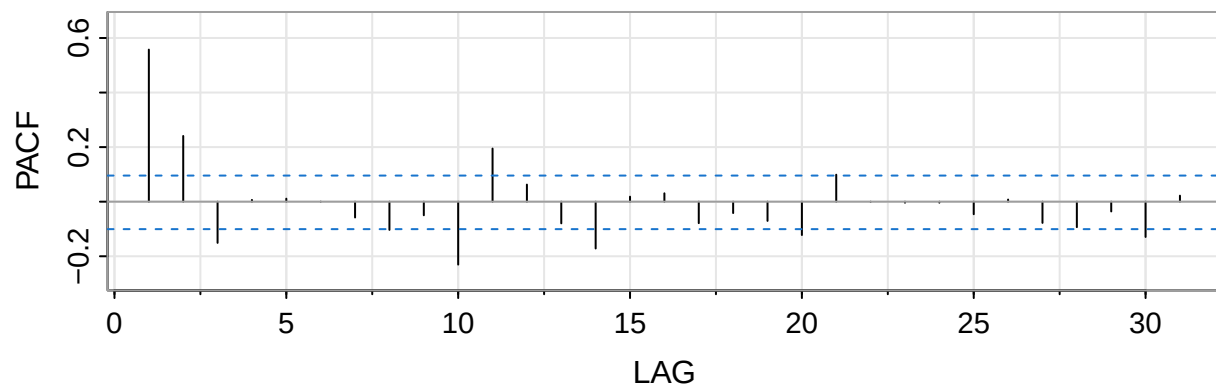
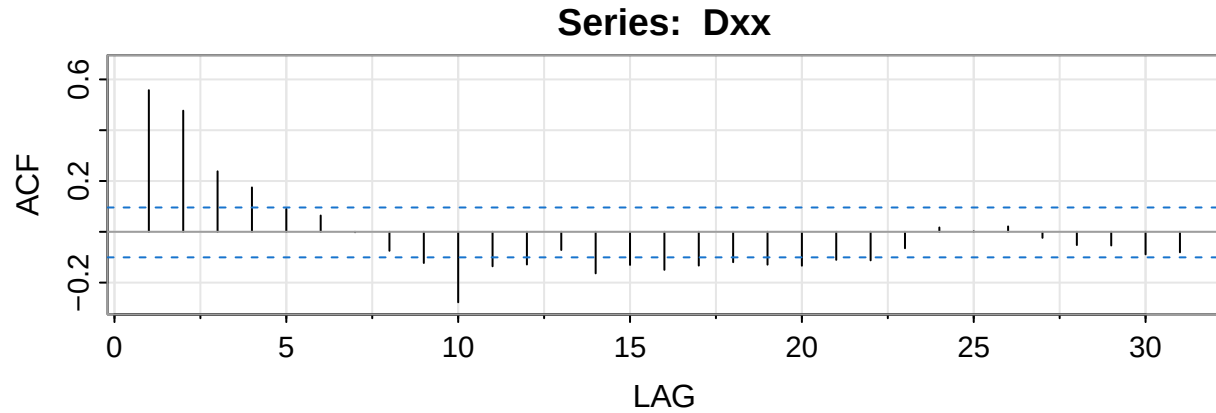
```
invisible(acf2(xx))
```

```
Dxx <- diff(xx, lag=10)  
plot(Dxx)
```



```
invisible(acf2(Dxx))
```

[illegible]

```

##      Estimate      SE t.value p.value
## xmean  0.0072 0.1021  0.071  0.9434
##
## sigma^2 estimated as 4.322589 on 414 degrees of freedom
##
## AIC = 4.31137  AICc = 4.311394  BIC = 4.330784
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sma1   -0.6125 0.0670 -9.1481  0.000
## xmean  -0.0056 0.0376 -0.1498  0.881
##
## sigma^2 estimated as 3.61335 on 413 degrees of freedom
##
## AIC = 4.148304  AICc = 4.148374  BIC = 4.177424
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sma1   -0.5926 0.0482 -12.2837 0.0000
## sma2   -0.3316 0.0458  -7.2347 0.0000
## xmean  -0.0063 0.0116  -0.5477 0.5842
##
## sigma^2 estimated as 3.202856 on 412 degrees of freedom
##
## AIC = 4.063797  AICc = 4.063938  BIC = 4.102624
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.2764 0.0470 -5.8850 0.0000
## xmean  -0.0006 0.0772 -0.0072 0.9943
##
## sigma^2 estimated as 3.983324 on 413 degrees of freedom
##
## AIC = 4.236366  AICc = 4.236436  BIC = 4.265486
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1    0.4157 0.0470  8.8365 0.0000
## sma1   -1.0000 0.0518 -19.3111 0.0000
## xmean  -0.0059 0.0113  -0.5235 0.6009
##
## sigma^2 estimated as 3.056551 on 412 degrees of freedom
##
## AIC = 4.044256  AICc = 4.044396  BIC = 4.083082
##
## <><><><><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1    0.3747 0.1129  3.3194  0.0010
## sma1   -0.9523 0.1311 -7.2626  0.0000
## sma2   -0.0477 0.1168 -0.4082  0.6833
## xmean  -0.0059 0.0111 -0.5305  0.5960
##
## sigma^2 estimated as 3.052573 on 411 degrees of freedom
##
## AIC = 4.048681  AICc = 4.048916  BIC = 4.097214
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.3436 0.0479 -7.1750  0.0000
## sar2   -0.2309 0.0476 -4.8471  0.0000
## xmean  -0.0051 0.0612 -0.0840  0.9331
##
## sigma^2 estimated as 3.76104 on 412 degrees of freedom
##
## AIC = 4.186445  AICc = 4.186586  BIC = 4.225272
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1    0.4227 0.0499  8.4739  0.0000
## sar2   -0.0209 0.0505 -0.4135  0.6795
## sma1   -0.9999 0.0608 -16.4550  0.0000
## xmean  -0.0059 0.0111 -0.5309  0.5958
##
## sigma^2 estimated as 3.052357 on 411 degrees of freedom
##
## AIC = 4.048663  AICc = 4.048898  BIC = 4.097197
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.3672   NaN    NaN    NaN
## sar2    0.3247   NaN    NaN    NaN
## sma1   -0.2166   NaN    NaN    NaN
## sma2   -0.7834   NaN    NaN    NaN
## xmean  -0.0059 0.0113 -0.524  0.6005
##
## sigma^2 estimated as 3.056478 on 410 degrees of freedom
##
## AIC = 4.053892  AICc = 4.054245  BIC = 4.112132
##

```

```

## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3517 0.0342 10.2987 0.0000
## xmean    0.0082 0.1241  0.0664 0.9471
##
## sigma^2 estimated as 3.499924 on 413 degrees of freedom
##
## AIC = 4.105394  AICc = 4.105464  BIC = 4.134514
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3554 0.0342 10.3852 0.0000
## sma1    -0.5889 0.0638 -9.2246 0.0000
## xmean   -0.0048 0.0484 -0.0989 0.9213
##
## sigma^2 estimated as 2.919141 on 412 degrees of freedom
##
## AIC = 3.939032  AICc = 3.939172  BIC = 3.977858
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3444 0.0353  9.7585 0.0000
## sma1    -0.5798 0.0486 -11.9205 0.0000
## sma2    -0.3133 0.0484  -6.4794 0.0000
## xmean   -0.0059 0.0165  -0.3616 0.7178
##
## sigma^2 estimated as 2.65883 on 411 degrees of freedom
##
## AIC = 3.875193  AICc = 3.875428  BIC = 3.923726
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3605 0.0339 10.6297 0.0000
## sar1    -0.2968 0.0469 -6.3224 0.0000
## xmean    0.0001 0.0924  0.0014 0.9989
##
## sigma^2 estimated as 3.186639 on 412 degrees of freedom
##
## AIC = 4.018678  AICc = 4.018819  BIC = 4.057505
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3489 0.0349 10.0108 0.0000
## sar1     0.4319 0.0466  9.2707 0.0000

```

```

## sma1    -1.0000 0.0666 -15.0229  0.0000
## xmean   -0.0054 0.0141  -0.3849  0.7005
##
## sigma^2 estimated as 2.499563 on 411 degrees of freedom
##
## AIC = 3.847316  AICc = 3.847551  BIC = 3.895849
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3500 0.0349 10.0335 0.0000
## sar1      0.4742 0.1079  4.3934 0.0000
## sma1     -1.0497 0.1295 -8.1049 0.0000
## sma2      0.0497 0.1179  0.4220 0.6732
## xmean    -0.0054 0.0144 -0.3768 0.7065
##
## sigma^2 estimated as 2.501273 on 410 degrees of freedom
##
## AIC = 3.851707  AICc = 3.85206  BIC = 3.909947
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3534 0.0341 10.3490 0.0000
## sar1     -0.3622 0.0484 -7.4770 0.0000
## sar2     -0.2112 0.0483 -4.3681 0.0000
## xmean    -0.0043 0.0745 -0.0582 0.9536
##
## sigma^2 estimated as 3.040576 on 411 degrees of freedom
##
## AIC = 3.978797  AICc = 3.979033  BIC = 4.027331
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3500 0.0349 10.0321 0.0000
## sar1      0.4243 0.0500  8.4906 0.0000
## sar2      0.0213 0.0506  0.4215 0.6736
## sma1     -1.0000 0.0544 -18.3924 0.0000
## xmean    -0.0054 0.0143  -0.3786 0.7052
##
## sigma^2 estimated as 2.501045 on 410 degrees of freedom
##
## AIC = 3.851706  AICc = 3.85206  BIC = 3.909946
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3498 0.0349 10.0132 0.0000
## sar1      0.0071 1.2595  0.0056 0.9955

```

```

## sar2    0.2011 0.5291 0.3800 0.7042
## sma1    -0.5827 1.2670 -0.4599 0.6458
## sma2    -0.4173 1.2660 -0.3296 0.7419
## xmean   -0.0054 0.0143 -0.3779 0.7057
##
## sigma^2 estimated as 2.500063 on 409 degrees of freedom
##
## AIC = 3.856457 AICc = 3.856953 BIC = 3.924404
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4129 0.0432  9.5511 0.0000
## ma2      0.4380 0.0458  9.5684 0.0000
## xmean    0.0126 0.1532  0.0823 0.9344
##
## sigma^2 estimated as 2.853686 on 412 degrees of freedom
##
## AIC = 3.906999 AICc = 3.90714 BIC = 3.945826
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3968 0.0472  8.4104 0.0000
## ma2      0.4266 0.0411 10.3891 0.0000
## sma1     -0.5520 0.0602 -9.1731 0.0000
## xmean    -0.0028 0.0637 -0.0441 0.9648
##
## sigma^2 estimated as 2.38014 on 411 degrees of freedom
##
## AIC = 3.739128 AICc = 3.739363 BIC = 3.787661
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.3929 0.0461  8.5186 0.0000
## ma2      0.4140 0.0418  9.9101 0.0000
## sma1     -0.5600 0.0554 -10.1059 0.0000
## sma2     -0.2852 0.0611  -4.6652 0.0000
## xmean    -0.0050 0.0254  -0.1968 0.8441
##
## sigma^2 estimated as 2.232289 on 410 degrees of freedom
##
## AIC = 3.697992 AICc = 3.698346 BIC = 3.756233
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4051 0.0453  8.9441 0.0000
## ma2      0.4435 0.0429 10.3472 0.0000

```



```

## sar1    -0.3137 0.0465 -6.7496  0.0000
## xmean    0.0033 0.1111  0.0296  0.9764
##
## sigma^2 estimated as 2.566324 on 411 degrees of freedom
##
## AIC = 3.808248  AICc = 3.808483  BIC = 3.856782
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4004 0.0452  8.8655  0.000
## ma2      0.4256 0.0427  9.9585  0.000
## sar1      0.4515 0.0462  9.7749  0.000
## sma1     -1.0000 0.0764 -13.0940  0.000
## xmean    -0.0044 0.0178  -0.2496  0.803
##
## sigma^2 estimated as 2.064911 on 410 degrees of freedom
##
## AIC = 3.66092  AICc = 3.661274  BIC = 3.71916
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4007 0.0454  8.8187  0.0000
## ma2      0.4287 0.0424 10.1104  0.0000
## sar1      0.5699 0.0959  5.9441  0.0000
## sma1     -1.1428 0.1178 -9.7047  0.0000
## sma2      0.1428 0.1101  1.2967  0.1955
## xmean    -0.0045 0.0190 -0.2350  0.8143
##
## sigma^2 estimated as 2.06404 on 409 degrees of freedom
##
## AIC = 3.661788  AICc = 3.662284  BIC = 3.729735
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4046 0.0457  8.8559  0.0000
## ma2      0.4338 0.0420 10.3294  0.0000
## sar1     -0.3776 0.0484 -7.8037  0.0000
## sar2     -0.1955 0.0483 -4.0443  0.0001
## xmean    -0.0017 0.0910 -0.0186  0.9852
##
## sigma^2 estimated as 2.464818 on 410 degrees of freedom
##
## AIC = 3.774582  AICc = 3.774936  BIC = 3.832822
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value

```

```

## ma1      0.4006 0.0454   8.8235  0.0000
## ma2      0.4290 0.0425  10.0972  0.0000
## sar1      0.4276 0.0499   8.5649  0.0000
## sar2      0.0633 0.0502   1.2608  0.2081
## sma1     -1.0000 0.0445 -22.4937  0.0000
## xmean    -0.0044 0.0189  -0.2358  0.8137
##
## sigma^2 estimated as 2.063082 on 409 degrees of freedom
##
## AIC = 3.661903  AICc = 3.662399  BIC = 3.72985
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4007 0.0455   8.8128  0.0000
## ma2      0.4281 0.0426  10.0518  0.0000
## sar1      0.6978 0.7948   0.8780  0.3805
## sar2     -0.0594 0.3731  -0.1593  0.8735
## sma1     -1.2698 0.7882  -1.6110  0.1080
## sma2      0.2698 0.7873   0.3427  0.7320
## xmean    -0.0045 0.0192  -0.2331  0.8158
##
## sigma^2 estimated as 2.065016 on 408 degrees of freedom
##
## AIC = 3.666556  AICc = 3.667219  BIC = 3.744209
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4584 0.0483   9.5014  0.0000
## ma2      0.4639 0.0453  10.2380  0.0000
## ma3      0.1182 0.0424   2.7892  0.0055
## xmean     0.0136 0.1673   0.0815  0.9350
##
## sigma^2 estimated as 2.800241 on 411 degrees of freedom
##
## AIC = 3.892905  AICc = 3.89314  BIC = 3.941438
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4422 0.0473   9.3530  0.0000
## ma2      0.4899 0.0433  11.3256  0.0000
## ma3      0.1826 0.0446   4.0943  0.0001
## sma1     -0.5771 0.0584  -9.8832  0.0000
## xmean    -0.0020 0.0685  -0.0298  0.9762
##
## sigma^2 estimated as 2.288649 on 410 degrees of freedom
##
## AIC = 3.705759  AICc = 3.706112  BIC = 3.763999
##

```

```

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4361 0.0478   9.1260 0.0000
## ma2      0.4634 0.0444  10.4275 0.0000
## ma3      0.1584 0.0467   3.3897 0.0008
## sma1     -0.5651 0.0565 -10.0036 0.0000
## sma2     -0.2415 0.0676  -3.5716 0.0004
## xmean    -0.0041 0.0335  -0.1218 0.9031
##
## sigma^2 estimated as 2.184032 on 409 degrees of freedom
##
## AIC = 3.676341  AICc = 3.676837  BIC = 3.744288
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4579 0.0477   9.6064 0.0000
## ma2      0.4939 0.0436  11.3162 0.0000
## ma3      0.1671 0.0438   3.8112 0.0002
## sar1     -0.3411 0.0470  -7.2525 0.0000
## xmean     0.0043 0.1226   0.0351 0.9721
##
## sigma^2 estimated as 2.478804 on 410 degrees of freedom
##
## AIC = 3.778867  AICc = 3.77922  BIC = 3.837107
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4454 0.0480   9.2750 0.0000
## ma2      0.4662 0.0446  10.4551 0.0000
## ma3      0.1435 0.0459   3.1286 0.0019
## sar1      0.4390 0.0564   7.7900 0.0000
## sma1     -0.9801 0.0718 -13.6421 0.0000
## xmean    -0.0040 0.0201  -0.2007 0.8411
##
## sigma^2 estimated as 2.050377 on 409 degrees of freedom
##
## AIC = 3.64232  AICc = 3.642816  BIC = 3.710267
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4484 0.0478   9.3878 0.0000
## ma2      0.4768 0.0445  10.7203 0.0000
## ma3      0.1560 0.0456   3.4237 0.0007
## sar1      0.6138 0.0928   6.6137 0.0000
## sma1     -1.2065 0.1187 -10.1683 0.0000
## sma2      0.2065 0.1103   1.8716 0.0620

```

```

## xmean -0.0040 0.0218 -0.1821 0.8556
##
## sigma^2 estimated as 2.010775 on 408 degrees of freedom
##
## AIC = 3.639184 AICc = 3.639847 BIC = 3.716837
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4502 0.0478  9.4287 0.0000
## ma2      0.4846 0.0437 11.0965 0.0000
## ma3      0.1620 0.0446  3.6332 0.0003
## sar1     -0.4066 0.0498 -8.1722 0.0000
## sar2     -0.1881 0.0486 -3.8695 0.0001
## xmean    -0.0008 0.1007 -0.0079 0.9937
##
## sigma^2 estimated as 2.38877 on 409 degrees of freedom
##
## AIC = 3.748393 AICc = 3.748889 BIC = 3.816339
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4492 0.0478  9.3966 0.0000
## ma2      0.4776 0.0447 10.6800 0.0000
## ma3      0.1561 0.0456  3.4215 0.0007
## sar1      0.4073 0.0507  8.0386 0.0000
## sar2      0.0923 0.0511  1.8067 0.0715
## sma1     -1.0000 0.0507 -19.7174 0.0000
## xmean    -0.0039 0.0214 -0.1831 0.8548
##
## sigma^2 estimated as 2.008855 on 408 degrees of freedom
##
## AIC = 3.639312 AICc = 3.639975 BIC = 3.716965
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4485 0.0479  9.3587 0.0000
## ma2      0.4770 0.0448 10.6413 0.0000
## ma3      0.1560 0.0456  3.4187 0.0007
## sar1      0.5960 0.7523  0.7922 0.4287
## sar2      0.0083 0.3466  0.0239 0.9809
## sma1     -1.1889 0.7486 -1.5883 0.1130
## sma2      0.1889 0.7473  0.2528 0.8005
## xmean    -0.0040 0.0217 -0.1823 0.8555
##
## sigma^2 estimated as 2.010582 on 407 degrees of freedom
##
## AIC = 3.644001 AICc = 3.644856 BIC = 3.731362
##

```

```

## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4599 0.0487  9.4408 0.0000
## ma2      0.5121 0.0519  9.8660 0.0000
## ma3      0.2269 0.0605  3.7474 0.0002
## ma4      0.1544 0.0519  2.9767 0.0031
## xmean     0.0173 0.1908  0.0908 0.9277
##
## sigma^2 estimated as 2.744361 on 410 degrees of freedom
##
## AIC = 3.877711  AICc = 3.878065  BIC = 3.935951
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4557 0.0486  9.3708 0.0000
## ma2      0.5361 0.0511 10.4971 0.0000
## ma3      0.2340 0.0510  4.5865 0.0000
## ma4      0.1203 0.0479  2.5096 0.0125
## sma1     -0.5576 0.0572 -9.7498 0.0000
## xmean    -0.0004 0.0787 -0.0050 0.9960
##
## sigma^2 estimated as 2.256522 on 409 degrees of freedom
##
## AIC = 3.695879  AICc = 3.696375  BIC = 3.763826
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4449 0.0487  9.1287 0.0000
## ma2      0.5085 0.0523  9.7152 0.0000
## ma3      0.2075 0.0536  3.8736 0.0001
## ma4      0.1059 0.0499  2.1211 0.0345
## sma1     -0.5338 0.0550 -9.7127 0.0000
## sma2     -0.2114 0.0663 -3.1875 0.0015
## xmean    -0.0026 0.0460 -0.0557 0.9556
##
## sigma^2 estimated as 2.172964 on 408 degrees of freedom
##
## AIC = 3.670821  AICc = 3.671484  BIC = 3.748474
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4652 0.0486  9.5794 0.0000
## ma2      0.5413 0.0507 10.6721 0.0000
## ma3      0.2410 0.0532  4.5327 0.0000
## ma4      0.1363 0.0484  2.8152 0.0051
## sar1     -0.3356 0.0466 -7.1995 0.0000

```

```

## xmean    0.0073 0.1371  0.0529  0.9578
##
## sigma^2 estimated as 2.433605 on 409 degrees of freedom
##
## AIC = 3.765366  AICc = 3.765863  BIC = 3.833313
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.4476 0.0487   9.1883 0.0000
## ma2      0.5025 0.0514   9.7813 0.0000
## ma3      0.1907 0.0532   3.5835 0.0004
## ma4      0.0977 0.0491   1.9884 0.0474
## sar1     0.4503 0.0573   7.8598 0.0000
## sma1    -0.9727 0.0569 -17.0908 0.0000
## xmean   -0.0035 0.0228  -0.1513 0.8798
##
## sigma^2 estimated as 2.042728 on 408 degrees of freedom
##
## AIC = 3.637893  AICc = 3.638556  BIC = 3.715547
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.4536 0.0487   9.3093 0.0000
## ma2      0.5174 0.0516  10.0193 0.0000
## ma3      0.2068 0.0528   3.9176 0.0001
## ma4      0.1058 0.0488   2.1690 0.0307
## sar1     0.6421 0.0887   7.2401 0.0000
## sma1    -1.2240 0.1146 -10.6764 0.0000
## sma2     0.2240 0.1068   2.0979 0.0365
## xmean   -0.0032 0.0248  -0.1295 0.8970
##
## sigma^2 estimated as 1.992022 on 407 degrees of freedom
##
## AIC = 3.63302  AICc = 3.633874  BIC = 3.72038
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.4575 0.0485   9.4336 0.0000
## ma2      0.5357 0.0510  10.4939 0.0000
## ma3      0.2354 0.0533   4.4148 0.0000
## ma4      0.1388 0.0483   2.8741 0.0043
## sar1    -0.4010 0.0491  -8.1729 0.0000
## sar2    -0.1893 0.0485  -3.9044 0.0001
## xmean    0.0017 0.1129   0.0149 0.9881
##
## sigma^2 estimated as 2.343604 on 408 degrees of freedom
##
## AIC = 3.73427  AICc = 3.734933  BIC = 3.811923

```

```

##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4539 0.0488  9.3036 0.0000
## ma2      0.5166 0.0517  9.9966 0.0000
## ma3      0.2050 0.0526  3.8944 0.0001
## ma4      0.1031 0.0487  2.1180 0.0348
## sar1      0.4202 0.0506  8.3088 0.0000
## sar2      0.0995 0.0508  1.9584 0.0509
## sma1     -1.0000 0.0514 -19.4498 0.0000
## xmean    -0.0032 0.0241  -0.1324 0.8947
##
## sigma^2 estimated as 1.990153 on 407 degrees of freedom
##
## AIC = 3.633633  AICc = 3.634488  BIC = 3.720993
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.4530 0.0488  9.2872 0.0000
## ma2      0.5168 0.0517  9.9940 0.0000
## ma3      0.2069 0.0528  3.9169 0.0001
## ma4      0.1069 0.0490  2.1834 0.0296
## sar1      0.7546 0.4582  1.6468 0.1004
## sar2     -0.0554 0.2282 -0.2427 0.8084
## sma1     -1.3342 0.4505 -2.9617 0.0032
## sma2      0.3342 0.4490  0.7445 0.4570
## xmean    -0.0032 0.0251 -0.1266 0.8993
##
## sigma^2 estimated as 1.993348 on 406 degrees of freedom
##
## AIC = 3.63772  AICc = 3.638791  BIC = 3.734787
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5577 0.0407 13.7140 0.000
## xmean     0.0146 0.1909  0.0766 0.939
##
## sigma^2 estimated as 2.975372 on 413 degrees of freedom
##
## AIC = 3.943602  AICc = 3.943672  BIC = 3.972722
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5639 0.0406 13.9008 0.0000
## sma1     -0.5551 0.0596 -9.3133 0.0000
## xmean    -0.0005 0.0810 -0.0062 0.9951

```

```

##
## sigma^2 estimated as 2.473842 on 412 degrees of freedom
##
## AIC = 3.772717 AICc = 3.772857 BIC = 3.811543
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5391 0.0422 12.7886 0.0000
## sma1     -0.5435 0.0561 -9.6845 0.0000
## sma2     -0.2567 0.0680 -3.7741 0.0002
## xmean    -0.0035 0.0376 -0.0942 0.9250
##
## sigma^2 estimated as 2.344579 on 411 degrees of freedom
##
## AIC = 3.736613 AICc = 3.736848 BIC = 3.785147
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5763 0.0402 14.3475 0.0000
## sar1     -0.3223 0.0467 -6.9062 0.0000
## xmean     0.0070 0.1434 0.0487 0.9612
##
## sigma^2 estimated as 2.663169 on 412 degrees of freedom
##
## AIC = 3.84028 AICc = 3.840421 BIC = 3.879107
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5445 0.0413 13.1923 0.0000
## sar1      0.4578 0.0460 9.9590 0.0000
## sma1     -1.0000 0.1504 -6.6476 0.0000
## xmean    -0.0035 0.0220 -0.1581 0.8744
##
## sigma^2 estimated as 2.160236 on 411 degrees of freedom
##
## AIC = 3.700476 AICc = 3.700711 BIC = 3.749009
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5530 0.0413 13.3926 0.0000
## sar1      0.5950 0.0925 6.4313 0.0000
## sma1     -1.1670 0.1160 -10.0565 0.0000
## sma2      0.1670 0.1082 1.5431 0.1236
## xmean    -0.0033 0.0243 -0.1360 0.8919
##
## sigma^2 estimated as 2.157418 on 410 degrees of freedom

```



```

##
## AIC = 3.699736 AICc = 3.700089 BIC = 3.757976
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5661 0.0406 13.9272 0.0000
## sar1     -0.3845 0.0488 -7.8734 0.0000
## sar2     -0.1871 0.0489 -3.8267 0.0002
## xmean     0.0012 0.1164 0.0099 0.9921
##
## sigma^2 estimated as 2.568413 on 411 degrees of freedom
##
## AIC = 3.810574 AICc = 3.810809 BIC = 3.859107
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5531 0.0414 13.3755 0.0000
## sar1      0.4283 0.0499 8.5850 0.0000
## sar2      0.0764 0.0507 1.5065 0.1327
## sma1     -1.0000 0.0458 -21.8194 0.0000
## xmean    -0.0033 0.0240 -0.1364 0.8916
##
## sigma^2 estimated as 2.156016 on 410 degrees of freedom
##
## AIC = 3.699821 AICc = 3.700175 BIC = 3.758061
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5527 0.0415 13.3165 0.0000
## sar1      0.6572 1.1564 0.5683 0.5701
## sar2     -0.0295 0.5479 -0.0538 0.9571
## sma1     -1.2287 1.1494 -1.0690 0.2857
## sma2      0.2287 1.1488 0.1991 0.8423
## xmean    -0.0033 0.0244 -0.1360 0.8919
##
## sigma^2 estimated as 2.158032 on 409 degrees of freedom
##
## AIC = 3.70454 AICc = 3.705036 BIC = 3.772487
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7415 0.0484 15.3261 0.0000
## ma1     -0.2636 0.0635 -4.1497 0.0000
## xmean     0.0235 0.2353 0.0999 0.9204
##
## sigma^2 estimated as 2.865068 on 412 degrees of freedom

```

```

##
## AIC = 3.910795 AICc = 3.910935 BIC = 3.949621
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7413 0.0480 15.4442 0.0000
## ma1     -0.2546 0.0621 -4.0979 0.0001
## sma1    -0.5467 0.0588 -9.2968 0.0000
## xmean    0.0046 0.1014  0.0451 0.9641
##
## sigma^2 estimated as 2.384221 on 411 degrees of freedom
##
## AIC = 3.740344 AICc = 3.740579 BIC = 3.788878
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7213 0.0518 13.9251 0.0000
## ma1     -0.2484 0.0649 -3.8274 0.0001
## sma1    -0.5294 0.0549 -9.6498 0.0000
## sma2    -0.2327 0.0674 -3.4530 0.0006
## xmean   -0.0006 0.0529 -0.0114 0.9909
##
## sigma^2 estimated as 2.277987 on 410 degrees of freedom
##
## AIC = 3.70921 AICc = 3.709564 BIC = 3.76745
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7531 0.0463 16.2642 0.0000
## ma1     -0.2603 0.0613 -4.2494 0.0000
## sar1    -0.3245 0.0465 -6.9798 0.0000
## xmean    0.0161 0.1775  0.0906 0.9279
##
## sigma^2 estimated as 2.558935 on 411 degrees of freedom
##
## AIC = 3.805289 AICc = 3.805524 BIC = 3.853823
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7246 0.0511 14.1910 0.0000
## ma1     -0.2507 0.0650 -3.8553 0.0001
## sar1     0.4646 0.0556  8.3588 0.0000
## sma1    -0.9914 0.1588 -6.2433 0.0000
## xmean   -0.0014 0.0272 -0.0502 0.9600
##
## sigma^2 estimated as 2.107912 on 410 degrees of freedom

```

```

##
## AIC = 3.672396 AICc = 3.672749 BIC = 3.730636
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7325 0.0499 14.6910 0.0000
## ma1     -0.2524 0.0639 -3.9515 0.0001
## sar1      0.6158 0.0885  6.9595 0.0000
## sma1     -1.1846 0.1126 -10.5214 0.0000
## sma2      0.1846 0.1048  1.7613 0.0789
## xmean    -0.0008 0.0304 -0.0256 0.9796
##
## sigma^2 estimated as 2.086367 on 409 degrees of freedom
##
## AIC = 3.670046 AICc = 3.670542 BIC = 3.737993
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7441 0.0476 15.6469 0.0000
## ma1     -0.2569 0.0619 -4.1514 0.0000
## sar1     -0.3858 0.0488 -7.9120 0.0000
## sar2     -0.1819 0.0488 -3.7253 0.0002
## xmean     0.0079 0.1438  0.0552 0.9560
##
## sigma^2 estimated as 2.472485 on 410 degrees of freedom
##
## AIC = 3.777341 AICc = 3.777694 BIC = 3.835581
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7328 0.0499 14.6878 0.0000
## ma1     -0.2525 0.0639 -3.9525 0.0001
## sar1      0.4314 0.0498  8.6561 0.0000
## sar2      0.0864 0.0505  1.7127 0.0875
## sma1     -1.0000 0.0458 -21.8165 0.0000
## xmean    -0.0007 0.0300 -0.0246 0.9804
##
## sigma^2 estimated as 2.084688 on 409 degrees of freedom
##
## AIC = 3.670145 AICc = 3.670641 BIC = 3.738092
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7324 0.0500 14.6598 0.0000
## ma1     -0.2523 0.0639 -3.9503 0.0001
## sar1      0.6058 1.0372  0.5841 0.5595

```

```

## sar2    0.0046 0.4973  0.0093  0.9926
## sma1   -1.1746 1.0336 -1.1364  0.2565
## sma2    0.1746 1.0328  0.1691  0.8658
## xmean  -0.0008 0.0304 -0.0260  0.9793
##
## sigma^2 estimated as 2.086299 on 408 degrees of freedom
##
## AIC = 3.674867  AICc = 3.67553  BIC = 3.752521
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5054 0.0970  5.2090  0.0000
## ma1     -0.0393 0.1030 -0.3813  0.7032
## ma2      0.2694 0.0589  4.5729  0.0000
## xmean    0.0185 0.2018  0.0918  0.9269
##
## sigma^2 estimated as 2.750792 on 411 degrees of freedom
##
## AIC = 3.875169  AICc = 3.875404  BIC = 3.923702
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5025 0.0848  5.9242  0.0000
## ma1     -0.0417 0.0867 -0.4811  0.6307
## ma2      0.3113 0.0569  5.4749  0.0000
## sma1    -0.5594 0.0568 -9.8502  0.0000
## xmean    0.0009 0.0852  0.0100  0.9920
##
## sigma^2 estimated as 2.252391 on 410 degrees of freedom
##
## AIC = 3.689275  AICc = 3.689628  BIC = 3.747515
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4689 0.0947  4.9539  0.0000
## ma1     -0.0175 0.0955 -0.1837  0.8544
## ma2      0.2992 0.0581  5.1535  0.0000
## sma1    -0.5404 0.0546 -9.8995  0.0000
## sma2    -0.2138 0.0680 -3.1457  0.0018
## xmean   -0.0021 0.0475 -0.0452  0.9639
##
## sigma^2 estimated as 2.168621 on 409 degrees of freedom
##
## AIC = 3.664673  AICc = 3.665169  BIC = 3.73262
##
## <><><><><><><><><><><>
##
## Coefficients:

```

```

##      Estimate      SE t.value p.value
## ar1      0.5224 0.0850  6.1444  0.0000
## ma1     -0.0487 0.0889 -0.5481  0.5839
## ma2      0.3023 0.0574  5.2691  0.0000
## sar1    -0.3419 0.0464 -7.3738  0.0000
## xmean    0.0094 0.1500  0.0629  0.9498
##
## sigma^2 estimated as 2.426737 on 410 degrees of freedom
##
## AIC = 3.757826  AICc = 3.75818  BIC = 3.816066
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4565 0.1006  4.5382  0.0000
## ma1      0.0006 0.1025  0.0060  0.9952
## ma2      0.2974 0.0605  4.9163  0.0000
## sar1     0.4469 0.0565  7.9086  0.0000
## sma1    -0.9728 0.0564 -17.2566  0.0000
## xmean   -0.0030 0.0242  -0.1250  0.9006
##
## sigma^2 estimated as 2.037676 on 409 degrees of freedom
##
## AIC = 3.630854  AICc = 3.631351  BIC = 3.698801
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4777 0.0946  5.0506  0.0000
## ma1     -0.0153 0.0965  -0.1586  0.8740
## ma2      0.3022 0.0593  5.0964  0.0000
## sar1     0.6412 0.0877  7.3117  0.0000
## sma1    -1.2264 0.1134 -10.8192  0.0000
## sma2     0.2264 0.1054  2.1484  0.0323
## xmean   -0.0026 0.0265  -0.0991  0.9211
##
## sigma^2 estimated as 1.986601 on 408 degrees of freedom
##
## AIC = 3.625556  AICc = 3.626219  BIC = 3.70321
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5122 0.0859  5.9651  0.0000
## ma1     -0.0458 0.0895 -0.5119  0.6090
## ma2      0.3018 0.0564  5.3546  0.0000
## sar1    -0.4072 0.0490 -8.3087  0.0000
## sar2    -0.1846 0.0486 -3.7969  0.0002
## xmean    0.0032 0.1225  0.0260  0.9792
##
## sigma^2 estimated as 2.34169 on 409 degrees of freedom

```

```

##
## AIC = 3.728634 AICc = 3.72913 BIC = 3.79658
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4762 0.0951  5.0058 0.0000
## ma1     -0.0131 0.0969 -0.1352 0.8925
## ma2      0.3027 0.0597  5.0714 0.0000
## sar1     0.4157 0.0500  8.3159 0.0000
## sar2     0.1031 0.0508  2.0298 0.0430
## sma1    -1.0000 0.0502 -19.9366 0.0000
## xmean   -0.0026 0.0259 -0.1002 0.9202
##
## sigma^2 estimated as 1.984303 on 408 degrees of freedom
##
## AIC = 3.625916 AICc = 3.626579 BIC = 3.703569
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4773 0.0946  5.0437 0.0000
## ma1     -0.0150 0.0966 -0.1548 0.8770
## ma2      0.3022 0.0593  5.0932 0.0000
## sar1     0.6580 0.5635  1.1677 0.2436
## sar2    -0.0082 0.2725 -0.0301 0.9760
## sma1    -1.2430 0.5590 -2.2236 0.0267
## sma2     0.2430 0.5576  0.4358 0.6632
## xmean   -0.0026 0.0266 -0.0987 0.9214
##
## sigma^2 estimated as 1.986625 on 407 degrees of freedom
##
## AIC = 3.630373 AICc = 3.631227 BIC = 3.717733
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6720 0.1168  5.7543 0.0000
## ma1     -0.2100 0.1270 -1.6534 0.0990
## ma2      0.2053 0.0667  3.0777 0.0022
## ma3     -0.1178 0.0788 -1.4959 0.1354
## xmean    0.0211 0.2164  0.0975 0.9224
##
## sigma^2 estimated as 2.738278 on 410 degrees of freedom
##
## AIC = 3.875473 AICc = 3.875827 BIC = 3.933714
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value

```

```

## ar1      0.5563 0.1566  3.5518  0.0004
## ma1     -0.0979 0.1644 -0.5956  0.5518
## ma2      0.2906 0.0777  3.7389  0.0002
## ma3     -0.0372 0.0966 -0.3851  0.7004
## sma1    -0.5564 0.0574 -9.6972  0.0000
## xmean    0.0013 0.0875  0.0149  0.9881
##
## sigma^2 estimated as 2.25183 on 409 degrees of freedom
##
## AIC = 3.693731  AICc = 3.694227  BIC = 3.761678
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5446 0.1786  3.0499  0.0024
## ma1     -0.0957 0.1855 -0.5155  0.6065
## ma2      0.2701 0.0846  3.1930  0.0015
## ma3     -0.0481 0.1039 -0.4631  0.6435
## sma1    -0.5371 0.0551 -9.7557  0.0000
## sma2    -0.2142 0.0681 -3.1430  0.0018
## xmean   -0.0018 0.0492 -0.0373  0.9703
##
## sigma^2 estimated as 2.167977 on 408 degrees of freedom
##
## AIC = 3.668968  AICc = 3.669631  BIC = 3.746622
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6317 0.1390  4.5449  0.0000
## ma1     -0.1637 0.1499 -1.0920  0.2755
## ma2      0.2595 0.0748  3.4677  0.0006
## ma3     -0.0802 0.0939 -0.8549  0.3931
## sar1    -0.3387 0.0465 -7.2760  0.0000
## xmean    0.0113 0.1577  0.0715  0.9430
##
## sigma^2 estimated as 2.422655 on 409 degrees of freedom
##
## AIC = 3.760906  AICc = 3.761402  BIC = 3.828853
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6020 0.1707  3.5268  0.0005
## ma1     -0.1496 0.1800 -0.8314  0.4062
## ma2      0.2410 0.0840  2.8681  0.0043
## ma3     -0.0907 0.1029 -0.8809  0.3789
## sar1     0.4541 0.0566  8.0260  0.0000
## sma1    -0.9789 0.0695 -14.0833  0.0000
## xmean   -0.0025 0.0250 -0.1011  0.9196
##

```

```

## sigma^2 estimated as 2.026589 on 408 degrees of freedom
##
## AIC = 3.633886 AICc = 3.634549 BIC = 3.711539
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5980 0.1642   3.6413 0.0003
## ma1     -0.1405 0.1739  -0.8078 0.4197
## ma2      0.2552 0.0822   3.1046 0.0020
## ma3     -0.0788 0.1016  -0.7752 0.4387
## sar1      0.6381 0.0872   7.3204 0.0000
## sma1     -1.2192 0.1124 -10.8495 0.0000
## sma2      0.2192 0.1048   2.0905 0.0372
## xmean    -0.0021 0.0277  -0.0771 0.9386
##
## sigma^2 estimated as 1.983585 on 407 degrees of freedom
##
## AIC = 3.628943 AICc = 3.629797 BIC = 3.716303
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6027 0.1460   4.1289 0.0000
## ma1     -0.1400 0.1550  -0.9030 0.3671
## ma2      0.2666 0.0759   3.5118 0.0005
## ma3     -0.0660 0.0957  -0.6894 0.4910
## sar1     -0.4036 0.0493  -8.1866 0.0000
## sar2     -0.1832 0.0487  -3.7642 0.0002
## xmean     0.0043 0.1273   0.0335 0.9733
##
## sigma^2 estimated as 2.339121 on 408 degrees of freedom
##
## AIC = 3.732281 AICc = 3.732944 BIC = 3.809934
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6013 0.1642   3.6618 0.0003
## ma1     -0.1434 0.1741  -0.8236 0.4107
## ma2      0.2540 0.0823   3.0873 0.0022
## ma3     -0.0813 0.1016  -0.8006 0.4238
## sar1      0.4196 0.0502   8.3651 0.0000
## sar2      0.1012 0.0508   1.9911 0.0471
## sma1     -1.0000 0.0473 -21.1347 0.0000
## xmean    -0.0021 0.0272  -0.0767 0.9389
##
## sigma^2 estimated as 1.981484 on 407 degrees of freedom
##
## AIC = 3.629211 AICc = 3.630066 BIC = 3.716572
##

```



```

## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5986 0.1644  3.6418 0.0003
## ma1     -0.1410 0.1739 -0.8110 0.4178
## ma2      0.2551 0.0822  3.1033 0.0020
## ma3     -0.0791 0.1018 -0.7772 0.4375
## sar1      0.6272 0.6055  1.0358 0.3009
## sar2      0.0053 0.2919  0.0181 0.9856
## sma1     -1.2084 0.6024 -2.0061 0.0455
## sma2      0.2084 0.6010  0.3467 0.7290
## xmean    -0.0021 0.0277 -0.0763 0.9392
##
## sigma^2 estimated as 1.983424 on 406 degrees of freedom
##
## AIC = 3.633762  AICc = 3.634833  BIC = 3.730829
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6193 0.1787  3.4660 0.0006
## ma1     -0.1607 0.1850 -0.8687 0.3855
## ma2      0.2354 0.0948  2.4824 0.0135
## ma3     -0.0966 0.1017 -0.9497 0.3428
## ma4      0.0325 0.0512  0.6356 0.5254
## xmean     0.0210 0.2146  0.0979 0.9221
##
## sigma^2 estimated as 2.735586 on 409 degrees of freedom
##
## AIC = 3.87932  AICc = 3.879816  BIC = 3.947267
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4354 0.3579  1.2166 0.2245
## ma1      0.0235 0.3612  0.0652 0.9481
## ma2      0.3488 0.1692  2.0609 0.0399
## ma3      0.0249 0.1870  0.1332 0.8941
## ma4      0.0400 0.0990  0.4042 0.6863
## sma1     -0.5556 0.0571 -9.7266 0.0000
## xmean     0.0010 0.0857  0.0119 0.9905
##
## sigma^2 estimated as 2.251033 on 408 degrees of freedom
##
## AIC = 3.69819  AICc = 3.698853  BIC = 3.775844
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4234 0.4026  1.0518 0.2935

```

```

## ma1      0.0255 0.4046 0.0631 0.9497
## ma2      0.3280 0.1878 1.7466 0.0815
## ma3      0.0110 0.2015 0.0545 0.9565
## ma4      0.0356 0.0979 0.3641 0.7159
## sma1     -0.5348 0.0549 -9.7434 0.0000
## sma2     -0.2129 0.0677 -3.1465 0.0018
## xmean    -0.0019 0.0489 -0.0397 0.9683
##
## sigma^2 estimated as 2.167866 on 407 degrees of freedom
##
## AIC = 3.673485  AICc = 3.67434  BIC = 3.760846
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5882 0.2740  2.1468 0.0324
## ma1     -0.1202 0.2798 -0.4296 0.6677
## ma2      0.2818 0.1397  2.0172 0.0443
## ma3     -0.0574 0.1555 -0.3691 0.7122
## ma4      0.0166 0.0821  0.2020 0.8400
## sar1    -0.3381 0.0466 -7.2540 0.0000
## xmean    0.0109 0.1557  0.0701 0.9442
##
## sigma^2 estimated as 2.422404 on 408 degrees of freedom
##
## AIC = 3.765624  AICc = 3.766287  BIC = 3.843277
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5621 0.2991  1.8794 0.0609
## ma1     -0.1103 0.3022 -0.3651 0.7153
## ma2      0.2609 0.1463  1.7834 0.0753
## ma3     -0.0717 0.1561 -0.4591 0.6464
## ma4      0.0137 0.0773  0.1770 0.8596
## sar1     0.4549 0.0568  8.0059 0.0000
## sma1    -0.9792 0.0705 -13.8944 0.0000
## xmean   -0.0026 0.0249 -0.1041 0.9172
##
## sigma^2 estimated as 2.026126 on 407 degrees of freedom
##
## AIC = 3.638629  AICc = 3.639483  BIC = 3.725989
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5553 0.3165  1.7544 0.0801
## ma1     -0.0980 0.3203 -0.3061 0.7597
## ma2      0.2765 0.1558  1.7745 0.0767
## ma3     -0.0574 0.1683 -0.3414 0.7330
## ma4      0.0142 0.0863  0.1650 0.8690

```

```

## sar1    0.6387 0.0873   7.3186  0.0000
## sma1   -1.2194 0.1125 -10.8416  0.0000
## sma2    0.2194 0.1050   2.0890  0.0373
## xmean  -0.0022 0.0275  -0.0810  0.9354
##
## sigma^2 estimated as 1.983494 on 406 degrees of freedom
##
## AIC = 3.633697  AICc = 3.634768  BIC = 3.730764
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4299 0.3803   1.1303  0.2590
## ma1      0.0329 0.3826   0.0860  0.9315
## ma2      0.3517 0.1816   1.9363  0.0535
## ma3      0.0245 0.2040   0.1202  0.9044
## ma4      0.0558 0.0998   0.5598  0.5760
## sar1     -0.4033 0.0491  -8.2178  0.0000
## sar2     -0.1861 0.0489  -3.8059  0.0002
## xmean     0.0034 0.1224   0.0277  0.9780
##
## sigma^2 estimated as 2.337169 on 407 degrees of freedom
##
## AIC = 3.736332  AICc = 3.737186  BIC = 3.823692
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5743 0.3028   1.8964  0.0586
## ma1     -0.1167 0.3067  -0.3804  0.7038
## ma2      0.2675 0.1500   1.7826  0.0754
## ma3     -0.0679 0.1618  -0.4198  0.6748
## ma4      0.0094 0.0845   0.1113  0.9114
## sar1      0.4201 0.0503   8.3501  0.0000
## sar2      0.1010 0.0508   1.9865  0.0477
## sma1     -1.0000 0.0472 -21.1683  0.0000
## xmean    -0.0021 0.0270  -0.0775  0.9383
##
## sigma^2 estimated as 1.981445 on 406 degrees of freedom
##
## AIC = 3.633999  AICc = 3.63507  BIC = 3.731066
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5545 0.3212   1.7264  0.0850
## ma1     -0.0973 0.3245  -0.2998  0.7645
## ma2      0.2768 0.1574   1.7583  0.0794
## ma3     -0.0570 0.1706  -0.3342  0.7384
## ma4      0.0144 0.0874   0.1652  0.8689
## sar1      0.6507 0.5846   1.1130  0.2664

```

```

## sar2    -0.0058 0.2837 -0.0205  0.9836
## sma1    -1.2311 0.5803 -2.1215  0.0345
## sma2     0.2312 0.5789  0.3994  0.6898
## xmean   -0.0023 0.0275 -0.0818  0.9348
##
## sigma^2 estimated as 1.983766 on 405 degrees of freedom
##
## AIC = 3.638516  AICc = 3.639828  BIC = 3.74529
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4222 0.0475  8.8870 0.0000
## ar2      0.2433 0.0476  5.1063 0.0000
## xmean    0.0269 0.2439  0.1105 0.9121
##
## sigma^2 estimated as 2.79886 on 412 degrees of freedom
##
## AIC = 3.887559  AICc = 3.8877  BIC = 3.926386
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4259 0.0475  8.9724 0.0000
## ar2      0.2475 0.0477  5.1854 0.0000
## sma1    -0.5437 0.0579 -9.3888 0.0000
## xmean    0.0065 0.1069  0.0608 0.9515
##
## sigma^2 estimated as 2.324073 on 411 degrees of freedom
##
## AIC = 3.714816  AICc = 3.715051  BIC = 3.763349
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4188 0.0477  8.7788 0.0000
## ar2      0.2325 0.0482  4.8194 0.0000
## sma1    -0.5232 0.0537 -9.7509 0.0000
## sma2    -0.2175 0.0656 -3.3133 0.0010
## xmean    0.0008 0.0598  0.0141 0.9888
##
## sigma^2 estimated as 2.231629 on 410 degrees of freedom
##
## AIC = 3.687132  AICc = 3.687486  BIC = 3.745373
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4317 0.0474  9.1072 0.0000
## ar2      0.2530 0.0476  5.3169 0.0000

```

```

## sar1    -0.3276 0.0463 -7.0739  0.0000
## xmean    0.0192 0.1850  0.1037  0.9175
##
## sigma^2 estimated as 2.492703 on 411 degrees of freedom
##
## AIC = 3.779271  AICc = 3.779506  BIC = 3.827804
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4203 0.0477   8.8148 0.0000
## ar2      0.2311 0.0480   4.8178 0.0000
## sar1      0.4651 0.0560   8.2991 0.0000
## sma1     -0.9832 0.0846 -11.6252 0.0000
## xmean    -0.0007 0.0289  -0.0230 0.9817
##
## sigma^2 estimated as 2.076852 on 410 degrees of freedom
##
## AIC = 3.650915  AICc = 3.651269  BIC = 3.709155
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4245 0.0476   8.9128 0.0000
## ar2      0.2375 0.0480   4.9511 0.0000
## sar1      0.6305 0.0866   7.2837 0.0000
## sma1     -1.1998 0.1112 -10.7940 0.0000
## sma2      0.1998 0.1035   1.9308 0.0542
## xmean     0.0002 0.0322   0.0049 0.9961
##
## sigma^2 estimated as 2.040533 on 409 degrees of freedom
##
## AIC = 3.647231  AICc = 3.647727  BIC = 3.715177
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4263 0.0475   8.9796 0.0000
## ar2      0.2496 0.0477   5.2375 0.0000
## sar1     -0.3894 0.0487  -8.0031 0.0000
## sar2     -0.1810 0.0487  -3.7180 0.0002
## xmean     0.0105 0.1504   0.0697 0.9444
##
## sigma^2 estimated as 2.408828 on 410 degrees of freedom
##
## AIC = 3.751444  AICc = 3.751798  BIC = 3.809684
##
## Warning in log(s2): NaNs produced
## <><><><><><><><><><><><><><>
##

```

```

## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4249 0.0477  8.9159 0.0000
## ar2      0.2372 0.0480  4.9453 0.0000
## sar1      0.4314 0.0498  8.6686 0.0000
## sar2      0.0936 0.0504  1.8576 0.0639
## sma1     -1.0000 0.0462 -21.6392 0.0000
## xmean      0.0002 0.0317  0.0054 0.9957
##
## sigma^2 estimated as 2.038644 on 409 degrees of freedom
##
## AIC = 3.647444 AICc = 3.64794 BIC = 3.715391
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4243 0.0477  8.8931 0.0000
## ar2      0.2375 0.0480  4.9504 0.0000
## sar1      0.6747 0.7047  0.9574 0.3389
## sar2     -0.0217 0.3465 -0.0627 0.9500
## sma1     -1.2435 0.6991 -1.7786 0.0760
## sma2      0.2435 0.6981  0.3488 0.7274
## xmean      0.0001 0.0324  0.0042 0.9967
##
## sigma^2 estimated as 2.040967 on 408 degrees of freedom
##
## AIC = 3.65204 AICc = 3.652703 BIC = 3.729693
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0332 0.1114  0.2977 0.7661
## ar2      0.4680 0.0645  7.2543 0.0000
## ma1      0.4148 0.1196  3.4688 0.0006
## xmean      0.0242 0.2295  0.1054 0.9161
##
## sigma^2 estimated as 2.746156 on 411 degrees of freedom
##
## AIC = 3.873498 AICc = 3.873734 BIC = 3.922032
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0601 0.1236  0.4863 0.6270
## ar2      0.4618 0.0707  6.5338 0.0000
## ma1      0.3897 0.1342  2.9040 0.0039
## sma1     -0.5371 0.0573 -9.3750 0.0000
## xmean      0.0049 0.1022  0.0476 0.9621
##
## sigma^2 estimated as 2.283395 on 410 degrees of freedom
##

```

```

## AIC = 3.701888 AICc = 3.702242 BIC = 3.760128
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0659 0.1251  0.5268  0.5986
## ar2      0.4351 0.0715  6.0850  0.0000
## ma1      0.3726 0.1335  2.7900  0.0055
## sma1     -0.5132 0.0526 -9.7495  0.0000
## sma2     -0.2062 0.0636 -3.2401  0.0013
## xmean     0.0005 0.0610  0.0077  0.9939
##
## sigma^2 estimated as 2.200262 on 409 degrees of freedom
##
## AIC = 3.676464 AICc = 3.67696 BIC = 3.744411
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0568 0.1186  0.4786  0.6325
## ar2      0.4763 0.0691  6.8950  0.0000
## ma1      0.4015 0.1295  3.1011  0.0021
## sar1     -0.3273 0.0463 -7.0707  0.0000
## xmean     0.0159 0.1737  0.0918  0.9269
##
## sigma^2 estimated as 2.445969 on 410 degrees of freedom
##
## AIC = 3.765306 AICc = 3.765659 BIC = 3.823546
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0668 0.1233  0.5420  0.5881
## ar2      0.4325 0.0699  6.1838  0.0000
## ma1      0.3722 0.1308  2.8443  0.0047
## sar1      0.4723 0.0565  8.3575  0.0000
## sma1     -0.9893 0.1329 -7.4445  0.0000
## xmean    -0.0013 0.0273 -0.0476  0.9620
##
## sigma^2 estimated as 2.035716 on 409 degrees of freedom
##
## AIC = 3.640383 AICc = 3.640879 BIC = 3.70833
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0687 0.1225  0.5608  0.5752
## ar2      0.4440 0.0705  6.3020  0.0000
## ma1      0.3764 0.1311  2.8719  0.0043
## sar1      0.6416 0.0851  7.5403  0.0000

```

```

## sma1    -1.2080 0.1092 -11.0593  0.0000
## sma2     0.2080 0.1025  2.0296  0.0430
## xmean    -0.0006 0.0311  -0.0198  0.9842
##
## sigma^2 estimated as 2.008817 on 408 degrees of freedom
##
## AIC = 3.635899  AICc = 3.636562  BIC = 3.713553
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0473 0.1186  0.3988  0.6902
## ar2      0.4722 0.0681  6.9298  0.0000
## ma1      0.4049 0.1290  3.1393  0.0018
## sar1     -0.3890 0.0486 -8.0068  0.0000
## sar2     -0.1818 0.0486 -3.7442  0.0002
## xmean     0.0082 0.1413  0.0577  0.9540
##
## sigma^2 estimated as 2.362565 on 409 degrees of freedom
##
## AIC = 3.737027  AICc = 3.737523  BIC = 3.804974
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0714 0.1227  0.5814  0.5613
## ar2      0.4425 0.0706  6.2664  0.0000
## ma1      0.3738 0.1311  2.8515  0.0046
## sar1      0.4352 0.0498  8.7414  0.0000
## sar2      0.0968 0.0504  1.9213  0.0554
## sma1     -1.0000 0.0434 -23.0588  0.0000
## xmean    -0.0006 0.0304  -0.0203  0.9838
##
## sigma^2 estimated as 2.007022 on 408 degrees of freedom
##
## AIC = 3.636313  AICc = 3.636977  BIC = 3.713967
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0682 0.1229  0.5548  0.5793
## ar2      0.4439 0.0705  6.3007  0.0000
## ma1      0.3765 0.1313  2.8682  0.0043
## sar1      0.7380 0.5282  1.3973  0.1631
## sar2     -0.0485 0.2675 -0.1812  0.8563
## sma1     -1.3030 0.5212 -2.4998  0.0128
## sma2      0.3030 0.5201  0.5826  0.5605
## xmean    -0.0007 0.0314  -0.0212  0.9831
##
## sigma^2 estimated as 2.009906 on 407 degrees of freedom
##

```



```

## AIC = 3.640655  AICc = 3.64151  BIC = 3.728015
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1759 0.2074  0.8481  0.3969
## ar2      0.2846 0.1858  1.5320  0.1263
## ma1      0.2828 0.2080  1.3594  0.1748
## ma2      0.1532 0.1199  1.2769  0.2024
## xmean    0.0217 0.2151  0.1009  0.9197
##
## sigma^2 estimated as 2.734079 on 410 degrees of freedom
##
## AIC = 3.873949  AICc = 3.874302  BIC = 3.932189
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4203 0.2025  2.0752  0.0386
## ar2      0.0752 0.1726  0.4360  0.6630
## ma1      0.0376 0.1983  0.1897  0.8496
## ma2      0.2789 0.0972  2.8695  0.0043
## sma1     -0.5555 0.0576 -9.6442  0.0000
## xmean    0.0014 0.0878  0.0158  0.9874
##
## sigma^2 estimated as 2.251686 on 409 degrees of freedom
##
## AIC = 3.693635  AICc = 3.694131  BIC = 3.761582
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3669 0.2070  1.7726  0.0770
## ar2      0.0907 0.1711  0.5303  0.5962
## ma1      0.0815 0.2033  0.4008  0.6888
## ma2      0.2608 0.0976  2.6718  0.0078
## sma1     -0.5357 0.0553 -9.6795  0.0000
## sma2     -0.2138 0.0679 -3.1468  0.0018
## xmean    -0.0018 0.0495 -0.0361  0.9712
##
## sigma^2 estimated as 2.167976 on 408 degrees of freedom
##
## AIC = 3.668842  AICc = 3.669505  BIC = 3.746496
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3538 0.1897  1.8653  0.0629
## ar2      0.1516 0.1637  0.9262  0.3549
## ma1      0.1136 0.1863  0.6094  0.5426

```

```

## ma2      0.2404 0.0975  2.4663  0.0141
## sar1     -0.3377 0.0467 -7.2393  0.0000
## xmean    0.0116 0.1567  0.0739  0.9411
##
## sigma^2 estimated as 2.422078 on 409 degrees of freedom
##
## AIC = 3.76066  AICc = 3.761156  BIC = 3.828607
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3053 0.1815   1.6820  0.0933
## ar2      0.1355 0.1514   0.8953  0.3712
## ma1      0.1468 0.1783   0.8229  0.4110
## ma2      0.2417 0.0947   2.5520  0.0111
## sar1      0.4546 0.0569   7.9932  0.0000
## sma1     -0.9783 0.0679 -14.4110  0.0000
## xmean    -0.0026 0.0248  -0.1060  0.9156
##
## sigma^2 estimated as 2.027477 on 408 degrees of freedom
##
## AIC = 3.633869  AICc = 3.634532  BIC = 3.711522
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3375 0.1854   1.8201  0.0695
## ar2      0.1264 0.1558   0.8116  0.4175
## ma1      0.1199 0.1817   0.6598  0.5098
## ma2      0.2498 0.0945   2.6438  0.0085
## sar1      0.6393 0.0869   7.3536  0.0000
## sma1     -1.2199 0.1121 -10.8791  0.0000
## sma2      0.2199 0.1046   2.1012  0.0362
## xmean    -0.0022 0.0276  -0.0804  0.9360
##
## sigma^2 estimated as 1.983565 on 407 degrees of freedom
##
## AIC = 3.628863  AICc = 3.629718  BIC = 3.716223
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3473 0.2011   1.7264  0.0850
## ar2      0.1449 0.1702   0.8517  0.3949
## ma1      0.1141 0.1982   0.5756  0.5652
## ma2      0.2428 0.0982   2.4720  0.0138
## sar1     -0.4020 0.0494  -8.1404  0.0000
## sar2     -0.1836 0.0486  -3.7763  0.0002
## xmean     0.0044 0.1275   0.0343  0.9726
##
## sigma^2 estimated as 2.338078 on 408 degrees of freedom

```

```

##
## AIC = 3.731824 AICc = 3.732487 BIC = 3.809477
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3371 0.1831  1.8412 0.0663
## ar2      0.1262 0.1540  0.8199 0.4128
## ma1      0.1208 0.1792  0.6741 0.5006
## ma2      0.2503 0.0942  2.6564 0.0082
## sar1     0.4203 0.0503  8.3557 0.0000
## sar2     0.1012 0.0508  1.9927 0.0470
## sma1    -1.0000 0.0474 -21.1047 0.0000
## xmean   -0.0022 0.0269  -0.0807 0.9357
##
## sigma^2 estimated as 1.981533 on 407 degrees of freedom
##
## AIC = 3.629188 AICc = 3.630042 BIC = 3.716548
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3372 0.1851  1.8214 0.0693
## ar2      0.1265 0.1555  0.8133 0.4165
## ma1      0.1201 0.1814  0.6619 0.5084
## ma2      0.2499 0.0945  2.6440 0.0085
## sar1     0.6539 0.5626  1.1622 0.2458
## sar2    -0.0071 0.2739 -0.0260 0.9793
## sma1    -1.2342 0.5582 -2.2110 0.0276
## sma2     0.2342 0.5568  0.4206 0.6743
## xmean   -0.0022 0.0276 -0.0788 0.9373
##
## sigma^2 estimated as 1.983736 on 406 degrees of freedom
##
## AIC = 3.633681 AICc = 3.634752 BIC = 3.730748
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0085 0.2967  0.0287 0.9771
## ar2      0.3564 0.1517  2.3497 0.0193
## ma1      0.4515 0.3004  1.5029 0.1336
## ma2      0.1602 0.1144  1.4001 0.1622
## ma3      0.0552 0.1084  0.5094 0.6107
## xmean    0.0206 0.2122  0.0973 0.9226
##
## sigma^2 estimated as 2.73288 on 409 degrees of freedom
##
## AIC = 3.878335 AICc = 3.878831 BIC = 3.946282
##
## <><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.2970 0.2029 -1.4637 0.1441
## ar2       0.4471 0.1083  4.1276 0.0000
## ma1       0.7554 0.2047  3.6895 0.0003
## ma2       0.2398 0.1144  2.0959 0.0367
## ma3       0.2154 0.0924  2.3307 0.0203
## sma1      -0.5578 0.0571 -9.7738 0.0000
## xmean      0.0013 0.0870  0.0150 0.9880
##
## sigma^2 estimated as 2.246998 on 408 degrees of freedom
##
## AIC = 3.696309  AICc = 3.696972  BIC = 3.773962
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.2226 0.5243 -0.4246 0.6714
## ar2       0.3643 0.2365  1.5400 0.1243
## ma1       0.6720 0.5264  1.2765 0.2025
## ma2       0.2542 0.1189  2.1379 0.0331
## ma3       0.1821 0.1825  0.9977 0.3190
## sma1      -0.5337 0.0550 -9.6964 0.0000
## sma2      -0.2069 0.0678 -3.0516 0.0024
## xmean     -0.0017 0.0506 -0.0338 0.9731
##
## sigma^2 estimated as 2.168682 on 407 degrees of freedom
##
## AIC = 3.673286  AICc = 3.674141  BIC = 3.760646
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.1621 0.2597 -0.6245 0.5327
## ar2       0.4188 0.1350  3.1029 0.0020
## ma1       0.6314 0.2629  2.4020 0.0168
## ma2       0.2172 0.1110  1.9564 0.0511
## ma3       0.1604 0.1071  1.4977 0.1350
## sar1      -0.3398 0.0468 -7.2608 0.0000
## xmean      0.0094 0.1543  0.0611 0.9513
##
## sigma^2 estimated as 2.41869 on 408 degrees of freedom
##
## AIC = 3.76409  AICc = 3.764753  BIC = 3.841744
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.4562 0.9984  0.4569 0.6480
## ar2       0.0701 0.4594  0.1525 0.8789

```

```

## ma1      -0.0043 0.9984 -0.0043 0.9966
## ma2       0.2384 0.0915  2.6043 0.0095
## ma3      -0.0486 0.3119 -0.1559 0.8762
## sar1      0.4548 0.0568  8.0067 0.0000
## sma1     -0.9791 0.0701 -13.9629 0.0000
## xmean    -0.0026 0.0249 -0.1028 0.9182
##
## sigma^2 estimated as 2.026305 on 407 degrees of freedom
##
## AIC = 3.638643  AICc = 3.639498  BIC = 3.726003
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3484 1.7796  0.1958 0.8449
## ar2      0.1214 0.8416  0.1443 0.8853
## ma1      0.1089 1.7817  0.0611 0.9513
## ma2      0.2498 0.0947  2.6394 0.0086
## ma3     -0.0035 0.5607 -0.0062 0.9951
## sar1      0.6392 0.0880  7.2608 0.0000
## sma1     -1.2197 0.1129 -10.8053 0.0000
## sma2      0.2197 0.1055  2.0834 0.0378
## xmean    -0.0022 0.0276 -0.0800 0.9363
##
## sigma^2 estimated as 1.983519 on 406 degrees of freedom
##
## AIC = 3.633683  AICc = 3.634754  BIC = 3.73075
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1     -0.1968 0.2637 -0.7464 0.4559
## ar2      0.4199 0.1338  3.1389 0.0018
## ma1      0.6595 0.2666  2.4736 0.0138
## ma2      0.2225 0.1101  2.0208 0.0440
## ma3      0.1700 0.1090  1.5601 0.1195
## sar1     -0.4054 0.0498 -8.1432 0.0000
## sar2     -0.1844 0.0486 -3.7943 0.0002
## xmean     0.0041 0.1256  0.0325 0.9741
##
## sigma^2 estimated as 2.334042 on 407 degrees of freedom
##
## AIC = 3.734934  AICc = 3.735789  BIC = 3.822294
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1     -0.5075 0.0977 -5.1922 0.0000
## ar2      0.4566 0.0969  4.7143 0.0000
## ma1      0.9777 0.1000  9.7733 0.0000
## ma2      0.3001 0.1404  2.1371 0.0332

```

```

## ma3      0.3081 0.0596   5.1704  0.0000
## sar1      0.4222 0.0508   8.3153  0.0000
## sar2      0.1054 0.0510   2.0653  0.0395
## sma1     -1.0000 0.0564 -17.7395  0.0000
## xmean    -0.0024 0.0262  -0.0922  0.9266
##
## sigma^2 estimated as 1.975755 on 406 degrees of freedom
##
## AIC = 3.633508  AICc = 3.634579  BIC = 3.730575
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.1522 0.5929 -0.2567  0.7975
## ar2       0.3518 0.2684  1.3107  0.1907
## ma1       0.6107 0.5962  1.0242  0.3063
## ma2       0.2491 0.1179  2.1135  0.0352
## ma3       0.1570 0.2059  0.7625  0.4462
## sar1       0.6894 0.5182  1.3304  0.1841
## sar2      -0.0214 0.2552 -0.0840  0.9331
## sma1      -1.2699 0.5128 -2.4766  0.0137
## sma2       0.2699 0.5114  0.5278  0.5979
## xmean     -0.0023 0.0275 -0.0849  0.9324
##
## sigma^2 estimated as 1.984572 on 405 degrees of freedom
##
## AIC = 3.638408  AICc = 3.63972  BIC = 3.745181
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.0123 0.3122  0.0394  0.9686
## ar2       0.3603 0.2059  1.7495  0.0810
## ma1       0.4476 0.3172  1.4112  0.1590
## ma2       0.1545 0.2149  0.7193  0.4724
## ma3       0.0513 0.1625  0.3155  0.7525
## ma4      -0.0031 0.1069 -0.0291  0.9768
## xmean     0.0209 0.2126  0.0983  0.9217
##
## sigma^2 estimated as 2.732875 on 408 degrees of freedom
##
## AIC = 3.883152  AICc = 3.883815  BIC = 3.960806
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.3338 0.2388 -1.3979  0.1629
## ar2       0.3945 0.1867  2.1126  0.0352
## ma1       0.7944 0.2429  3.2712  0.0012
## ma2       0.3086 0.2345  1.3163  0.1888
## ma3       0.2618 0.1662  1.5752  0.1160

```

```

## ma4      0.0359 0.1026 0.3502 0.7264
## sma1     -0.5593 0.0573 -9.7542 0.0000
## xmean    0.0011 0.0852 0.0125 0.9900
##
## sigma^2 estimated as 2.246234 on 407 degrees of freedom
##
## AIC = 3.700841 AICc = 3.701695 BIC = 3.788201
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2412    NaN      NaN      NaN
## ar2      0.1051    NaN      NaN      NaN
## ma1      0.2077    NaN      NaN      NaN
## ma2      0.3048    NaN      NaN      NaN
## ma3      0.0582    NaN      NaN      NaN
## ma4      0.0245    NaN      NaN      NaN
## sma1     -0.5342 0.0308 -17.3196 0.0000
## sma2     -0.2118    NaN      NaN      NaN
## xmean    -0.0019 0.0491  -0.0380 0.9697
##
## sigma^2 estimated as 2.168113 on 406 degrees of freedom
##
## AIC = 3.678289 AICc = 3.67936 BIC = 3.775356
##
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, :
## possible convergence problem: optim gave code = 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.1387 0.3039 -0.4565 0.6483
## ar2      0.4057 0.2161  1.8777 0.0611
## ma1      0.6084 0.3093  1.9670 0.0499
## ma2      0.2198 0.2257  0.9735 0.3309
## ma3      0.1543 0.1732  0.8905 0.3737
## ma4      0.0028 0.1109  0.0254 0.9798
## sar1     -0.3393 0.0468 -7.2459 0.0000
## xmean    0.0098 0.1547  0.0632 0.9496
##
## sigma^2 estimated as 2.418725 on 407 degrees of freedom
##
## AIC = 3.768925 AICc = 3.76978 BIC = 3.856286
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7053 3.2093  0.2198 0.8262

```

```

## ar2      -0.0882 1.9305  -0.0457  0.9636
## ma1      -0.2535 3.2103  -0.0790  0.9371
## ma2       0.2845 0.5040   0.5645  0.5727
## ma3      -0.1056 0.7875  -0.1341  0.8934
## ma4       0.0279 0.3049   0.0916  0.9271
## sar1      0.4550 0.0569   8.0025  0.0000
## sma1     -0.9794 0.0713 -13.7326  0.0000
## xmean    -0.0026 0.0248  -0.1047  0.9166
##
## sigma^2 estimated as 2.025804 on 406 degrees of freedom
##
## AIC = 3.643437  AICc = 3.644508  BIC = 3.740504
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.7265 3.6808   0.1974  0.8436
## ar2     -0.0978 2.0980  -0.0466  0.9629
## ma1     -0.2692 3.6808  -0.0731  0.9417
## ma2      0.2957 0.4381   0.6749  0.5001
## ma3     -0.1032 0.9911  -0.1041  0.9171
## ma4      0.0246 0.2365   0.1042  0.9170
## sar1      0.6384 0.0875   7.2941  0.0000
## sma1     -1.2191 0.1126 -10.8249  0.0000
## sma2      0.2191 0.1052   2.0830  0.0379
## xmean    -0.0022 0.0275  -0.0809  0.9355
##
## sigma^2 estimated as 1.983483 on 405 degrees of freedom
##
## AIC = 3.638532  AICc = 3.639845  BIC = 3.745306
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1     -0.2120 0.2792  -0.7591  0.4482
## ar2      0.3815 0.2122   1.7979  0.0729
## ma1      0.6758 0.2830   2.3882  0.0174
## ma2      0.2679 0.2249   1.1913  0.2342
## ma3      0.1983 0.1649   1.2026  0.2298
## ma4      0.0256 0.1080   0.2366  0.8131
## sar1     -0.4058 0.0499  -8.1275  0.0000
## sar2     -0.1848 0.0486  -3.8024  0.0002
## xmean     0.0037 0.1241   0.0302  0.9759
##
## sigma^2 estimated as 2.333664 on 406 degrees of freedom
##
## AIC = 3.739615  AICc = 3.740686  BIC = 3.836682
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value

```



```

## ar1      -0.3498 0.1650 -2.1199 0.0346
## ar2       0.6034 0.1546  3.9020 0.0001
## ma1       0.8150 0.1745  4.6710 0.0000
## ma2       0.0837 0.2341  0.3574 0.7210
## ma3       0.1501 0.1649  0.9104 0.3632
## ma4      -0.1003 0.1016 -0.9876 0.3239
## sar1      0.4282 0.0511  8.3813 0.0000
## sar2      0.1031 0.0510  2.0204 0.0440
## sma1     -1.0000 0.0526 -19.0175 0.0000
## xmean    -0.0017 0.0279 -0.0626 0.9501
##
## sigma^2 estimated as 1.971217 on 405 degrees of freedom
##
## AIC = 3.636032  AICc = 3.637344  BIC = 3.742805
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.1584 0.4927 -0.3215 0.7480
## ar2       0.4092 0.3256  1.2568 0.2095
## ma1       0.6158 0.4955  1.2428 0.2146
## ma2       0.1946 0.2470  0.7880 0.4311
## ma3       0.1325 0.1992  0.6651 0.5064
## ma4      -0.0288 0.1174 -0.2458 0.8060
## sar1      0.6681 0.5539  1.2061 0.2285
## sar2     -0.0122 0.2700 -0.0451 0.9640
## sma1     -1.2488 0.5491 -2.2741 0.0235
## sma2      0.2488 0.5478  0.4543 0.6499
## xmean    -0.0022 0.0277 -0.0796 0.9366
##
## sigma^2 estimated as 1.983797 on 404 degrees of freedom
##
## AIC = 3.643074  AICc = 3.644652  BIC = 3.759554
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.4591 0.0484  9.4868 0.0000
## ar2       0.3077 0.0514  5.9905 0.0000
## ar3      -0.1521 0.0486 -3.1315 0.0019
## xmean     0.0205 0.2099  0.0976 0.9223
##
## sigma^2 estimated as 2.733793 on 411 degrees of freedom
##
## AIC = 3.869026  AICc = 3.869261  BIC = 3.917559
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       0.4663 0.0483  9.6552 0.0000
## ar2       0.3164 0.0513  6.1698 0.0000

```

```

## ar3      -0.1645 0.0486 -3.3846  0.0008
## sma1     -0.5435 0.0569 -9.5477  0.0000
## xmean    0.0020 0.0906  0.0221  0.9823
##
## sigma^2 estimated as 2.260858 on 410 degrees of freedom
##
## AIC = 3.692394  AICc = 3.692748  BIC = 3.750634
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4554 0.0485  9.3904  0.0000
## ar2      0.2980 0.0518  5.7495  0.0000
## ar3     -0.1566 0.0487 -3.2169  0.0014
## sma1    -0.5207 0.0530 -9.8229  0.0000
## sma2    -0.2086 0.0652 -3.1998  0.0015
## xmean   -0.0013 0.0531 -0.0242  0.9807
##
## sigma^2 estimated as 2.178412 on 409 degrees of freedom
##
## AIC = 3.667315  AICc = 3.667811  BIC = 3.735262
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4727 0.0483  9.7768  0.0000
## ar2      0.3224 0.0514  6.2775  0.0000
## ar3     -0.1619 0.0486 -3.3331  0.0009
## sar1    -0.3318 0.0462 -7.1771  0.0000
## xmean    0.0115 0.1569  0.0735  0.9414
##
## sigma^2 estimated as 2.426816 on 410 degrees of freedom
##
## AIC = 3.757665  AICc = 3.758019  BIC = 3.815906
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4555 0.0484  9.4097  0.0000
## ar2      0.2946 0.0515  5.7211  0.0000
## ar3     -0.1534 0.0487 -3.1526  0.0017
## sar1     0.4646 0.0563  8.2509  0.0000
## sma1    -0.9834 0.0868 -11.3317  0.0000
## xmean   -0.0025 0.0248  -0.1012  0.9194
##
## sigma^2 estimated as 2.027258 on 409 degrees of freedom
##
## AIC = 3.63206  AICc = 3.632556  BIC = 3.700007
##
## <><><><><><><><><><><>
##

```

```

## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4619 0.0484   9.5369 0.0000
## ar2      0.3037 0.0516   5.8837 0.0000
## ar3     -0.1573 0.0487  -3.2302 0.0013
## sar1      0.6425 0.0859   7.4828 0.0000
## sma1     -1.2152 0.1103 -11.0176 0.0000
## sma2      0.2152 0.1034   2.0802 0.0381
## xmean    -0.0021 0.0279  -0.0738 0.9412
##
## sigma^2 estimated as 1.990742 on 408 degrees of freedom
##
## AIC = 3.627209  AICc = 3.627872  BIC = 3.704862
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4678 0.0484   9.6752 0.0000
## ar2      0.3198 0.0513   6.2373 0.0000
## ar3     -0.1662 0.0486  -3.4223 0.0007
## sar1     -0.3953 0.0486  -8.1342 0.0000
## sar2     -0.1846 0.0486  -3.7995 0.0002
## xmean     0.0045 0.1266   0.0359 0.9714
##
## sigma^2 estimated as 2.341645 on 409 degrees of freedom
##
## AIC = 3.728425  AICc = 3.728921  BIC = 3.796372
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4620 0.0485   9.5300 0.0000
## ar2      0.3029 0.0516   5.8680 0.0000
## ar3     -0.1563 0.0487  -3.2117 0.0014
## sar1      0.4292 0.0497   8.6315 0.0000
## sar2      0.0986 0.0505   1.9544 0.0513
## sma1     -1.0000 0.0446 -22.4045 0.0000
## xmean    -0.0020 0.0273  -0.0739 0.9411
##
## sigma^2 estimated as 1.988992 on 408 degrees of freedom
##
## AIC = 3.627704  AICc = 3.628367  BIC = 3.705358
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4615 0.0485   9.5186 0.0000
## ar2      0.3036 0.0516   5.8835 0.0000
## ar3     -0.1575 0.0487  -3.2340 0.0013
## sar1      0.7289 0.5008   1.4555 0.1463
## sar2     -0.0432 0.2515  -0.1716 0.8638

```

```

## sma1    -1.3002  0.4945 -2.6295  0.0089
## sma2     0.3002  0.4932  0.6087  0.5431
## xmean   -0.0021  0.0281 -0.0731  0.9418
##
## sigma^2 estimated as 1.991712 on 407 degrees of freedom
##
## AIC = 3.631967  AICc = 3.632821  BIC = 3.719327
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4189 0.3015  1.3892  0.1655
## ar2      0.3247 0.1344  2.4154  0.0162
## ar3     -0.1423 0.0888 -1.6024  0.1098
## ma1      0.0412 0.3044  0.1354  0.8924
## xmean    0.0198 0.2112  0.0939  0.9252
##
## sigma^2 estimated as 2.73368 on 410 degrees of freedom
##
## AIC = 3.873805  AICc = 3.874158  BIC = 3.932045
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6277 0.2134  2.9416  0.0035
## ar2      0.2484 0.1073  2.3138  0.0212
## ar3     -0.2058 0.0666 -3.0900  0.0021
## ma1     -0.1657 0.2157 -0.7684  0.4427
## sma1    -0.5470 0.0570 -9.6037  0.0000
## xmean    0.0014 0.0868  0.0156  0.9876
##
## sigma^2 estimated as 2.257565 on 409 degrees of freedom
##
## AIC = 3.695941  AICc = 3.696437  BIC = 3.763888
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6170 0.2303  2.6783  0.0077
## ar2      0.2306 0.1129  2.0430  0.0417
## ar3     -0.1955 0.0673 -2.9060  0.0039
## ma1     -0.1657 0.2330 -0.7112  0.4774
## sma1    -0.5249 0.0535 -9.8050  0.0000
## sma2    -0.2095 0.0658 -3.1817  0.0016
## xmean   -0.0017 0.0505 -0.0327  0.9739
##
## sigma^2 estimated as 2.175099 on 408 degrees of freedom
##
## AIC = 3.671013  AICc = 3.671676  BIC = 3.748666
##
## <><><><><><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5823 0.2417  2.4091 0.0164
## ar2      0.2754 0.1178  2.3374 0.0199
## ar3     -0.1901 0.0739 -2.5710 0.0105
## ma1     -0.1124 0.2445 -0.4598 0.6459
## sar1    -0.3333 0.0463 -7.1988 0.0000
## xmean    0.0108 0.1535  0.0704 0.9439
##
## sigma^2 estimated as 2.425528 on 409 degrees of freedom
##
## AIC = 3.762003  AICc = 3.762499  BIC = 3.82995
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5760 0.2497  2.3072 0.0215
## ar2      0.2442 0.1188  2.0553 0.0405
## ar3     -0.1819 0.0712 -2.5538 0.0110
## ma1     -0.1234 0.2524 -0.4891 0.6250
## sar1     0.4612 0.0567  8.1300 0.0000
## sma1    -0.9803 0.0745 -13.1561 0.0000
## xmean   -0.0027 0.0244 -0.1112 0.9115
##
## sigma^2 estimated as 2.03033 on 408 degrees of freedom
##
## AIC = 3.636323  AICc = 3.636986  BIC = 3.713977
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5935 0.2369  2.5050 0.0126
## ar2      0.2483 0.1152  2.1549 0.0318
## ar3     -0.1894 0.0700 -2.7051 0.0071
## ma1     -0.1348 0.2394 -0.5632 0.5736
## sar1     0.6433 0.0862  7.4655 0.0000
## sma1    -1.2184 0.1109 -10.9838 0.0000
## sma2     0.2184 0.1039  2.1013 0.0362
## xmean   -0.0023 0.0272 -0.0850 0.9323
##
## sigma^2 estimated as 1.989189 on 407 degrees of freedom
##
## AIC = 3.631308  AICc = 3.632163  BIC = 3.718668
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5965 0.2327  2.5634 0.0107
## ar2      0.2654 0.1137  2.3338 0.0201
## ar3     -0.1990 0.0711 -2.7980 0.0054

```

```

## ma1      -0.1322 0.2357 -0.5611  0.5750
## sar1      -0.3977 0.0487 -8.1578  0.0000
## sar2      -0.1853 0.0486 -3.8120  0.0002
## xmean     0.0039 0.1232  0.0315  0.9749
##
## sigma^2 estimated as 2.33982 on 408 degrees of freedom
##
## AIC = 3.732541  AICc = 3.733204  BIC = 3.810195
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5893 0.2383  2.4736  0.0138
## ar2      0.2493 0.1156  2.1557  0.0317
## ar3     -0.1874 0.0704 -2.6641  0.0080
## ma1     -0.1304 0.2406 -0.5420  0.5881
## sar1      0.4269 0.0499  8.5609  0.0000
## sar2      0.0995 0.0505  1.9708  0.0494
## sma1     -1.0000 0.0456 -21.9255  0.0000
## xmean    -0.0023 0.0266 -0.0852  0.9322
##
## sigma^2 estimated as 1.987359 on 407 degrees of freedom
##
## AIC = 3.631828  AICc = 3.632683  BIC = 3.719189
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5937 0.2372  2.5034  0.0127
## ar2      0.2480 0.1152  2.1518  0.0320
## ar3     -0.1897 0.0700 -2.7098  0.0070
## ma1     -0.1353 0.2397 -0.5647  0.5726
## sar1      0.7242 0.4984  1.4530  0.1470
## sar2     -0.0404 0.2492 -0.1620  0.8714
## sma1     -1.2980 0.4924 -2.6360  0.0087
## sma2      0.2980 0.4911  0.6068  0.5444
## xmean    -0.0023 0.0274 -0.0849  0.9324
##
## sigma^2 estimated as 1.99005 on 406 degrees of freedom
##
## AIC = 3.636066  AICc = 3.637137  BIC = 3.733133
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3273 0.5131  0.6378  0.5240
## ar2      0.3014 0.1781  1.6929  0.0912
## ar3     -0.0857 0.2727 -0.3141  0.7536
## ma1      0.1325 0.5124  0.2586  0.7961
## ma2      0.0675 0.3049  0.2215  0.8248
## xmean     0.0209 0.2123  0.0985  0.9215

```

```

##
## sigma^2 estimated as 2.733393 on 409 degrees of freedom
##
## AIC = 3.87852  AICc = 3.879016  BIC = 3.946467
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5092 0.2877  1.7702  0.0774
## ar2      0.0988 0.1799  0.5493  0.5831
## ar3     -0.0633 0.1619 -0.3907  0.6962
## ma1     -0.0508 0.2855 -0.1781  0.8588
## ma2      0.2170 0.1937  1.1204  0.2632
## sma1    -0.5548 0.0574 -9.6722  0.0000
## xmean    0.0011 0.0864  0.0127  0.9899
##
## sigma^2 estimated as 2.250921 on 408 degrees of freedom
##
## AIC = 3.698105  AICc = 3.698768  BIC = 3.775758
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4490 0.3484  1.2885  0.1983
## ar2      0.1015 0.1753  0.5790  0.5629
## ar3     -0.0481 0.1759 -0.2732  0.7848
## ma1     -0.0001 0.3463 -0.0004  0.9997
## ma2      0.2150 0.2016  1.0666  0.2868
## sma1    -0.5345 0.0552 -9.6899  0.0000
## sma2    -0.2122 0.0676 -3.1406  0.0018
## xmean   -0.0019 0.0493 -0.0378  0.9698
##
## sigma^2 estimated as 2.168066 on 407 degrees of freedom
##
## AIC = 3.673493  AICc = 3.674348  BIC = 3.760854
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4234 0.3373  1.2555  0.2100
## ar2      0.1665 0.1766  0.9430  0.3463
## ar3     -0.0472 0.1929 -0.2449  0.8066
## ma1      0.0444 0.3348  0.1325  0.8947
## ma2      0.1939 0.2189  0.8858  0.3762
## sar1    -0.3374 0.0466 -7.2354  0.0000
## xmean    0.0106 0.1550  0.0681  0.9458
##
## sigma^2 estimated as 2.421708 on 408 degrees of freedom
##
## AIC = 3.765325  AICc = 3.765988  BIC = 3.842979
##

```

```

## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2279 0.4799   0.4750 0.6351
## ar2      0.1254 0.1607   0.7802 0.4357
## ar3      0.0440 0.2536   0.1733 0.8625
## ma1      0.2239 0.4770   0.4693 0.6391
## ma2      0.2859 0.2693   1.0618 0.2890
## sar1     0.4547 0.0567   8.0260 0.0000
## sma1    -0.9796 0.0717  -13.6564 0.0000
## xmean   -0.0026 0.0249   -0.1024 0.9185
##
## sigma^2 estimated as 2.025512 on 407 degrees of freedom
##
## AIC = 3.638618  AICc = 3.639473  BIC = 3.725978
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3371 0.1021   3.3035 0.0010
## ar2      0.1248 0.1042   1.1980 0.2316
## ar3      0.0013   NaN      NaN      NaN
## ma1      0.1203 0.1018   1.1808 0.2384
## ma2      0.2516 0.0896   2.8087 0.0052
## sar1     0.6392 0.0460  13.8936 0.0000
## sma1    -1.2196 0.0678  -17.9977 0.0000
## sma2     0.2197 0.0581   3.7792 0.0002
## xmean   -0.0022 0.0276   -0.0791 0.9370
##
## sigma^2 estimated as 1.983581 on 406 degrees of freedom
##
## AIC = 3.633683  AICc = 3.634754  BIC = 3.73075
##
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4908 0.3108   1.5792 0.1151
## ar2      0.1654 0.1784   0.9268 0.3546
## ar3     -0.0899 0.1709  -0.5261 0.5991
## ma1     -0.0285 0.3096  -0.0920 0.9268
## ma2      0.1587 0.2025   0.7838 0.4336
## sar1    -0.4021 0.0493  -8.1623 0.0000
## sar2    -0.1850 0.0487  -3.8011 0.0002
## xmean    0.0038 0.1242   0.0307 0.9756
##
## sigma^2 estimated as 2.336663 on 407 degrees of freedom
##
## AIC = 3.736076  AICc = 3.736931  BIC = 3.823437

```



```

##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3148 0.4143  0.7599 0.4478
## ar2      0.1234 0.1625  0.7594 0.4480
## ar3      0.0131 0.2210  0.0591 0.9529
## ma1      0.1429 0.4110  0.3476 0.7283
## ma2      0.2631 0.2351  1.1192 0.2637
## sar1      0.4202 0.0504  8.3444 0.0000
## sar2      0.1010 0.0509  1.9857 0.0477
## sma1     -1.0000 0.0473 -21.1329 0.0000
## xmean    -0.0021 0.0270 -0.0785 0.9375
##
## sigma^2 estimated as 1.981458 on 406 degrees of freedom
##
## AIC = 3.633999  AICc = 3.63507  BIC = 3.731066
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2862    NaN    NaN    NaN
## ar2      0.1816    NaN    NaN    NaN
## ar3     -0.0086    NaN    NaN    NaN
## ma1      0.1661    NaN    NaN    NaN
## ma2      0.2119    NaN    NaN    NaN
## sar1     -0.1555    NaN    NaN    NaN
## sar2      0.3374 0.0192 17.5599 0.0000
## sma1     -0.3891    NaN    NaN    NaN
## sma2     -0.5850    NaN    NaN    NaN
## xmean    -0.0025 0.0262 -0.0938 0.9253
##
## sigma^2 estimated as 2.015028 on 405 degrees of freedom
##
## AIC = 3.644014  AICc = 3.645327  BIC = 3.750788
##
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, :
## possible convergence problem: optim gave code = 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0322 1.0583  0.0304 0.9757
## ar2      0.3530 0.1831  1.9275 0.0546
## ar3     -0.0071 0.3715 -0.0190 0.9848
## ma1      0.4279 1.0573  0.4047 0.6859
## ma2      0.1526 0.3917  0.3897 0.6970
## ma3      0.0518 0.1562  0.3315 0.7404

```

```

## xmean    0.0211 0.2122 0.0997 0.9207
##
## sigma^2 estimated as 2.732888 on 408 degrees of freedom
##
## AIC = 3.883157  AICc = 3.88382  BIC = 3.96081
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.2076 0.3869 -0.5366 0.5918
## ar2     0.4458 0.1155 3.8580 0.0001
## ar3    -0.0529 0.1836 -0.2879 0.7736
## ma1     0.6676 0.3832 1.7419 0.0823
## ma2     0.1998 0.1852 1.0787 0.2813
## ma3     0.2212 0.0987 2.2406 0.0256
## sma1    -0.5590 0.0572 -9.7689 0.0000
## xmean    0.0010 0.0855 0.0121 0.9904
##
## sigma^2 estimated as 2.246438 on 407 degrees of freedom
##
## AIC = 3.700921  AICc = 3.701776  BIC = 3.788281
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.1945 0.8930 -0.2178 0.8277
## ar2     0.3564 0.2840 1.2552 0.2101
## ar3    -0.0048 0.2294 -0.0208 0.9834
## ma1     0.6439 0.8910 0.7227 0.4703
## ma2     0.2494 0.2302 1.0838 0.2791
## ma3     0.1760 0.2090 0.8421 0.4003
## sma1    -0.5334 0.0551 -9.6805 0.0000
## sma2    -0.2069 0.0684 -3.0267 0.0026
## xmean   -0.0017 0.0506 -0.0338 0.9731
##
## sigma^2 estimated as 2.168717 on 406 degrees of freedom
##
## AIC = 3.678109  AICc = 3.679179  BIC = 3.775175
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.1396 0.5110 -0.2733 0.7848
## ar2     0.4164 0.1431 2.9094 0.0038
## ar3    -0.0103 0.2072 -0.0495 0.9605
## ma1     0.6091 0.5086 1.1976 0.2318
## ma2     0.2093 0.2045 1.0232 0.3068
## ma3     0.1594 0.1107 1.4398 0.1507
## sar1    -0.3399 0.0468 -7.2567 0.0000
## xmean    0.0106 0.1540 0.0687 0.9453
##

```

```

## sigma^2 estimated as 2.418672 on 407 degrees of freedom
##
## AIC = 3.768905  AICc = 3.769759  BIC = 3.856265
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1    -0.7251 0.1815  -3.9955 0.0001
## ar2     0.4168 0.0969   4.3006 0.0000
## ar3     0.1792 0.1549   1.1570 0.2480
## ma1     1.1830 0.1800   6.5714 0.0000
## ma2     0.4272 0.1536   2.7814 0.0057
## ma3     0.2277 0.0993   2.2923 0.0224
## sar1     0.4633 0.0589   7.8722 0.0000
## sma1    -0.9716 0.0577 -16.8495 0.0000
## xmean   -0.0023 0.0262  -0.0896 0.9287
##
## sigma^2 estimated as 2.025369 on 406 degrees of freedom
##
## AIC = 3.64037  AICc = 3.641441  BIC = 3.737437
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1    -0.2941 0.7082  -0.4153 0.6782
## ar2     0.3742 0.2102   1.7800 0.0758
## ar3     0.0479 0.2066   0.2321 0.8166
## ma1     0.7520 0.7062   1.0649 0.2876
## ma2     0.2922 0.2139   1.3658 0.1728
## ma3     0.1736 0.1652   1.0506 0.2941
## sar1     0.6436 0.0879   7.3232 0.0000
## sma1    -1.2248 0.1134 -10.8028 0.0000
## sma2     0.2248 0.1062   2.1177 0.0348
## xmean   -0.0023 0.0275  -0.0817 0.9349
##
## sigma^2 estimated as 1.983613 on 405 degrees of freedom
##
## AIC = 3.638298  AICc = 3.639611  BIC = 3.745072
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1    -0.1036 0.5721  -0.1811 0.8564
## ar2     0.4146 0.1501   2.7616 0.0060
## ar3    -0.0455 0.2317  -0.1962 0.8445
## ma1     0.5671 0.5700   0.9949 0.3204
## ma2     0.1848 0.2239   0.8255 0.4096
## ma3     0.1681 0.1179   1.4261 0.1546
## sar1    -0.4059 0.0499  -8.1348 0.0000
## sar2    -0.1848 0.0487  -3.7991 0.0002
## xmean    0.0038 0.1243   0.0308 0.9754

```

```

##
## sigma^2 estimated as 2.33376 on 406 degrees of freedom
##
## AIC = 3.739652  AICc = 3.740723  BIC = 3.836719
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2539 9.5827  0.0265  0.9789
## ar2      0.1454 3.5480  0.0410  0.9673
## ar3      0.0192 0.9133  0.0210  0.9832
## ma1      0.2038 9.5880  0.0213  0.9831
## ma2      0.2690 0.8801  0.3057  0.7600
## ma3      0.0162 2.5755  0.0063  0.9950
## sar1     0.4202 0.0504  8.3325  0.0000
## sar2     0.1010 0.0509  1.9831  0.0480
## sma1    -1.0000 0.0473 -21.1280  0.0000
## xmean   -0.0022 0.0270  -0.0799  0.9363
##
## sigma^2 estimated as 1.98146 on 405 degrees of freedom
##
## AIC = 3.638821  AICc = 3.640133  BIC = 3.745594
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.6957 0.1808 -3.8488  0.0001
## ar2      0.4410 0.0918  4.8046  0.0000
## ar3      0.1705 0.1555  1.0963  0.2736
## ma1      1.1595 0.1792  6.4698  0.0000
## ma2      0.4081 0.1493  2.7327  0.0066
## ma3      0.2361 0.0978  2.4129  0.0163
## sar1      0.6867 0.3913  1.7551  0.0800
## sar2     -0.0192 0.2020 -0.0950  0.9244
## sma1     -1.2557 0.3878 -3.2376  0.0013
## sma2      0.2558 0.3856  0.6632  0.5076
## xmean    -0.0018 0.0286 -0.0640  0.9490
##
## sigma^2 estimated as 1.972935 on 404 degrees of freedom
##
## AIC = 3.640097  AICc = 3.641676  BIC = 3.756578
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2855 1.9316  0.1478  0.8826
## ar2      0.3622 0.2528  1.4328  0.1527
## ar3     -0.0965 0.6975 -0.1383  0.8901
## ma1      0.1743 1.9310  0.0903  0.9281
## ma2      0.0269 0.8252  0.0327  0.9740
## ma3      0.0043 0.2874  0.0150  0.9880

```

```

## ma4      -0.0213 0.1024 -0.2083 0.8351
## xmean    0.0210 0.2133 0.0983 0.9217
##
## sigma^2 estimated as 2.732826 on 407 degrees of freedom
##
## AIC = 3.887954 AICc = 3.888808 BIC = 3.975314
##

## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, :
## possible convergence problem: optim gave code = 1
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, : NaNs
## produced
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, : NaNs
## produced

## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.2562    NaN     NaN    NaN
## ar2       0.4132    NaN     NaN    NaN
## ar3      -0.0267    NaN     NaN    NaN
## ma1       0.7167    NaN     NaN    NaN
## ma2       0.2545    NaN     NaN    NaN
## ma3       0.2373    NaN     NaN    NaN
## ma4       0.0191    NaN     NaN    NaN
## sma1     -0.5585 0.0572 -9.7693 0.0000
## xmean     0.0010 0.0856 0.0119 0.9905
##
## sigma^2 estimated as 2.246372 on 406 degrees of freedom
##
## AIC = 3.705711 AICc = 3.706782 BIC = 3.802778
##

## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced

## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.0033    NaN     NaN    NaN
## ar2       0.4198    NaN     NaN    NaN
## ar3      -0.0820    NaN     NaN    NaN
## ma1       0.4527    NaN     NaN    NaN
## ma2       0.1002    NaN     NaN    NaN
## ma3       0.1254    NaN     NaN    NaN
## ma4      -0.0424    NaN     NaN    NaN
## sma1     -0.5337 0.0551 -9.6859 0.0000
## sma2     -0.2068 0.0680 -3.0397 0.0025
## xmean    -0.0017 0.0506 -0.0337 0.9731
##
## sigma^2 estimated as 2.168679 on 405 degrees of freedom
##
## AIC = 3.682924 AICc = 3.684237 BIC = 3.789698
##

```

```
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1204    NaN     NaN    NaN
## ar2      0.4814    NaN     NaN    NaN
## ar3     -0.1169    NaN     NaN    NaN
## ma1      0.3481    NaN     NaN    NaN
## ma2      0.0221    NaN     NaN    NaN
## ma3      0.0885    NaN     NaN    NaN
## ma4     -0.0557    NaN     NaN    NaN
## sar1     -0.3398 0.0468 -7.2621 0.0000
## xmean     0.0108 0.1555 0.0696 0.9446
##
## sigma^2 estimated as 2.418627 on 406 degrees of freedom
##
## AIC = 3.773697  AICc = 3.774768  BIC = 3.870764
##

## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, :
## possible convergence problem: optim gave code = 1

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -1.2027 0.3076 -3.9093 0.0001
## ar2      0.1589 0.2446 0.6498 0.5162
## ar3      0.3800 0.1371 2.7723 0.0058
## ma1      1.6637 0.3107 5.3545 0.0000
## ma2      0.9075 0.3774 2.4048 0.0166
## ma3      0.3908 0.2224 1.7573 0.0796
## ma4      0.1551 0.1233 1.2576 0.2093
## sar1      0.4670 0.0606 7.7001 0.0000
## sma1     -0.9686 0.0547 -17.6939 0.0000
## xmean    -0.0024 0.0263 -0.0910 0.9276
##
## sigma^2 estimated as 2.02711 on 405 degrees of freedom
##
## AIC = 3.644323  AICc = 3.645635  BIC = 3.751096
##

## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.1484 0.1286 -1.1541 0.2491
## ar2      0.3847    NaN     NaN    NaN
## ar3     -0.0211 0.0877 -0.2404 0.8101
## ma1      0.6121 0.1352 4.5282 0.0000
```

```

## ma2      0.2353      NaN      NaN      NaN
## ma3      0.1828 0.1147  1.5936  0.1118
## ma4      0.0171 0.0331  0.5154  0.6065
## sar1     -0.4053 0.0487 -8.3273  0.0000
## sar2     -0.1848 0.0475 -3.8890  0.0001
## xmean    0.0043 0.1241  0.0348  0.9723
##
## sigma^2 estimated as 2.333717 on 405 degrees of freedom
##
## AIC = 3.744455  AICc = 3.745768  BIC = 3.851229
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.5075 0.0913   5.5612  0.0000
## ar2     -0.9869 0.0432 -22.8680  0.0000
## ar3      0.4501 0.0915   4.9174  0.0000
## ma1     -0.0489 0.0899  -0.5440  0.5867
## ma2      1.2924 0.0680  18.9974  0.0000
## ma3     -0.0276 0.0859  -0.3213  0.7481
## ma4      0.3307 0.0602   5.4938  0.0000
## sar1      0.4447 0.0534   8.3330  0.0000
## sar2      0.0778 0.0530   1.4686  0.1427
## sma1     -1.0000 0.0557 -17.9679  0.0000
## xmean    -0.0025 0.0261  -0.0945  0.9248
##
## sigma^2 estimated as 1.96318 on 404 degrees of freedom
##
## AIC = 3.635249  AICc = 3.636828  BIC = 3.75173
##
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, :
## possible convergence problem: optim gave code = 1
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, : NaNs
## produced
## Warning in arima(xdata, order = c(p, d, q), seasonal = list(order = c(P, : NaNs
## produced
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -1.2752      NaN      NaN      NaN
## ar2      0.1488      NaN      NaN      NaN
## ar3      0.4367 0.0940  4.6456  0.0000
## ma1      1.7437      NaN      NaN      NaN
## ma2      0.9777      NaN      NaN      NaN
## ma3      0.4173      NaN      NaN      NaN
## ma4      0.1880      NaN      NaN      NaN
## sar1      0.7083 0.2170  3.2641  0.0012
## sar2     -0.0196 0.1111 -0.1766  0.8599
## sma1     -1.2762 0.2170 -5.8822  0.0000
## sma2      0.2764 0.2153  1.2838  0.2000
## xmean    -0.0018 0.0290 -0.0607  0.9516

```

```
##
## sigma^2 estimated as 1.969508 on 403 degrees of freedom
##
## AIC = 3.642214  AICc = 3.644084  BIC = 3.768401
##

print("AICcs")

## [1] "AICcs"
## [1] "AICcs"
for (ii in 1:maxAR) {
  for (jj in 1:maxMA) {
    for (kk in 1:maxSAR) {
      for (ll in 1:maxSMA) {
        if(!(ii==4 && jj == 5 && kk == 2 && ll == 3)){ #Condition because (3,4)x(1,2) caused an error
          AICdelta <- ((fits_all[[ii, jj, kk, ll]])$ICs["AICc"] - AICmin) * nn
          if (AICdelta < 10) {
            print(paste0("(", ii - 1, ", ", jj - 1, ") x (", kk - 1, ", ", ll - 1, ")"))
            print(AICdelta)
          }
        }
      }
    }
  }
}

## [1] "(0, 3) x (1, 1)"
##      AICc
## 7.883637
## [1] "(0, 3) x (1, 2)"
##      AICc
## 6.473117
## [1] "(0, 3) x (2, 1)"
##      AICc
## 6.533841
## [1] "(0, 3) x (2, 2)"
##      AICc
## 8.852472
## [1] "(0, 4) x (1, 1)"
##      AICc
## 5.860093
## [1] "(0, 4) x (1, 2)"
##      AICc
## 3.636166
## [1] "(0, 4) x (2, 1)"
##      AICc
## 3.92754
## [1] "(0, 4) x (2, 2)"
##      AICc
## 5.971487
## [1] "(1, 2) x (1, 1)"
##      AICc
## 2.43733
```



```

## [1] "(1, 2) x (1, 2)"
## AICc
## 0
## [1] "(1, 2) x (2, 1)"
## AICc
## 0.1707033
## [1] "(1, 2) x (2, 2)"
## AICc
## 2.378803
## [1] "(1, 3) x (1, 1)"
## AICc
## 3.956456
## [1] "(1, 3) x (1, 2)"
## AICc
## 1.699481
## [1] "(1, 3) x (2, 1)"
## AICc
## 1.827154
## [1] "(1, 3) x (2, 2)"
## AICc
## 4.091502
## [1] "(1, 4) x (1, 1)"
## AICc
## 6.300441
## [1] "(1, 4) x (1, 2)"
## AICc
## 4.060774
## [1] "(1, 4) x (2, 1)"
## AICc
## 4.204162
## [1] "(1, 4) x (2, 2)"
## AICc
## 6.464185
## [1] "(2, 1) x (1, 1)"
## AICc
## 6.963496
## [1] "(2, 1) x (1, 2)"
## AICc
## 4.912816
## [1] "(2, 1) x (2, 1)"
## AICc
## 5.109645
## [1] "(2, 1) x (2, 2)"
## AICc
## 7.26297
## [1] "(2, 2) x (1, 1)"
## AICc
## 3.948499
## [1] "(2, 2) x (1, 2)"
## AICc
## 1.661717
## [1] "(2, 2) x (2, 1)"
## AICc
## 1.815874

```

```

## [1] "(2, 2) x (2, 2)"
##      AICc
## 4.052918
## [1] "(2, 3) x (1, 1)"
##      AICc
## 6.307167
## [1] "(2, 3) x (1, 2)"
##      AICc
## 4.053916
## [1] "(2, 3) x (2, 1)"
##      AICc
## 3.97087
## [1] "(2, 3) x (2, 2)"
##      AICc
## 6.412722
## [1] "(2, 4) x (1, 1)"
##      AICc
## 8.687217
## [1] "(2, 4) x (1, 2)"
##      AICc
## 6.471949
## [1] "(2, 4) x (2, 1)"
##      AICc
## 5.284066
## [1] "(2, 4) x (2, 2)"
##      AICc
## 8.755604
## [1] "(3, 0) x (1, 1)"
##      AICc
## 3.00993
## [1] "(3, 0) x (1, 2)"
##      AICc
## 0.7850133
## [1] "(3, 0) x (2, 1)"
##      AICc
## 1.020183
## [1] "(3, 0) x (2, 2)"
##      AICc
## 3.135869
## [1] "(3, 1) x (1, 1)"
##      AICc
## 5.114312
## [1] "(3, 1) x (1, 2)"
##      AICc
## 2.823175
## [1] "(3, 1) x (2, 1)"
##      AICc
## 3.070278
## [1] "(3, 1) x (2, 2)"
##      AICc
## 5.185817
## [1] "(3, 2) x (1, 1)"
##      AICc
## 6.295416

```

```

## [1] "(3, 2) x (1, 2)"
##      AICc
## 4.054001
## [1] "(3, 2) x (2, 1)"
##      AICc
## 4.204054
## [1] "(3, 2) x (2, 2)"
##      AICc
## 9.075899
## [1] "(3, 3) x (1, 1)"
##      AICc
## 7.23045
## [1] "(3, 3) x (1, 2)"
##      AICc
## 6.360809
## [1] "(3, 3) x (2, 1)"
##      AICc
## 6.608925
## [1] "(3, 3) x (2, 2)"
##      AICc
## 7.34185
## [1] "(3, 4) x (1, 1)"
##      AICc
## 9.222438
## [1] "(3, 4) x (2, 1)"
##      AICc
## 5.039024
## [1] "(3, 4) x (2, 2)"
##      AICc
## 8.485563

print("BICs")

## [1] "BICs"

## [1] "BICs"
for (ii in 1:maxAR) {
  for (jj in 1:maxMA) {
    for (kk in 1:maxSAR) {
      for (ll in 1:maxSMA) {
        if(!(ii==4 && jj == 5 && kk == 2 && ll == 3)){ #Condition because (3,4)x(1,2) caused an error
          BICdelta <- ((fits_all[[ii, jj, kk, ll]])$ICs["BIC"] - BICmin) * nn
          if (BICdelta < 14) {
            print(paste0("(", ii - 1, ", ", jj - 1, ") x (", kk - 1, ", ", ll - 1, ")"))
            print(BICdelta)
          }
        }
      }
    }
  }
}

## [1] "(0, 2) x (1, 1)"
##      BIC

```

```

## 9.670495
## [1] "(0, 3) x (1, 1)"
##      BIC
## 5.446307
## [1] "(0, 3) x (1, 2)"
##      BIC
## 8.567146
## [1] "(0, 3) x (2, 1)"
##      BIC
## 8.627869
## [1] "(0, 4) x (1, 1)"
##      BIC
## 7.954121
## [1] "(0, 4) x (1, 2)"
##      BIC
## 10.24989
## [1] "(0, 4) x (2, 1)"
##      BIC
## 10.54126
## [1] "(1, 2) x (1, 1)"
## BIC
##      0
## [1] "(1, 2) x (1, 2)"
##      BIC
## 2.094028
## [1] "(1, 2) x (2, 1)"
##      BIC
## 2.264732
## [1] "(1, 2) x (2, 2)"
##      BIC
## 8.992522
## [1] "(1, 3) x (1, 1)"
##      BIC
## 6.050485
## [1] "(1, 3) x (1, 2)"
##      BIC
## 8.313201
## [1] "(1, 3) x (2, 1)"
##      BIC
## 8.440874
## [1] "(1, 4) x (1, 1)"
##      BIC
## 12.91416
## [1] "(2, 0) x (1, 1)"
##      BIC
## 4.918086
## [1] "(2, 0) x (1, 2)"
##      BIC
## 7.778666
## [1] "(2, 0) x (2, 1)"
##      BIC
## 7.880067
## [1] "(2, 1) x (1, 1)"
##      BIC

```

```

## 4.526166
## [1] "(2, 1) x (1, 2)"
##      BIC
## 7.006845
## [1] "(2, 1) x (2, 1)"
##      BIC
## 7.203673
## [1] "(2, 1) x (2, 2)"
##      BIC
## 13.87669
## [1] "(2, 2) x (1, 1)"
##      BIC
## 6.042527
## [1] "(2, 2) x (1, 2)"
##      BIC
## 8.275437
## [1] "(2, 2) x (2, 1)"
##      BIC
## 8.429594
## [1] "(2, 3) x (1, 1)"
##      BIC
## 12.92089
## [1] "(3, 0) x (1, 1)"
##      BIC
## 0.5725994
## [1] "(3, 0) x (1, 2)"
##      BIC
## 2.879042
## [1] "(3, 0) x (2, 1)"
##      BIC
## 3.114211
## [1] "(3, 0) x (2, 2)"
##      BIC
## 9.749589
## [1] "(3, 1) x (1, 1)"
##      BIC
## 7.20834
## [1] "(3, 1) x (1, 2)"
##      BIC
## 9.436895
## [1] "(3, 1) x (2, 1)"
##      BIC
## 9.683998
## [1] "(3, 2) x (1, 1)"
##      BIC
## 12.90914

```

SARMA(1,2)x(1,1) on seasonally differenced data:

```

sarima(Dxx, p = 1, d = 0, q = 2, P = 1, D = 0,
Q = 1, S = 10,no.constant=F,details=T)

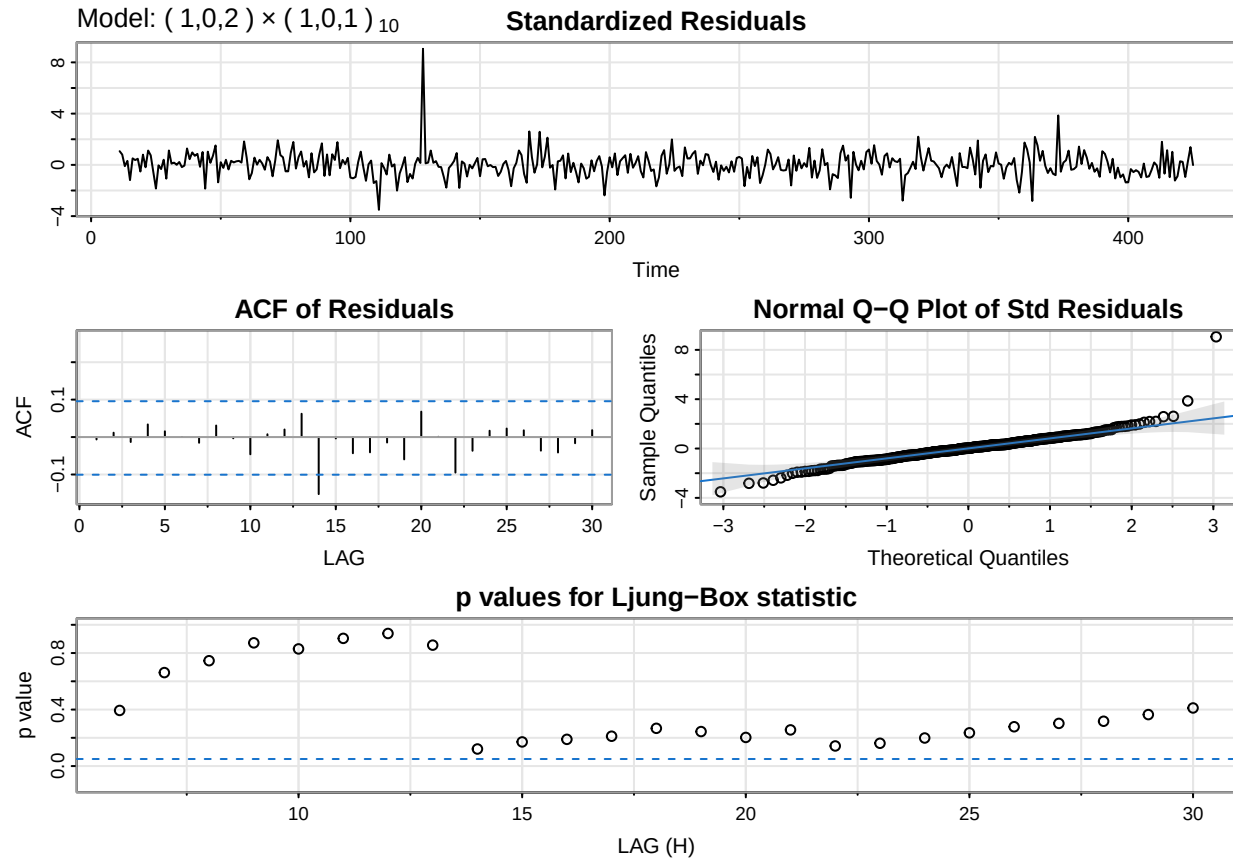
```

```

## initial  value 0.739706
## iter    2 value 0.653641
## iter    3 value 0.444605

```

```
## iter 4 value 0.434881
## iter 5 value 0.425588
## iter 6 value 0.396391
## iter 7 value 0.391841
## iter 8 value 0.390219
## iter 9 value 0.389542
## iter 10 value 0.389190
## iter 11 value 0.388424
## iter 12 value 0.388037
## iter 13 value 0.387878
## iter 14 value 0.387782
## iter 15 value 0.387770
## iter 16 value 0.387762
## iter 17 value 0.387757
## iter 18 value 0.387754
## iter 19 value 0.387752
## iter 20 value 0.387751
## iter 21 value 0.387751
## iter 21 value 0.387751
## iter 21 value 0.387751
## final value 0.387751
## converged
## initial value 0.384937
## iter 2 value 0.382363
## iter 3 value 0.380148
## iter 4 value 0.379891
## iter 5 value 0.379669
## iter 6 value 0.379638
## iter 7 value 0.379634
## iter 8 value 0.379630
## iter 9 value 0.379622
## iter 10 value 0.379621
## iter 11 value 0.379621
## iter 11 value 0.379621
## iter 11 value 0.379621
## final value 0.379621
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1      0.4565 0.1006   4.5382 0.0000
## ma1      0.0006 0.1025   0.0060 0.9952
## ma2      0.2974 0.0605   4.9163 0.0000
## sar1     0.4469 0.0565   7.9086 0.0000
## sma1    -0.9728 0.0564  -17.2566 0.0000
## xmean   -0.0030 0.0242  -0.1250 0.9006
##
## sigma^2 estimated as 2.037676 on 409 degrees of freedom
##
## AIC = 3.630854 AICc = 3.631351 BIC = 3.698801
##
```



SARMA(1,2)x(1,2) on seasonally differenced data:

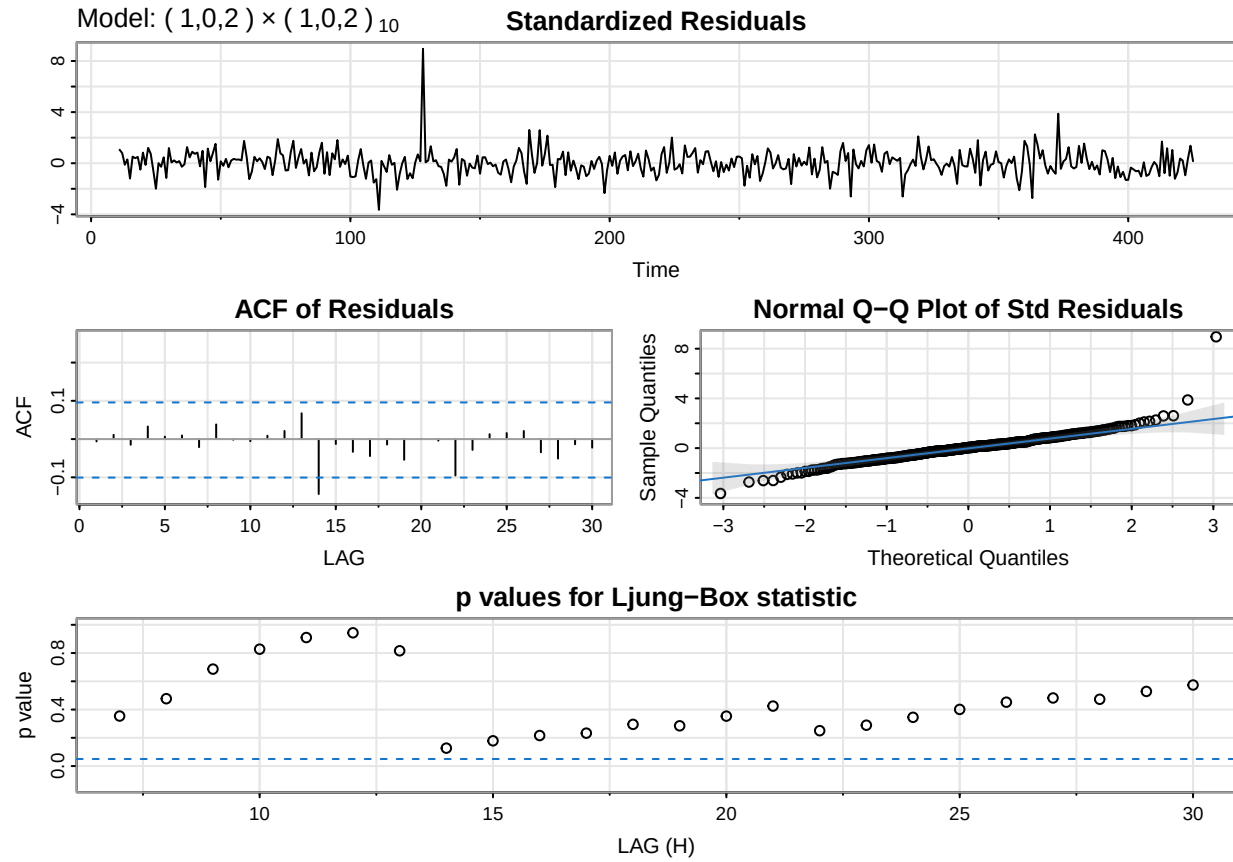
```
sarima(Dxx, p = 1, d = 0, q = 2, P = 1, D = 0,
Q = 2, S = 10,no.constant=F,details=T)
```

```
## initial value 0.739706
## iter 2 value 0.631691
## iter 3 value 0.420199
## iter 4 value 0.411322
## iter 5 value 0.406670
## iter 6 value 0.404857
## iter 7 value 0.401537
## iter 8 value 0.397428
## iter 9 value 0.395333
## iter 10 value 0.393564
## iter 11 value 0.391162
## iter 12 value 0.389451
## iter 13 value 0.388646
## iter 14 value 0.387858
## iter 15 value 0.387754
## iter 16 value 0.387745
## iter 17 value 0.387744
## iter 18 value 0.387744
## iter 19 value 0.387744
## iter 20 value 0.387744
## iter 21 value 0.387744
## iter 22 value 0.387744
```

```

## iter 22 value 0.387744
## iter 22 value 0.387744
## final value 0.387744
## converged
## initial value 0.385014
## iter 2 value 0.384016
## iter 3 value 0.383865
## iter 4 value 0.381967
## iter 5 value 0.380416
## iter 6 value 0.379149
## iter 7 value 0.377295
## iter 8 value 0.375941
## iter 9 value 0.374914
## iter 10 value 0.374775
## iter 11 value 0.374644
## iter 12 value 0.374599
## iter 13 value 0.374590
## iter 14 value 0.374570
## iter 15 value 0.374564
## iter 16 value 0.374563
## iter 17 value 0.374562
## iter 17 value 0.374562
## iter 17 value 0.374562
## final value 0.374562
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4777 0.0946  5.0506 0.0000
## ma1     -0.0153 0.0965 -0.1586 0.8740
## ma2      0.3022 0.0593  5.0964 0.0000
## sar1     0.6412 0.0877  7.3117 0.0000
## sma1    -1.2264 0.1134 -10.8192 0.0000
## sma2     0.2264 0.1054  2.1484 0.0323
## xmean   -0.0026 0.0265 -0.0991 0.9211
##
## sigma^2 estimated as 1.986601 on 408 degrees of freedom
##
## AIC = 3.625556  AICc = 3.626219  BIC = 3.70321
##

```

SARMA(1,2)x(2,1) on seasonally differenced data:

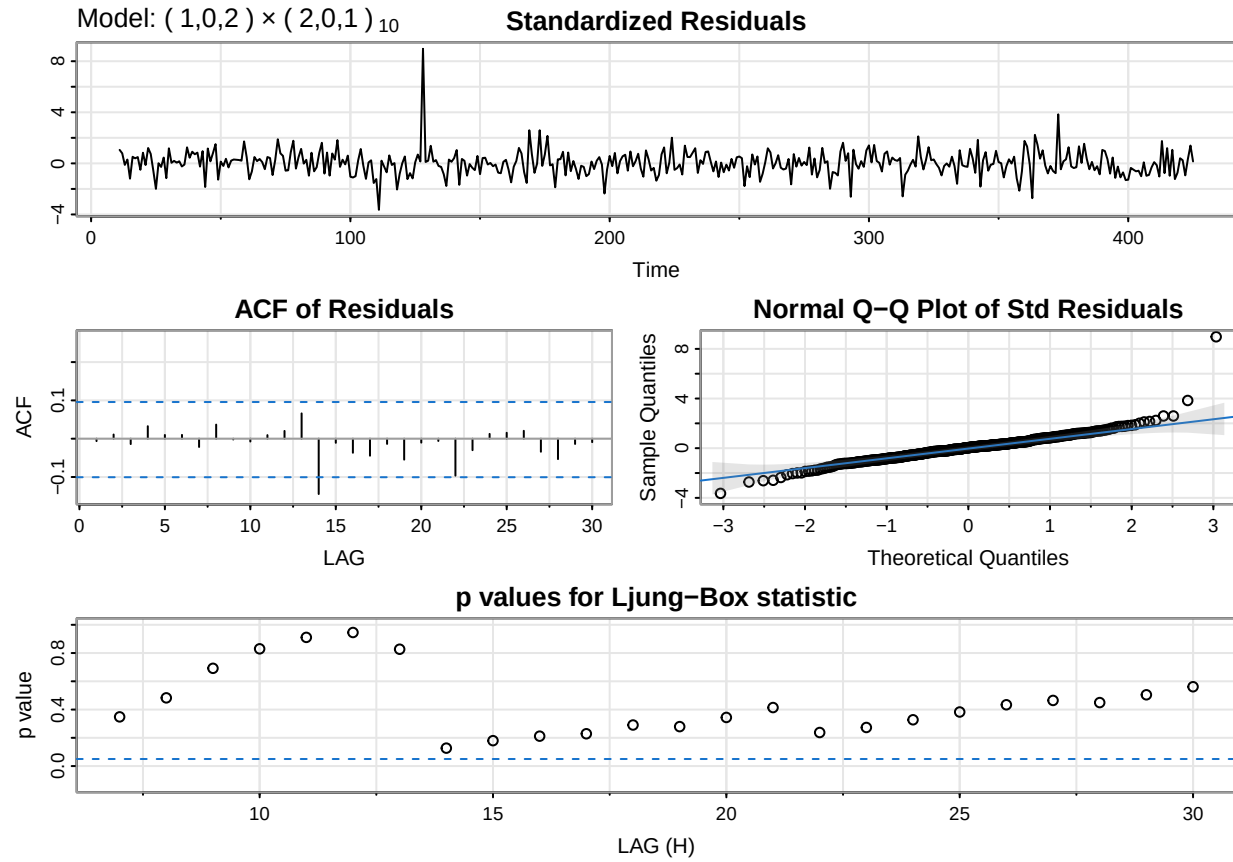
```
sarima(Dxx, p = 1, d = 0, q = 2, P = 2, D = 0,
Q = 1, S = 10,no.constant=F,details=T)
```

```
## initial value 0.746372
## iter 2 value 0.649949
## iter 3 value 0.438955
## iter 4 value 0.433438
## iter 5 value 0.429618
## iter 6 value 0.424113
## iter 7 value 0.418200
## iter 8 value 0.415358
## iter 9 value 0.413186
## iter 10 value 0.412411
## iter 11 value 0.411324
## iter 12 value 0.411007
## iter 13 value 0.410958
## iter 14 value 0.410954
## iter 15 value 0.410954
## iter 16 value 0.410954
## iter 16 value 0.410954
## final value 0.410954
## converged
## initial value 0.397900
## iter 2 value 0.395699
## iter 3 value 0.384648
```

```

## iter    4 value 0.379395
## iter    5 value 0.375745
## iter    6 value 0.375233
## iter    7 value 0.374803
## iter    8 value 0.374767
## iter    9 value 0.374755
## iter   10 value 0.374752
## iter   11 value 0.374745
## iter   12 value 0.374743
## iter   13 value 0.374742
## iter   14 value 0.374742
## iter   14 value 0.374742
## iter   14 value 0.374742
## final   value 0.374742
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4762 0.0951  5.0058 0.0000
## ma1     -0.0131 0.0969 -0.1352 0.8925
## ma2      0.3027 0.0597  5.0714 0.0000
## sar1      0.4157 0.0500  8.3159 0.0000
## sar2      0.1031 0.0508  2.0298 0.0430
## sma1     -1.0000 0.0502 -19.9366 0.0000
## xmean    -0.0026 0.0259 -0.1002 0.9202
##
## sigma^2 estimated as 1.984303 on 408 degrees of freedom
##
## AIC = 3.625916  AICc = 3.626579  BIC = 3.703569
##

```



SARMA(2,0)x(1,1) on seasonally differenced data:

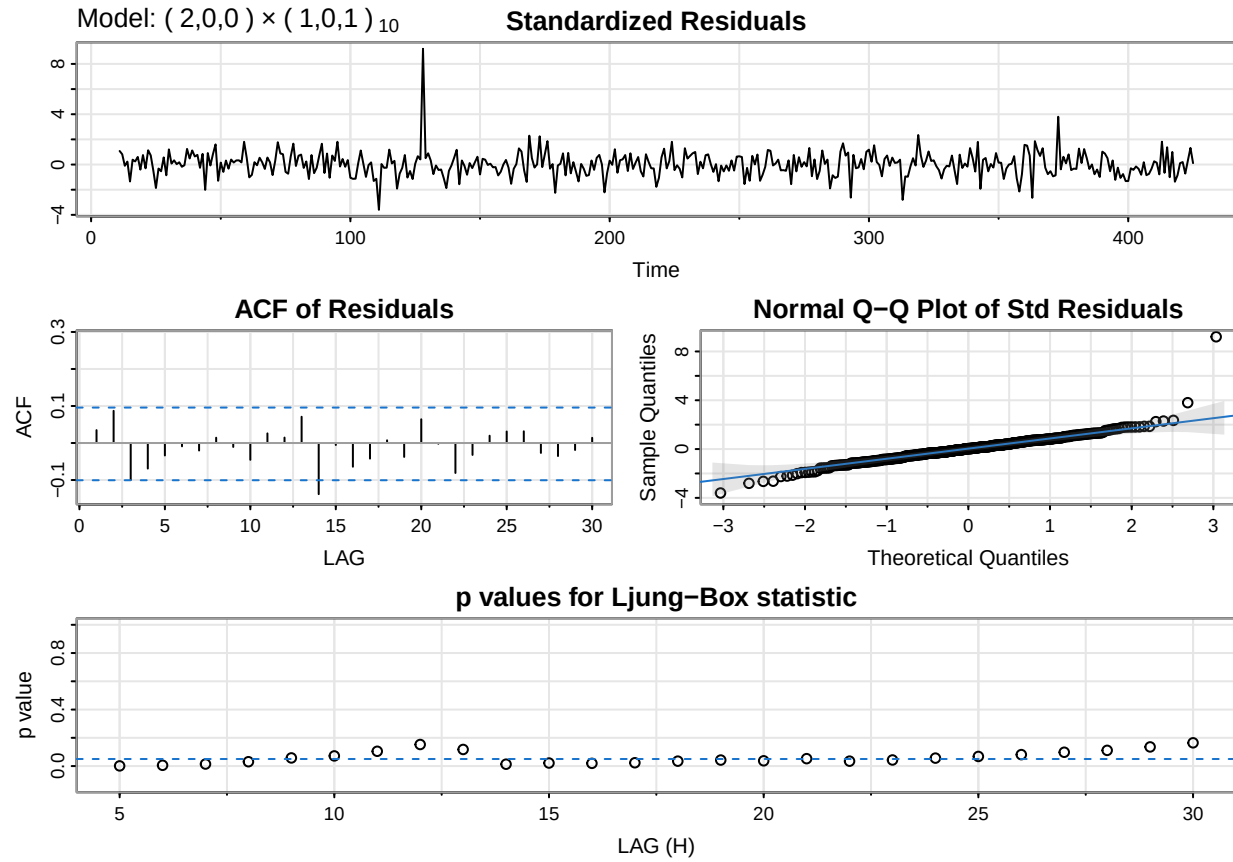
```
sarima(Dxx, p = 2, d = 0, q = 0, P = 1, D = 0,
Q = 1, S = 10,no.constant=F,details=T)
```

```
## initial value 0.740334
## iter 2 value 0.530507
## iter 3 value 0.452308
## iter 4 value 0.442682
## iter 5 value 0.437071
## iter 6 value 0.434049
## iter 7 value 0.424336
## iter 8 value 0.416737
## iter 9 value 0.408794
## iter 10 value 0.407812
## iter 11 value 0.406911
## iter 12 value 0.406697
## iter 13 value 0.406692
## iter 14 value 0.406689
## iter 15 value 0.406689
## iter 16 value 0.406689
## iter 17 value 0.406689
## iter 17 value 0.406689
## iter 17 value 0.406689
## final value 0.406689
## converged
## initial value 0.401473
```

```

## iter 2 value 0.397607
## iter 3 value 0.395458
## iter 4 value 0.392589
## iter 5 value 0.392327
## iter 6 value 0.392189
## iter 7 value 0.392124
## iter 8 value 0.392121
## iter 9 value 0.392071
## iter 10 value 0.392066
## iter 11 value 0.392065
## iter 12 value 0.392065
## iter 13 value 0.392064
## iter 14 value 0.392062
## iter 15 value 0.392061
## iter 16 value 0.392061
## iter 17 value 0.392061
## iter 17 value 0.392061
## iter 17 value 0.392061
## final value 0.392061
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4203 0.0477  8.8148 0.0000
## ar2      0.2311 0.0480  4.8178 0.0000
## sar1      0.4651 0.0560  8.2991 0.0000
## sma1     -0.9832 0.0846 -11.6252 0.0000
## xmean    -0.0007 0.0289  -0.0230 0.9817
##
## sigma^2 estimated as 2.076852 on 410 degrees of freedom
##
## AIC = 3.650915  AICc = 3.651269  BIC = 3.709155
##

```



SARMA(2,1)x(1,1) on seasonally differenced data:

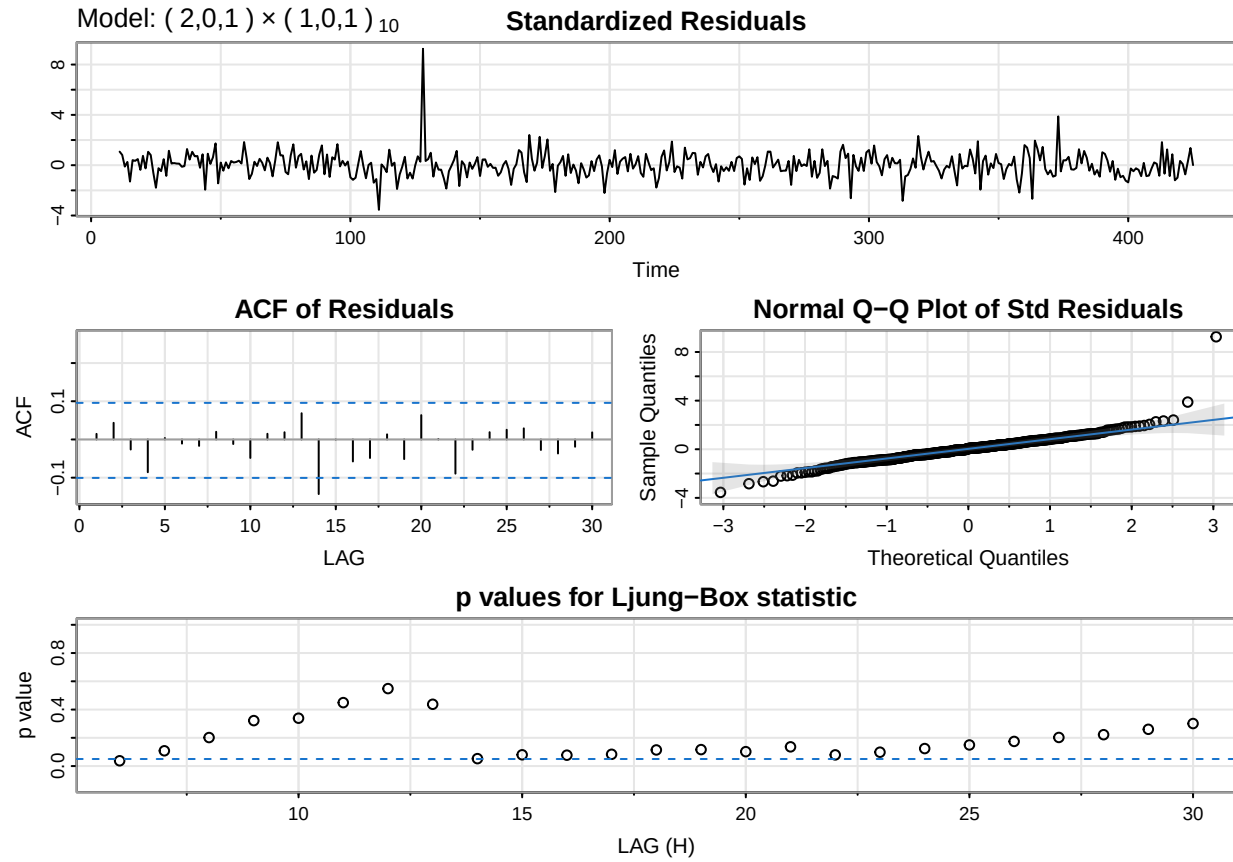
```
sarima(Dxx, p = 2, d = 0, q = 1, P = 1, D = 0,
Q = 1, S = 10,no.constant=F,details=T)
```

```
## initial value 0.740334
## iter 2 value 0.679786
## iter 3 value 0.494603
## iter 4 value 0.494550
## iter 5 value 0.433766
## iter 6 value 0.429387
## iter 7 value 0.416586
## iter 8 value 0.408241
## iter 9 value 0.406841
## iter 10 value 0.402308
## iter 11 value 0.402279
## iter 12 value 0.402148
## iter 13 value 0.402123
## iter 14 value 0.402086
## iter 15 value 0.402078
## iter 16 value 0.402059
## iter 17 value 0.402037
## iter 18 value 0.402032
## iter 19 value 0.402024
## iter 20 value 0.402022
## iter 21 value 0.402022
## iter 21 value 0.402022
```

```

## iter 21 value 0.402022
## final value 0.402022
## converged
## initial value 0.394928
## iter 2 value 0.391272
## iter 3 value 0.386093
## iter 4 value 0.385870
## iter 5 value 0.385489
## iter 6 value 0.384994
## iter 7 value 0.384820
## iter 8 value 0.384631
## iter 9 value 0.384581
## iter 10 value 0.384548
## iter 11 value 0.384491
## iter 12 value 0.384443
## iter 13 value 0.384398
## iter 14 value 0.384388
## iter 15 value 0.384386
## iter 16 value 0.384386
## iter 17 value 0.384386
## iter 18 value 0.384386
## iter 18 value 0.384386
## iter 18 value 0.384386
## final value 0.384386
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0668 0.1233  0.5420 0.5881
## ar2      0.4325 0.0699  6.1838 0.0000
## ma1      0.3722 0.1308  2.8443 0.0047
## sar1     0.4723 0.0565  8.3575 0.0000
## sma1    -0.9893 0.1329 -7.4445 0.0000
## xmean   -0.0013 0.0273 -0.0476 0.9620
##
## sigma^2 estimated as 2.035716 on 409 degrees of freedom
##
## AIC = 3.640383  AICc = 3.640879  BIC = 3.70833
##

```



SARMA(3,0)x(1,1) on seasonally differenced data:

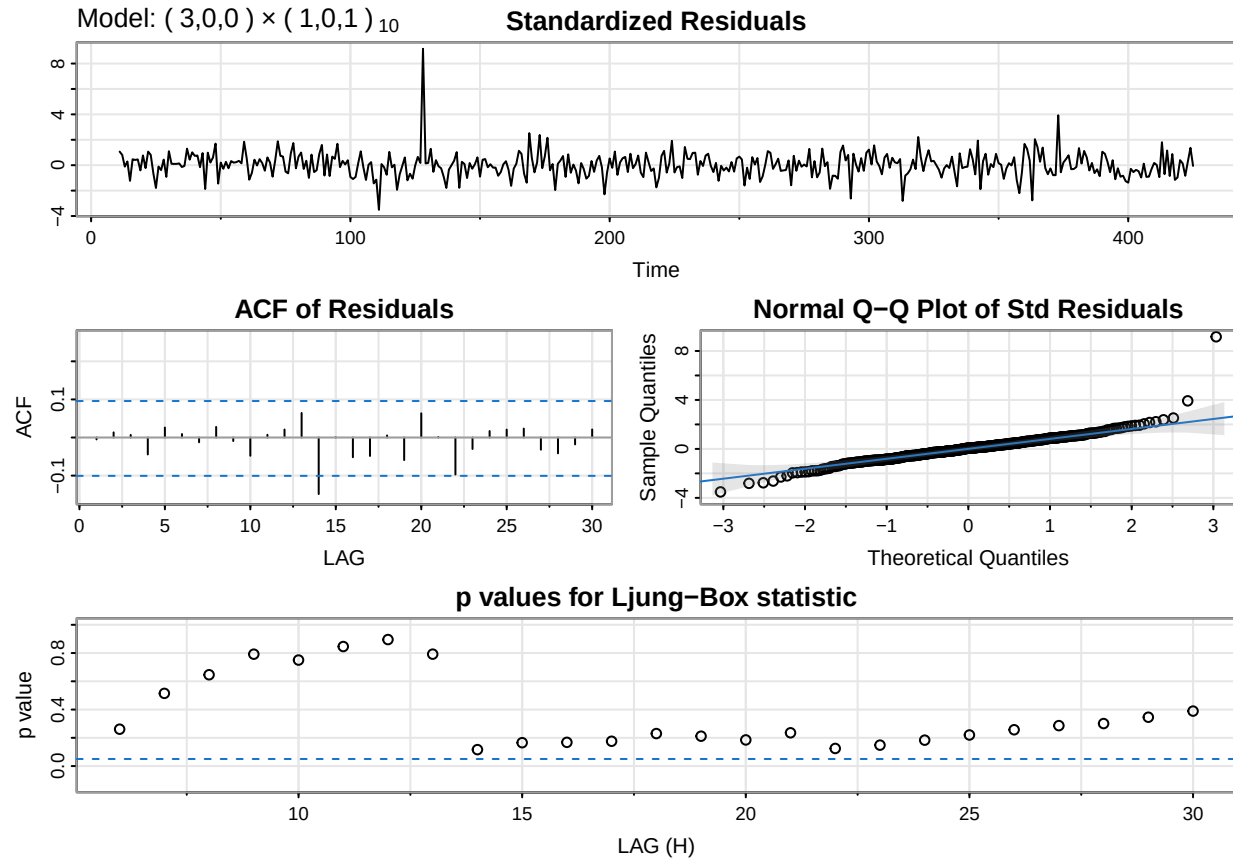
```
sarima(Dxx, p = 3, d = 0, q = 0, P = 1, D = 0,
Q = 1, S = 10,no.constant=F,details=T)
```

```
## initial value 0.740763
## iter 2 value 0.655813
## iter 3 value 0.444419
## iter 4 value 0.429339
## iter 5 value 0.423511
## iter 6 value 0.413245
## iter 7 value 0.407827
## iter 8 value 0.403288
## iter 9 value 0.399369
## iter 10 value 0.396539
## iter 11 value 0.395844
## iter 12 value 0.394191
## iter 13 value 0.393594
## iter 14 value 0.393507
## iter 15 value 0.393501
## iter 16 value 0.393491
## iter 17 value 0.393491
## iter 18 value 0.393490
## iter 19 value 0.393490
## iter 19 value 0.393490
## iter 19 value 0.393490
## final value 0.393490
```

```

## converged
## initial value 0.388803
## iter 2 value 0.385194
## iter 3 value 0.383299
## iter 4 value 0.380668
## iter 5 value 0.380327
## iter 6 value 0.380251
## iter 7 value 0.380236
## iter 8 value 0.380233
## iter 9 value 0.380231
## iter 10 value 0.380231
## iter 11 value 0.380228
## iter 12 value 0.380226
## iter 13 value 0.380224
## iter 14 value 0.380224
## iter 15 value 0.380224
## iter 16 value 0.380224
## iter 17 value 0.380224
## iter 17 value 0.380224
## iter 17 value 0.380224
## final value 0.380224
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4555 0.0484  9.4097 0.0000
## ar2      0.2946 0.0515  5.7211 0.0000
## ar3     -0.1534 0.0487 -3.1526 0.0017
## sar1      0.4646 0.0563  8.2509 0.0000
## sma1     -0.9834 0.0868 -11.3317 0.0000
## xmean    -0.0025 0.0248 -0.1012 0.9194
##
## sigma^2 estimated as 2.027258 on 409 degrees of freedom
##
## AIC = 3.63206  AICc = 3.632556  BIC = 3.700007
##

```

SARMA(3,0)x(1,2) on seasonally differenced data:

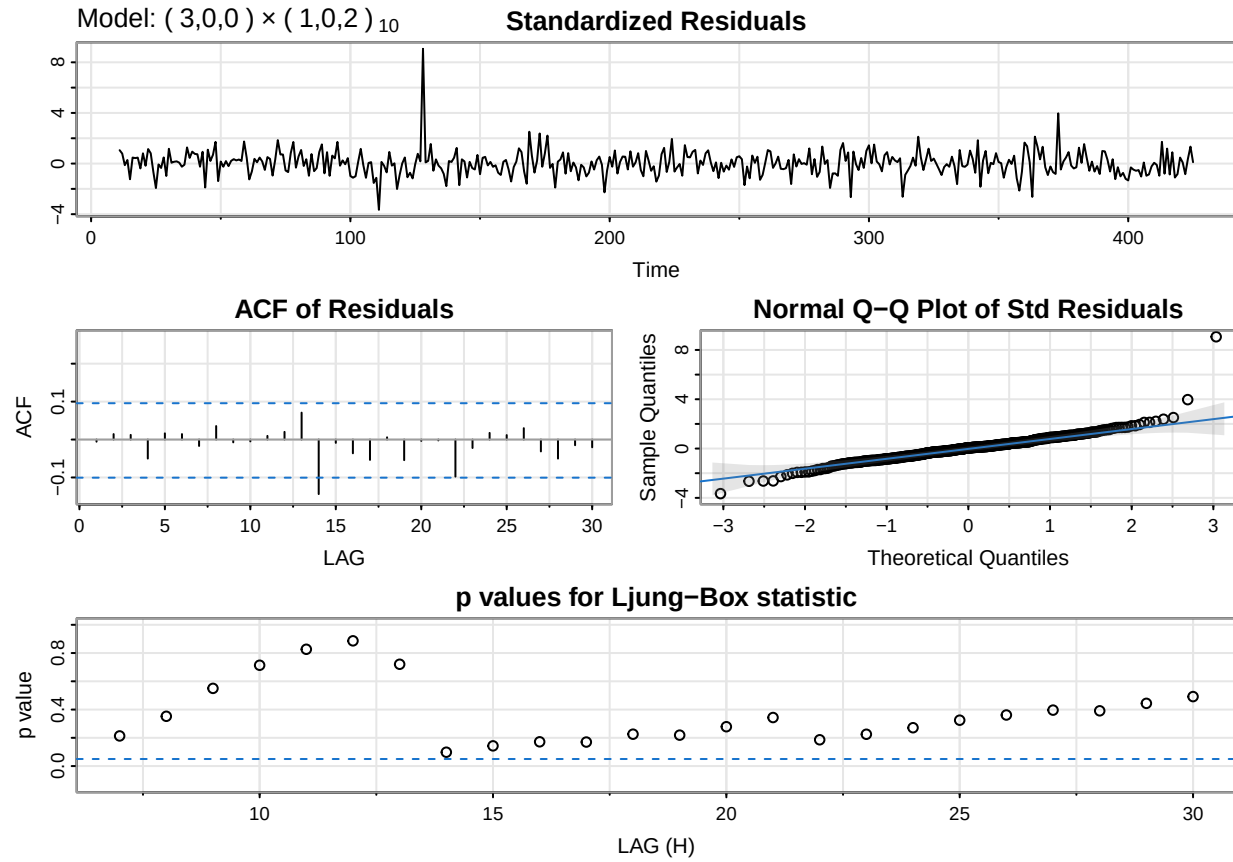
```
sarima(Dxx, p = 3, d = 0, q = 0, P = 1, D = 0,
Q = 2, S = 10,no.constant=F,details=T)
```

```
## initial value 0.740763
## iter 2 value 0.631071
## iter 3 value 0.413234
## iter 4 value 0.403959
## iter 5 value 0.403438
## iter 6 value 0.402810
## iter 7 value 0.402159
## iter 8 value 0.399867
## iter 9 value 0.397029
## iter 10 value 0.394949
## iter 11 value 0.394807
## iter 12 value 0.393320
## iter 13 value 0.393308
## iter 14 value 0.393307
## iter 15 value 0.393306
## iter 16 value 0.393306
## iter 16 value 0.393306
## final value 0.393306
## converged
## initial value 0.389219
## iter 2 value 0.386134
## iter 3 value 0.384846
```

```

## iter    4 value 0.383903
## iter    5 value 0.382182
## iter    6 value 0.379259
## iter    7 value 0.377305
## iter    8 value 0.375569
## iter    9 value 0.375405
## iter   10 value 0.375393
## iter   11 value 0.375389
## iter   12 value 0.375389
## iter   13 value 0.375389
## iter   13 value 0.375389
## iter   13 value 0.375389
## final   value 0.375389
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4619 0.0484   9.5369 0.0000
## ar2      0.3037 0.0516   5.8837 0.0000
## ar3     -0.1573 0.0487  -3.2302 0.0013
## sar1      0.6425 0.0859   7.4828 0.0000
## sma1     -1.2152 0.1103 -11.0176 0.0000
## sma2      0.2152 0.1034   2.0802 0.0381
## xmean    -0.0021 0.0279  -0.0738 0.9412
##
## sigma^2 estimated as 1.990742 on 408 degrees of freedom
##
## AIC = 3.627209  AICc = 3.627872  BIC = 3.704862
##

```



SARMA(3,0)x(2,1) on seasonally differenced data:

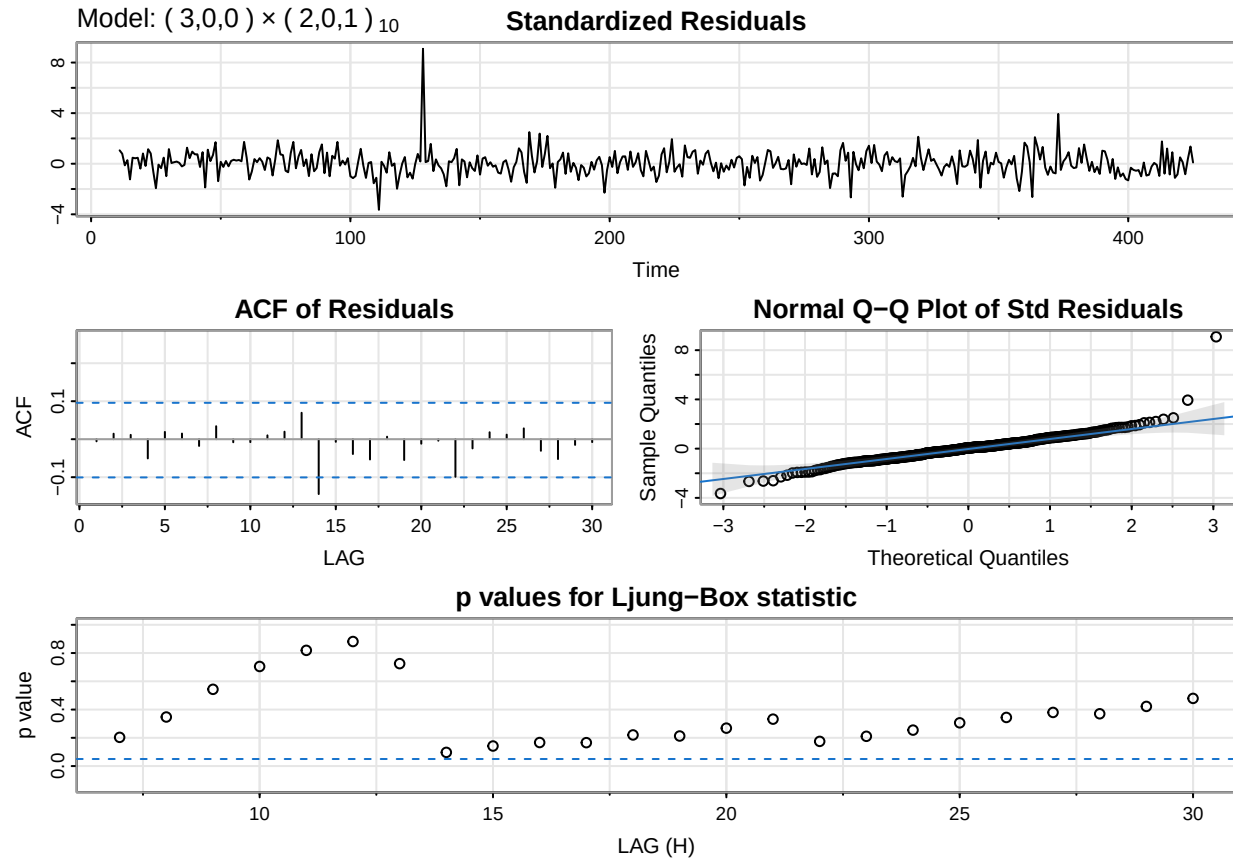
```
sarima(Dxx, p = 3, d = 0, q = 0, P = 2, D = 0,
Q = 1, S = 10,no.constant=F,details=T)
```

```
## initial value 0.748845
## iter 2 value 0.648994
## iter 3 value 0.439399
## iter 4 value 0.429293
## iter 5 value 0.427450
## iter 6 value 0.426197
## iter 7 value 0.419091
## iter 8 value 0.417478
## iter 9 value 0.416058
## iter 10 value 0.414738
## iter 11 value 0.413745
## iter 12 value 0.413627
## iter 13 value 0.413611
## iter 14 value 0.413608
## iter 15 value 0.413608
## iter 16 value 0.413608
## iter 16 value 0.413608
## final value 0.413608
## converged
## initial value 0.396859
## iter 2 value 0.394629
## iter 3 value 0.381433
```

```

## iter    4 value 0.380451
## iter    5 value 0.379896
## iter    6 value 0.378865
## iter    7 value 0.377748
## iter    8 value 0.376804
## iter    9 value 0.375935
## iter   10 value 0.375704
## iter   11 value 0.375650
## iter   12 value 0.375639
## iter   13 value 0.375637
## iter   14 value 0.375636
## iter   14 value 0.375636
## iter   14 value 0.375636
## final   value 0.375636
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4620 0.0485   9.5300 0.0000
## ar2      0.3029 0.0516   5.8680 0.0000
## ar3     -0.1563 0.0487  -3.2117 0.0014
## sar1      0.4292 0.0497   8.6315 0.0000
## sar2      0.0986 0.0505   1.9544 0.0513
## sma1     -1.0000 0.0446 -22.4045 0.0000
## xmean    -0.0020 0.0273  -0.0739 0.9411
##
## sigma^2 estimated as 1.988992 on 408 degrees of freedom
##
## AIC = 3.627704  AICc = 3.628367  BIC = 3.705358
##

```



Final choice: SARMA(1,2)x(1,1) on seasonally differenced data with no intercept term:

```
sarima(Dxx, p = 1, d = 0, q = 2, P = 1, D = 0,
Q = 1, S = 10,no.constant=T,details=T)
```

```
## initial value 0.739679
## iter 2 value 0.653620
## iter 3 value 0.444598
## iter 4 value 0.434880
## iter 5 value 0.425594
## iter 6 value 0.396457
## iter 7 value 0.391811
## iter 8 value 0.390192
## iter 9 value 0.389506
## iter 10 value 0.389178
## iter 11 value 0.388430
## iter 12 value 0.388061
## iter 13 value 0.387978
## iter 14 value 0.387876
## iter 15 value 0.387811
## iter 16 value 0.387792
## iter 17 value 0.387785
## iter 18 value 0.387784
## iter 19 value 0.387784
## iter 19 value 0.387784
## iter 19 value 0.387784
## final value 0.387784
```

```

## converged
## initial value 0.384921
## iter 2 value 0.382347
## iter 3 value 0.380113
## iter 4 value 0.379664
## iter 5 value 0.379658
## iter 6 value 0.379654
## iter 7 value 0.379643
## iter 8 value 0.379641
## iter 9 value 0.379640
## iter 10 value 0.379640
## iter 10 value 0.379640
## iter 10 value 0.379640
## final value 0.379640
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.4570 0.1005   4.5483 0.0000
## ma1      0.0002 0.1024   0.0020 0.9984
## ma2      0.2973 0.0605   4.9145 0.0000
## sar1     0.4470 0.0565   7.9050 0.0000
## sma1    -0.9727 0.0562  -17.3005 0.0000
##
## sigma^2 estimated as 2.037939 on 410 degrees of freedom
##
## AIC = 3.626073  AICc = 3.626426  BIC = 3.684313
##

```

